

Bit Complexity of LS'15 under Fixed Point

Abstract

We show that the linear programming runtime of $\tilde{O}(d^{1/2}(\text{nnz}(A) + d^2))$ arithmetic operations by [Lee-Sidford FOCS'15] holds under bit complexity: on inputs with B bits the method takes $\tilde{O}(d^{1/2}(\text{nnz}(A) + d^2)B^3)$ bit operations, where $\tilde{O}$ hides factors polylogarithmic in n .

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1 Introduction

Let $1 \leq d \leq n \leq d^{10}$, let $A \in \mathbb{R}^{n \times d}$ have full column rank and no zero row, and consider the linear program

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & c^\top x \\ \text{subject to} \quad & A^\top x = b, \\ & l \leq x \leq u, \end{aligned} \tag{primal}$$

whose dual is

$$\begin{aligned} \max_{y \in \mathbb{R}^d, sl, su \in \mathbb{R}^n} \quad & b^\top y + l^\top sl - u^\top su \\ \text{subject to} \quad & Ay + sl - su = c, \\ & sl, su \geq 0. \end{aligned} \tag{dual}$$

We refer to these two programs as the primal/dual pair.

We work in the fixed point model of [GPV23]: the input is given, and every number the algorithm forms is kept, as a signed binary fraction with B bits before and after the decimal point, each arithmetic operation on such numbers costs $\tilde{O}(B)$ bit operations, and every intermediate result is truncated back to this precision. The model is made precise in Section 2.1, and all costs below are bit complexities in it. Throughout, $\tilde{O}(\cdot)$ hides factors polynomial in $\log n$ only; every power of B is written out. Since $B \leq d^{1/2} \leq n$, factors polynomial in $\log B$, such as the $\log B$ of fast integer multiplication, are hidden as well. We assume throughout that $B \leq d^{1/2}$.

Theorem 1. *Given a primal/dual pair A, b, c, l, u and a point $x^{(0)}$ with $B \leq d^{1/2}$ bits before/after decimal place in fixed point, such that $\|A^\top A\|, \|(A^\top A)^{-1}\| \leq 2^B$, and $l + 2^{-B} < x^{(0)} < u - 2^{-B}$ and $A^\top x^{(0)} = b$, we can compute $x \in [l, u]$ satisfying $\|A^\top x - b\|_2 \leq 2^{-B}$ and $c^\top x \leq \text{opt} + 2^{-B}$, where opt is the optimum of the primal, in bit complexity $\tilde{O}(d^{1/2}(\text{nnz}(A) + d^2)B^3)$ with high probability.*

The objective guarantee is one sided, as in [LS15, vdBLL⁺21]: the returned x is only 2^{-B} -feasible, and a lower bound on $c^\top x$ in terms of opt would need the sensitivity of the optimum to b , which is not controlled here. The point $x^{(0)}$ is used only to know that the primal is feasible.

Why the bound does not follow from LS15. Theorem 1 is the fixed point counterpart of the running time $\tilde{O}(d^{1/2}(\text{nnz}(A) + d^2) \log(1/\epsilon))$ of [LS15], and it is not obtained from that paper by inspection. The count there is of arithmetic operations on real numbers: the inverse maintenance keeps a $d \times d$ inverse of a sampled matrix, corrects it by low rank updates, and uses it as a preconditioner for an ϵ -approximate solve, and none of the growth of the entries of the maintained matrices, the compounding of truncation errors over the $\tilde{O}(d^{1/2})$ updates, or the loss of the spectral guarantee of a row sample chosen from truncated weights, is accounted for. Three further points have to be settled before the arithmetic count converts. The path following analysis is stated for exact Newton directions computed from approximate weights, so a direction that is itself the truncated output of an approximate solver needs its own perturbation argument (Section 5); the sampled inverse of [LS15] is valid only against weights that do not depend on its random choices, and that paper hides the sample from the path following method by adding Gaussian noise to a real-valued solve, an argument that has to be redone when the noise is discrete, the solve is truncated, and everything the caller derives from the output is itself truncated (Section 3.3); and the magnitudes of the weights, the inverses and the iterates along the path have to be bounded by $2^{O(B)}$ in terms of the input so that the fixed point numbers do not overflow (Section 6). The path following analysis that [LS15] invokes is that of [LS14], and this analysis has itself been revised substantially since: its arXiv version [LS13] went from 48 pages in December 2013 to 60 pages over three versions ending in March 2015, and was then merged with its companion paper into [LS19], 65 to 69 pages over two versions in 2019 and 2020, which by its own description contains new results beyond the earlier submissions, among them a nearly optimal self-concordant barrier and its relation to Lewis weights. The path following statements imported below are therefore taken from the robust formulation of [vdBLL⁺21] rather than from [LS14], and the constants of those statements that we could not rederive from their proofs are listed in the issues above.

The same bound via the robust interior point method. A second route to the bound of Theorem 1 goes through the robust interior point method of [vdBLL⁺21], which is stated against three abstract data structures, a heavy hitter, a heavy sampler and an inverse maintenance, each with an initialization cost P , a cost $c_i \geq \text{nnz}(a_i)$ for changing the scaling of row i , and a query cost Q . Theorem 6.1 and Lemma 6.7 there bound the total time of path following, in our dimensions, by

$$\tilde{O} \left((P \|c\|_1)^{1/2} + d^{1/2} \left(Q + d \max_i \text{nnz}(a_i) \right) \right)$$

per factor of $\log(\mu_0/\mu_f)$. For a general linear program that paper uses the sketching based heavy hitter of [vdBLSS20] with $c_i = \tilde{O}(d)$ for every row and a heavy sampler with $Q = \tilde{O}(d^2 + nd^{1/2})$ (Lemmas B.1 and B.2 and Corollary B.6 there), and it is these two, $c_i = \tilde{O}(d)$ in place of $\text{nnz}(a_i)$ and $nd^{1/2}$ in place of $\text{nnz}(A)$, that turn the bound into $\tilde{O}(nd + d^{2.5})$ in its Lemma 8.2. The brute force alternative computes the vector Ah explicitly at every query and reads the heavy entries and the sampling probabilities off it: this is a heavy hitter and a heavy sampler with $P = \text{nnz}(A)$, $c_i = \text{nnz}(a_i)$ and $Q = \text{nnz}(A)$, while the inverse maintenance of Lemma B.2 there has $P = \text{nnz}(A) + \text{MATMULT}(d, d, d)$, $c_i = O(1)$ and $Q = \tilde{O}(d^2)$. Substituting, with $\max_i \text{nnz}(a_i) \leq d$ and $(\text{nnz}(A) \text{MATMULT}(d, d, d))^{1/2} \leq \text{nnz}(A) d^{1/2} + \text{MATMULT}(d, d, d) d^{-1/2}$, gives $\tilde{O}(d^{1/2}(\text{nnz}(A) + d^2))$, formally. We have not carried this out. The substitution itself needs only that the interfaces

(Definitions 3.1, 6.2 and 6.3 there) accept a query cost Q as large as $\text{nnz}(A)$, which they do since Q enters the accounting additively per step; the work is that the whole of [vdBLL⁺21] is in exact arithmetic, so the magnitude, truncation and randomness accounting of this paper would have to be repeated for its primal, gradient, dual and Lewis weight maintenance structures, instead of for the single SOLVE of Section 3.

Difficulties in reducing the $d^{2.5}$ term. The term $d^{1/2} \cdot d^2$ is $d^{1/2}$ path steps times the cost of applying a dense $d \times d$ maintained inverse to a vector inside SOLVE, and row sampling does not lower it: the sample only brings the number of rows down to $\tilde{O}(d)$. In the square dense case the corresponding $n^{2.5}$ has been beaten by combining a robust central path, which accepts directions accurate only to a constant in an ℓ_∞ -type norm, with implicit inverse maintenance, in which both the inverse and the vector it is applied to are updated lazily in batches [CLS21, vdB20, JSWZ21]. Those running times are stated in terms of the exponent of square matrix multiplication and its rectangular and dual variants, and the exponent has moved over the past five years, from 2.37286 [AVW21] to 2.371866 [DWZ23], 2.371552 [VWXXZ24], 2.371339 [ADVW⁺25] and 2.371177 [DEK⁺26], the last three also improving the rectangular exponents. For tall sparse A , however, the exponent is not what decides whether $d^{2.5}$ can be lowered. The inverse side, Woodbury layers with periodic restarts and batched folds, already costs less than d^2 per path step under the assumption $\text{MATMULT}(d, d, d) = \tilde{O}(d^{1+\sqrt{2}})$ of this paper, and any of the bounds above only lowers that; the robust path itself accepts a fixed constant-accurate inverse with a perturbation argument and no change of exponent. What blocks the improvement is that the inverse has to be applied to a fresh Newton right hand side at every step, a vector that changes in all d coordinates. The square case avoids this by maintaining the right hand side as a slowly changing vector, and none of the known ways of doing so extends from the square, or dense, case to tall sparse A without further assumptions on the sparsity of the rows.

Organization. Section 2 (Prelims) fixes the fixed point model, the notation, and the imported facts on matrix inversion with truncation and on leverage scores. Section 3 (Dynamic Solver via Row Sampling) gives the data structure behind INITIALIZE, UPDATE and SOLVE: exact inverse maintenance and its dependence on matrix multiplication, the row sampled approximate inverse, the hiding of its randomness from an adaptive caller, and the accounting of the solver against leverage stable weight sequences. Section 4 (Lewis Weights and Potential) collects everything about regularized Lewis weights, their estimation from a solver, the barrier and its parameters, and the centrality potential. Section 5 (Effect of One Step) analyzes one step of the method in fixed point: the step itself, direction finding without a bisection, the effect of solve and truncation errors on centrality and feasibility, the progress per step, and the stability of the weights handed to the data structure. Section 6 (Initialization and Overall Guarantees) bounds the magnitudes and bit lengths along the path, verifies the stability hypothesis of the data structure, constructs the initial point, and assembles the proof of Theorem 1.

2 Prelims

For positive vectors, all powers, quotients, logarithms, and comparisons are componentwise. Logarithms are natural, so that $e^{-c \log n} = n^{-c}$.

For vectors x and y and some $\epsilon > 0$, we use $x \approx_\epsilon y$ to denote $e^{-\epsilon} x_i \leq y_i \leq e^\epsilon x_i$ for every i . For symmetric positive definite matrices X and Y of the same size, we use $X \approx_\epsilon Y$ to denote

$e^{-\epsilon}Y \preceq X \preceq e^{\epsilon}Y$. The subscript of $\approx$ is a generic accuracy and takes whatever value is written there; the unsubscripted ϵ of Sections 4 to 6 is the fixed centrality parameter of Equation (11).

The matrix A is assumed to have no zero rows, so that $n \leq \text{nnz}(A)$: a zero row constrains nothing and can be dropped in a single pass.

Definition 2.1. *We say invertible matrix X has magnitude ρ if $\|X\|, \|X^{-1}\| < \rho$, and ρ is more than both dimension of X .*

$\text{MATMULT}(n_1, n_2, n_3)$ denotes the arithmetic cost of multiplying a $n_1 \times n_2$ matrix by an $n_2 \times n_3$ matrix. We state all dense costs through MATMULT . Dense products arise only in the exact inverse maintenance of Section 3.1, in the static estimate that initializes it, and in the restarts built on it, so the assumption under which they are simplified is stated there rather than here.

2.1 Arithmetic under Roundoffs

We work with fixed point numbers. A *number with ℓ bits* is a signed binary number with at most ℓ bits before and ℓ bits after the decimal point, that is, an integer multiple of $2^{-\ell}$ of absolute value less than 2^{ℓ} . For a real x with at most ℓ bits before the decimal point, $|x| < 2^{\ell}$, we write $\text{Trunc}(x, \ell)$ for x truncated toward zero to ℓ bits after the decimal point; it is a number with ℓ bits, has the sign of x , and satisfies

$$|\text{Trunc}(x, \ell)| \leq |x|, \quad |x - \text{Trunc}(x, \ell)| < 2^{-\ell}.$$

$\text{Trunc}(x, \ell)$ is only ever applied to x with at most ℓ bits before the decimal point; wherever it appears below, the argument is bounded in magnitude by a power of $\rho = 2^{O(B)}$ that is far below 2^{ℓ} , and we do not repeat this check. Applied to a vector or a matrix, Trunc acts entrywise. We say a matrix has entries with ℓ bits if all its entries are numbers with ℓ bits.

This is the fixed point model of [GPV23]: an arithmetic operation on numbers with ℓ bits, one of $+$, $-$, $\times$, $\div$ and the square root, returns the exact result truncated by $\text{Trunc}(\cdot, \ell)$, and comparisons are exact.

Lemma 2.1. *Arithmetic in numbers with ℓ bits can be done at cost $\tilde{O}(\ell)$.*

Two operations lose nothing when given room: the sum or difference of two numbers with ℓ bits is a number with $\ell + 1$ bits, and their product is a number with 2ℓ bits, so both are exact when carried out on numbers with 2ℓ bits. This is how the comparisons of ratios in Figure 6 are made exact.

The error of a longer computation is bounded by counting operations and the number of multiplicative operations on a path. We say a quantity is computed by a formula with m operations if it is obtained from numbers with ℓ bits by m arithmetic operations in which every intermediate result is used once, a result used k times being counted k times; the multiplicative depth of the formula is the largest number of operations among $\times$, $\div$ and square root on a path from an input to the output.

Lemma 2.2. *Let a quantity be computed by a formula with m operations and multiplicative depth q on numbers with ℓ bits, whose inputs are within $e_1, \dots, e_k$ of the exact input values, each use of an input counted separately, and let $M \geq 2$ be such that, in exact arithmetic, every input and every intermediate result has absolute value at most M , every divisor and every radicand has absolute value at least $1/M$, and $M \leq 2^{\ell-1}$. Write*

$$E \triangleq \left(m2^{-\ell} + \sum_{j=1}^k e_j \right) (2M^2)^q.$$

If $E \leq 1/(2M)$, then the computed value differs from the exact one by at most E .

The proof is in Appendix A. Every formula of Sections 5 and 6 is evaluated on numbers with $40 \log \rho$ bits at $M = \rho$, and Equation (12) there is the resulting standing bound of ρ^{-20} on its error; each proof that uses it records the operation count and the depth of the formula in question.

2.2 Matrix Inversion with Fixed Point Roundoffs

The first imported result, Lemma 3.1 of [GPV23], is that a small perturbation of a matrix perturbs its inverse in proportion to the square of its magnitude.

Imported result 2.1. *For invertible matrices X and Y such that Y has magnitude ρ_Y , and $\|X - Y\|_F \leq \epsilon$ for some $0 < \epsilon < 1/(2\rho_Y)$, we have:*

1. X has magnitude at most $2\rho_Y$.
2. $\|X^{-1} - Y^{-1}\|_F \leq 2\rho_Y^2\epsilon$.

The second, Theorem 3.2 of [GPV23], is an approximate dense inverse in fixed point.

Imported result 2.2. *For $d \times d$ matrix X with magnitude ρ , and error $0 < \epsilon < 0.1$, we can compute explicit matrix Z with such that*

$$\|Z - X^{-1}\|_F \leq \epsilon.$$

in bit complexity $\tilde{O}(\text{MATMULT}(d, d, d) \log(\rho/\epsilon))$.

2.3 Leverage scores

The one weight vector the data structure of Section 3 works with is the classical one. The leverage score of a row is its diagonal entry in the orthogonal projection onto the column space, so it lies in $[0, 1]$, the scores sum to the rank d , and a row with a small score is one whose removal barely changes $A^\top A$. Sampling rows with probabilities proportional to leverage scores gives a spectral approximation of $A^\top A$ with $\tilde{O}(d)$ rows, which is how Section 3 keeps its sample small, and the same scores measure how much a change of weights moves the sample, which is the stability notion of Definition 3.2. The regularized Lewis weights of Section 4 are defined through the leverage scores of a reweighting of A by a power of the weights themselves, and Section 5 passes between the two through Lemma 5.12.

Definition 2.2. *For a full-column-rank $A \in \mathbb{R}^{n \times d}$, write $\tau(A)$ for the diagonal of the orthogonal projection $A(A^\top A)^{-1}A^\top$, and $\tau_i(A)$ for its i th entry.*

The concentration inequality behind leverage score sampling is the matrix Chernoff inequality, Theorem 1.1 of [Tro12], stated with its constants.

Imported result 2.3. *Let $X_1, \dots, X_N$ be independent random symmetric positive semidefinite $d \times d$ matrices with $\lambda_{\max}(X_j) \leq R$ almost surely, and let $\mu_{\min} \triangleq \lambda_{\min}(\sum_j \mathbb{E}X_j)$ and $\mu_{\max} \triangleq \lambda_{\max}(\sum_j \mathbb{E}X_j)$. Then for every $\delta \in [0, 1]$,*

$$\Pr \left[\lambda_{\min} \left(\sum_{j=1}^N X_j \right) \leq (1 - \delta)\mu_{\min} \right] \leq d \left(\frac{e^{-\delta}}{(1 - \delta)^{1-\delta}} \right)^{\mu_{\min}/R},$$

and for every $\delta \geq 0$,

$$\Pr \left[\lambda_{\max} \left(\sum_{j=1}^N X_j \right) \geq (1 + \delta)\mu_{\max} \right] \leq d \left(\frac{e^{\delta}}{(1 + \delta)^{1+\delta}} \right)^{\mu_{\max}/R}.$$

2.4 Notation

Figure 1 lists the symbols used throughout. Symbols local to one part are introduced at the start of that part, in a table of the same form with a pointer to where each symbol is defined and used: Figure 2 for the data structure of Section 3, Figure 3 for the weights and potential of Section 4, and Figure 4 for the step and initialization of Sections 5 and 6.

Symbol	Meaning
B	bits before and after the decimal point, $B \leq d^{1/2}$
$\tilde{O}(\cdot)$	hides factors polynomial in $\log n$; powers of B are explicit
$x \approx_\epsilon y$	$e^{-\epsilon}x_i \leq y_i \leq e^\epsilon x_i$ for every i
ρ	magnitude of X : $\ X\ , \ X^{-1}\ < \rho$; along the path $\rho = 2^{O(B)}$
$\text{MATMULT}(n_1, n_2, n_3)$	arithmetic cost of multiplying $n_1 \times n_2$ by $n_2 \times n_3$
A	full-column-rank $n \times d$ matrix, $n > d$, no zero rows, so $n \leq \text{nnz}(A)$
$\text{nnz}(A)$	number of nonzero entries of A
$\tau(A)$	leverage scores of A : diagonal of $A(A^\top A)^{-1}A^\top$
$\text{diag}(x)$	diagonal matrix with x on the diagonal
w	positive weight vector; along the path $w_i = (\tau_i \phi_i''(x_i))^{-1}$
W	$W = \text{diag}(w)$; the inverse of $A^\top W A$ is maintained, never exposed
ϵ	target accuracy
b, c, l, u	LP data: $\min c^\top x$ over $A^\top x = b, l \leq x \leq u$
$\text{Trunc}(x, \ell)$	x truncated toward zero to ℓ bits after the decimal point, Section 2.1
r	number of rounds, $r = \tilde{O}(d^{1/2} \log(\mu_0/\mu_f))$
κ	per-step contraction of the potential Ψ , $\kappa = \alpha^2 \lambda_\gamma / (32C d^{1/2})$
$\mathbb{E}$	expectation
$\mathcal{S}, T_{\mathcal{S}}$	a linear solver handed to a routine, and its bit complexity per call
INITIALIZE	with UPDATE, SOLVE: the calls of Definition 3.1

Figure 1: Global variables.

3 Dynamic Solver via Row Sampling

Symbol	Meaning
$w^{(k)}$	weights handed to the data structure at round k , $W^{(k)} = \text{diag}(w^{(k)})$; Definition 3.2
$\sigma^{(k)}$	leverage scores $\tau((W^{(k)})^{1/2}A)$ of the reweighted matrix at round k ; Definition 3.2, Corollary 3.4
$\hat{w}$	the new weight vector handed to UPDATE; Definition 3.1, Lemma 3.1
ζ	stability parameter of Definition 3.2; Theorem 2, Lemma 3.2
$S^{(k)}$	the private sampling diagonal of round k ; c_k of its rows change; Lemma 3.2 and its proof, proof of Corollary 3.4
c_s	sampling constant, $c_s \triangleq 10^6$; proof of Lemma 3.2
r_0	length of a window between restarts, $r_0 = d^{1/2}$; Lemma 3.2, proof of Corollary 3.4
$t_{i,j}, (\eta, \eta')$	refresh rounds of coordinate i , and the trigger thresholds of a refresh schedule; Corollary 3.4
Q	total number of SOLVE calls, by the caller and inside the structure; Theorem 2, Lemma 3.2, Corollary 3.3

Figure 2: Variables local to the dynamic solver of Section 3.

This section gives the data structure that the interior point method of Section 5 runs against. The interior point method sees the data structure only through the three calls below. In particular the maintained inverse, and the row sample it is built from, never leave the data structure: the only thing that leaves it is the output of SOLVE, a vector in $\mathbb{R}^d$. Whatever else the caller needs, it computes itself from that output.

The operations of the data structure are the following.

Definition 3.1. *An inverse maintenance data structure for an $n \times d$ matrix A of column rank $d \leq n$ holds a positive vector $w \in \mathbb{R}_{>0}^n$, with diagonal $W = \text{diag}(w)$, and supports:*

1. *INITIALIZE(A, w), which stores A and the initial w , together with whatever internal approximation of $(A^\top W A)^{-1}$ the implementation maintains. Nothing about how the caller arrived at w is passed: the only weights the structure ever forms are leverage scores of $W^{1/2}A$.*
2. *UPDATE($\hat{w}$), which given the full vector $\hat{w} \in \mathbb{R}_{>0}^n$ of new weights replaces w by $\hat{w}$ and restores the internal approximation for the new W . The caller is not asked to say which coordinates moved: reading $\hat{w}$ costs $O(n)$, below the $\text{nnz}(A)$ that every round pays anyway, and the data structure decides for itself which of its rows to refresh.*
3. *SOLVE(b), which given a fixed point $b \in \mathbb{R}^d$ with $20 \log \rho$ bits returns a fixed point $y \in \mathbb{R}^d$ with $40 \log \rho$ bits and*

$$\left\| y - (A^\top W A)^{-1} b \right\|_{A^\top W A} \leq \rho^{-10} \left\| (A^\top W A)^{-1} b \right\|_{A^\top W A}. \quad (1)$$

There is no accuracy parameter: every solve is at the high accuracy ρ^{-10} . That this accuracy, and the quadratic form it is an accuracy for, are what the two callers of SOLVE need is checked in Section 5, where the calls are made.

This interface is deliberately smaller than the ones of [vdBLL⁺21], which in addition to a solve expose a call returning the ℓ_2 norm of the projected vector. Such a call is there because those data

structures maintain the step implicitly, in a sketched or coordinate wise lazy representation, and so cannot afford to write down the vector whose norm the potential argument needs. We are in the earlier regime, where every quantity the caller handles is written out explicitly in $\mathbb{R}^d$ or $\mathbb{R}^n$: SOLVE returns all d entries of y , the caller forms Ay in $\text{nnz}(A)$ time, and any norm of the projected vector is then computed directly, within the $\text{nnz}(A) + d^2$ that the round pays anyway. Only the sample and the inverse built from it stay inside; the vectors do not.

The weights handed to UPDATE are not arbitrary either. They are the sequence produced by the interior point method of Section 5, and the only property of that sequence the data structure uses is the following, which is the σ stability assumption of [LS15, Definition 4]: the multiplicative movement of the weights is measured in the norm induced by the leverage scores of the reweighted matrix. How the caller arrives at its weights is its own business; Section 6 converts its control of them into the condition below.

Definition 3.2. A weight sequence $w^{(0)}, \dots, w^{(r)} \in \mathbb{R}_{>0}^n$ with diagonals $W^{(k)}$ and induced leverage scores

$$\sigma^{(k)} \triangleq \tau \left((W^{(k)})^{1/2} A \right) = \text{diag} \left((W^{(k)})^{1/2} A \left(A^\top W^{(k)} A \right)^{-1} A^\top (W^{(k)})^{1/2} \right)$$

is ζ -leverage stable, for $0 < \zeta < 0.1$ and magnitude bound ρ , if for every $k \in [r]$:

1. the operators stay well conditioned: $w^{(k)} \in [\rho^{-1}, \rho]^n$ and $A^\top W^{(k)} A$ has magnitude at most ρ ;
2. coordinate wise things are stable:

$$w^{(k)} \approx_\zeta w^{(k-1)};$$

3. the leverage score norm of the relative change is small:

$$\sum_i \sigma_i^{(k)} \log^2 \left(\frac{w_i^{(k)}}{w_i^{(k-1)}} \right) \leq \zeta^2.$$

Condition 1 is the analogue of the β containment of [LS15, Definitions 3 and 4], and is what lets W , the maintained inverse, and the intermediate quantities be stored with $O(\log \rho)$ bits. Condition 3 is weaker than the ℓ_2 condition $\|\log w^{(k)} - \log w^{(k-1)}\|_2 \leq \zeta$ that a short step method gives for free, because $\sigma_i^{(k)} \leq 1$; the point of measuring in the leverage score norm is that rows of small leverage are allowed to move often, and those are exactly the rows that a leverage score sampled sparsifier rarely keeps. Two facts about $\sigma^{(k)}$ are used repeatedly: $\sum_i \sigma_i^{(k)} = d$, and for positive diagonals $W \approx_\delta W'$ the scores satisfy $\tau(W^{1/2} A) \approx_{2\delta} \tau(W'^{1/2} A)$, since $\tau_i(W^{1/2} A) = w_i a_i^\top (A^\top W A)^{-1} a_i$ and both factors move by at most $e^{\pm\delta}$.

The caller of Section 5 does not hand over an exactly stable sequence but an estimate of one, so the definition is used with the following slack. A sequence $\hat{w}^{(0)}, \dots, \hat{w}^{(r)}$ is ζ -leverage stable up to noise if there is a ζ -leverage stable $w^{(0)}, \dots, w^{(r)}$ with

$$\hat{w}^{(k)} \approx_{\zeta/10} w^{(k)} \quad \text{for every } k.$$

The noisy sequence then satisfies condition 1 with $e^{0.02}\rho$ in place of ρ and condition 2 with 1.2ζ in place of ζ , its leverage scores agree with those of $w^{(k)}$ within $e^{\pm\zeta/5}$, and condition 3, which need not hold for $\hat{w}^{(k)}$ itself, enters the analysis only through the counting of Corollary 3.4, which is stated to absorb the noise. Everything below is written for an exactly stable sequence and holds for a sequence stable up to noise with these constants.

The main guarantee of this section is that the operations of Definition 3.1 can be supported along any such sequence, even one generated adaptively from the outputs of the data structure itself, at an amortized cost per round of one pass over A plus one dense $d \times d$ operation.

Theorem 2. *Given $n \times d$ full rank matrix A where $A^\top A$ of magnitude at most ρ , and $w^{(0)}, \dots, w^{(r)}$, $r \leq n^2$ an adaptive ζ -leverage stable sequence with magnitude bound ρ , for some $0 < \zeta < 0.1$, possibly up to noise, and let Q be the total number of SOLVE calls made during these rounds, by the caller and by the structure itself. There is a data structure supporting the operations of Definition 3.1 along this sequence, with probability at least $1 - n^{-5} - 2Qn^{-3}$, at bit complexity of:*

1. $\tilde{O}((\text{nnz}(A) + \text{MATMULT}(d, d, d)) \log \rho)$ per INITIALIZE,
2. Amortized cost of $\tilde{O}((\text{nnz}(A) + d^2) \log^2 \rho)$ per UPDATE($\hat{w}$),
3. $\tilde{O}((d^2 + \text{nnz}(A)) \log^2 \rho)$ per SOLVE(b).

The proof occupies the rest of the section. We implement the operations twice: once without sampling, by maintaining the inverse of the full $d \times d$ matrix $A^\top W A$ explicitly (Section 3.1), and once on a leverage score sampled row subset, which is what brings the amortized cost per round down to one pass over A plus one dense $d \times d$ operation (Section 3.2). The sampled version is only sound against an adversary that does not see the samples, so we then inject noise into the solver’s output, state the conditions under which the sampling stays hidden, and reduce adaptively generated updates to oblivious ones (Section 3.3). Section 3.4 bounds the leverage score mass of the coordinates that the lazy refresh rule touches over a window, which is the only way the stability of the sequence enters the cost, and combines the three with a restart schedule to prove the overall guarantee.

3.1 Exact inverse maintenance and role of matrix multiplication exponent

We start with the guarantees of the square unsampled version, which holds the inverse of $A^\top W A$ itself. These are mostly a restatement of [GPV23].

This is the only part of the algorithm that multiplies dense matrices, so it is where the cost of fast matrix multiplication enters, and we record here the assumption under which those costs are simplified,

$$\text{MATMULT}(d, d, d) = \tilde{O}\left(d^{1+\sqrt{2}}\right), \quad (2)$$

which is the assumption made by [LS15], whose inverse maintenance we follow: the restart balance closing Section 3.1 there, on page 8, uses exactly this exponent to conclude that the preprocessing term dominates the cost of the periodic recomputations. It is weaker than what the algorithm of [Vai89] asks for: the remark closing Section 1 there, on page 333, notes that the algorithm “does not use the fastest known matrix multiplication algorithm; an algorithm that can multiply two $n \times n$ matrices in $O(n^{2.4})$ operations suffices”, and Section 3 there invokes that bound, for both the product of two $n \times n$ matrices and the inverse of a non-singular $n \times n$ matrix, when recomputing $(A^\top D A)^{-1}$ at the start of each period. Both bounds are met by the fast matrix multiplication algorithms available then and now. We use two standard consequences of Equation (2): costs are superadditive in the middle dimension,

$$\text{MATMULT}(d, c_1, d) + \text{MATMULT}(d, c_2, d) \leq \text{MATMULT}(d, c_1 + c_2, d),$$

and a thin product is a c/d fraction of a square one, so that for $1 \leq c \leq d$,

$$\text{MATMULT}(d, c, d) = \tilde{O}\left(\frac{c}{d} \text{MATMULT}(d, d, d)\right) = \tilde{O}\left(c d^{\sqrt{2}}\right).$$

In particular $\text{MATMULT}(d, d, d) = \tilde{O}(d^{2.5})$ and $\text{MATMULT}(d, d^{1/2}, d) = \tilde{O}(d^{1/2+\sqrt{2}}) = \tilde{O}(d^{1.92})$, both inside the budget of Theorem 1. Where the two are balanced against each other, and why $d^{1+\sqrt{2}}$ is exactly what that balance can afford, is Section 3.4.

The unsampled inverse maintenance is the following.

Lemma 3.1. *Let A be an $n \times d$ matrix with column rank $d \leq n$, such that $A^\top A$ has magnitude at most $\rho > 10n$, and let every weight vector handed to the structure lie in $[\rho^{-1}, \rho]^n$ coordinatewise. There is a deterministic data structure that maintains $(A^\top W A)^{-1}$ explicitly to error $\rho^{-O(1)}$ whose cost is:*

1. *Initialize: $\tilde{O}((\text{nnz}(A) + \text{MATMULT}(n, d, d)) \log \rho)$,*
2. *Update from w to $\hat{w}$: $\tilde{O}((n + \text{MATMULT}(d, \|w - \hat{w}\|_0, d)) \log \rho)$, where $\|w - \hat{w}\|_0$ is the number of coordinates on which $\hat{w}$ differs from w .*

The cost of SOLVE is not an item here. It is the same for this structure as for the sampled one, since both serve a solve by Richardson iteration against whatever inverse they hold, and it is accounted once, in Section 3.4.

Proof. Throughout, every number is kept to $B = \Theta(\log \rho)$ bits, so by Lemma 2.1 each arithmetic operation costs $\tilde{O}(\log \rho)$ and the arithmetic counts below become bit counts after multiplying by that factor. The weights are positive with $w_i \in [\rho^{-1}, \rho]$, so $A^\top W A$ has magnitude $\rho^{O(1)}$, and the same holds of every matrix formed below.

INITIALIZE. Scaling the rows of A by $w_i^{1/2}$ takes $\text{nnz}(A)$ operations, and forming the Gram matrix of the scaled matrix takes $\text{MATMULT}(n, d, d)$. Inverting it with Imported result 2.2 at accuracy ρ^{-C} , for a constant C fixed below, takes $\tilde{O}(\text{MATMULT}(d, d, d) \log \rho) \leq \tilde{O}(\text{MATMULT}(n, d, d) \log \rho)$. This is Item 3.1(1).

UPDATE. Let $K \subseteq [n]$ be the coordinates on which $\hat{w}$ differs from w and $k = |K|$, let A_K be the corresponding k rows of A , and let $D \triangleq \text{diag}(\hat{w}_K - w_K)$, so that

$$A^\top \widehat{W} A = A^\top W A + A_K^\top D A_K.$$

Writing $M \triangleq A^\top W A$, the Woodbury identity gives the new inverse as

$$\left(M + A_K^\top D A_K\right)^{-1} = M^{-1} - M^{-1} A_K^\top \left(D^{-1} + A_K M^{-1} A_K^\top\right)^{-1} A_K M^{-1}. \quad (3)$$

Reading $\hat{w}$ and locating K takes $O(n)$ operations. Forming $A_K M^{-1}$, a $k \times d$ matrix, takes $\text{MATMULT}(k, d, d)$; forming and inverting the $k \times k$ matrix in the middle takes $\text{MATMULT}(k, d, k) + \text{MATMULT}(k, k, k)$; and the final product of a $d \times k$, a $k \times k$, and a $k \times d$ matrix takes $\text{MATMULT}(d, k, d)$. All of these are $\tilde{O}(\text{MATMULT}(d, k, d))$, and when $k \geq d$ one may instead recompute the Gram matrix of the changed rows and invert from scratch, at $\text{MATMULT}(d, k, d) + \text{MATMULT}(d, d, d) = \tilde{O}(\text{MATMULT}(d, k, d))$. This is Item 3.1(2). The middle matrix is invertible with magnitude $\rho^{O(1)}$: D^{-1} has entries bounded by ρ in absolute value since $w, \hat{w} \in [\rho^{-1}, \rho]$ coordinatewise, and $A_K M^{-1} A_K^\top$ is positive semidefinite with norm $\rho^{O(1)}$.

Accumulation of error. The one point that is not a count of operations is that Equation (3) is applied to the maintained inverse rather than to the exact one, so rounding errors are fed back into the structure. What makes $B = \Theta(\log \rho)$ bits enough is that the recurrence is backward stable, in the following sense. Let Z be the maintained matrix before an update and $N \triangleq A^\top W A$ the matrix it approximates, and measure the error backward, as $\|Z^{-1} - N\|_F$. Equation (3) is an algebraic identity valid for every invertible Z in place of M^{-1} , as long as the middle matrix is invertible, so the exact Woodbury update of Z is

$$Z' = \left(Z^{-1} + A_K^\top D A_K\right)^{-1}, \quad \text{hence} \quad (Z')^{-1} - N' = Z^{-1} - N$$

for the new matrix $N' = N + A_K^\top D A_K$: exact Woodbury carries the backward error over unchanged. What the B -bit computation returns is $\tilde{Z}'$ with $\|\tilde{Z}' - Z'\|_F \leq \rho^{O(1)} 2^{-B}$, since every intermediate has magnitude $\rho^{O(1)}$, there are $O(1)$ rounding stages, and the inverse of the middle matrix is computed by Imported result 2.2 to accuracy 2^{-B} . Imported result 2.1, applied to $\tilde{Z}'$ and Z' , converts this into $\|(\tilde{Z}')^{-1} - (Z')^{-1}\|_F \leq \rho^{O(1)} 2^{-B}$, so each update adds $\rho^{O(1)} 2^{-B}$ to the backward error rather than multiplying it. Starting from the backward error $\rho^{O(1)} \rho^{-C}$ of INITIALIZE, after at most $\text{poly}(n) \leq \rho$ updates the backward error is at most $\rho^{O(1)} (\rho^{-C} + 2^{-B})$, which taking the constants in C and in $B = \Theta(\log \rho)$ large enough is below $\rho^{-C'-30}$ for any target C' , and Imported result 2.1 turns this into a forward error $\rho^{-C'}$ on the maintained inverse, which is the accuracy claimed. The magnitude of the maintained matrix stays $\rho^{O(1)}$ throughout by Item 2.1(1). $\square$

Charging UPDATE a rank one update for every changed coordinate is what the sampled version of the next subsection avoids: there, a change to w_i only costs anything when row i is in the sample.

3.2 Maintaining a sparse approximate inverse

Running Lemma 3.1 on A itself is too expensive in two places: INITIALIZE pays $\text{MATMULT}(n, d, d)$, and UPDATE pays a $d \times d$ multiplication for every changed coordinate no matter how little that coordinate matters. Both are fixed by running it on a leverage score sampled row subset of A instead. What the sampled structure holds is never returned; it is used only as the preconditioner inside SOLVE, and inside the refresh test of its own UPDATE, so the following lemma is stated in terms of what it maintains and what the three calls cost, against an adversary that fixes the weight sequence in advance.

Lemma 3.2. *Let A , ρ be as in Lemma 3.1 and let $w^{(0)}, \dots, w^{(r_0)}$ be a ζ -leverage stable sequence chosen by an oblivious adversary, that is, one fixed independently of the data structure's random coins, and let Q be the total number of SOLVE calls made during these rounds, by the caller and by the structure itself. There is a data structure which, with probability at least $1 - n^{-7} - 2Qn^{-3}$, maintains at every round k a diagonal $S^{(k)} \succeq 0$ supported on $\tilde{O}(d \log n)$ rows, together with the exact inverse of $A^\top S^{(k)} W^{(k)} A$, such that*

$$A^\top S^{(k)} W^{(k)} A \approx_{0.1} A^\top W^{(k)} A, \quad (4)$$

with initialization bit complexity $\tilde{O}((\text{nnz}(A) + \text{MATMULT}(d, d, d)) \log \rho)$ and amortized update bit complexity

$$\tilde{O}((\text{nnz}(A) + d^2) \log^2 \rho).$$

Proof. Keeping row i independently with probability $p_i = \min\{1, c_s \sigma_i^{(k)} \log n\}$, $c_s \triangleq 10^6$, and setting $S_{ii}^{(k)} = 1/p_i$ on the kept rows gives Equation (4) with probability $1 - n^{-9}$, with $\tilde{O}(d \log n)$ rows retained, since $\|\sigma^{(k)}\|_1 = d$; this is leverage score sampling, [SS11], and the computation at the end of this proof, from the matrix Chernoff inequality of Imported result 2.3, shows that the concentration bound needs only that the rows are kept independently with p_i between $\min\{1, (c_s/100) \sigma_i^{(k)} \log n\}$ and $\min\{1, 100 c_s \sigma_i^{(k)} \log n\}$. It is here that obliviousness is used, since the $\sigma^{(k)}$ defining p_i must not depend on the coins that produced the earlier samples.

Leverage scores are estimated, here and below, by the standard Johnson–Lindenstrauss argument: since

$$\sigma_i^{(k)} = \left\| (W^{(k)})^{1/2} A \left(A^\top W^{(k)} A \right)^{-1} a_i (w_i^{(k)})^{1/2} \right\|_2^2,$$

a Gaussian G with $O(\log n)$ rows gives $\|G(W^{(k)})^{1/2}A(A^\top W^{(k)}A)^{-1}a_i(w_i^{(k)})^{1/2}\|_2^2 \approx_{0.1} \sigma_i^{(k)}$ for all i with probability $1 - n^{-9}$, and the n numbers on the left are obtained from $O(\log n)$ solves in $A^\top W^{(k)}A$, one per row of $G(W^{(k)})^{1/2}A$, followed by one multiplication of A against the $O(\log n)$ solutions. With a solver at accuracy ρ^{-10} costing $\tilde{O}((\text{nnz}(A)+d^2)\log^2 \rho)$ per call, this is $\tilde{O}((\text{nnz}(A)+d^2)\log^2 \rho)$ bit operations, all numbers having $\tilde{O}(\log \rho)$ bits by Item 3.2(1).

The probabilities needed to build the first sample are computed this way with a one-off solver: a sparse embedding [CW17] of $(W^{(0)})^{1/2}A$ with $\tilde{O}(d)$ rows, applied at $\tilde{O}(\text{nnz}(A))$ bit operations and a constant factor spectral approximation of $A^\top W^{(0)}A$ by the embedding guarantee, preconditions $\tilde{O}(\log \rho)$ Richardson steps with the dense inverse of its Gram matrix from Imported result 2.2, at cost $\tilde{O}((\text{nnz}(A) + \text{MATMULT}(d, d, d)) \log \rho)$; this is the only place the full n -row matrix is touched by anything other than a matrix vector product. The inverse of the sampled $d \times d$ matrix is then formed once, inside the INITIALIZE of Lemma 3.1 applied to the $\tilde{O}(d \log n)$ sampled rows, which is

$$\tilde{O}(\text{MATMULT}(d, d, d) \log \rho) \quad \text{rather than} \quad \tilde{O}(\text{MATMULT}(n, d, d) \log \rho).$$

Item 3.2(1) keeps the sampled matrix of magnitude $\rho^{O(1)}$ throughout by Equation (4), so $\tilde{O}(\log \rho)$ bits suffice everywhere, and Item 3.2(2) keeps the sampling probabilities themselves accurate between refreshes, by the $\tau(W^{1/2}A) \approx_{2\delta} \tau(W'^{1/2}A)$ fact recorded after Definition 3.2.

An UPDATE is served in two parts. First the new sampling probabilities are recomputed: the test below needs the current $\sigma^{(k)}$, at constant accuracy only, so it is one Johnson–Lindenstrauss estimate as above with the structure’s own SOLVE as the solver, at

$$\tilde{O}((\text{nnz}(A) + d^2) \log^2 \rho),$$

which also subsumes the $O(n)$ spent reading $\hat{w}$. Second, rows are resampled lazily, as in [LS15, Algorithm 3]: coordinate i is refreshed only when w_i , or the estimate of σ_i , has moved by more than a factor of 2 since it was last refreshed, and a refresh changes the maintained matrix only when it changes the kept status or the scaling of a sampled row. Each such change is a rank one update to $A^\top SWA$: the structure of Lemma 3.1 is run on the kept rows with weights $s_i w_i \in [\rho^{-1}, n\rho]$, and a row entering or leaving the sample is a weight change between 0 and $s_i w_i$, for which Equation (3) holds verbatim, the middle matrix being invertible because the updated matrix is positive definite. So c_k such changes cost $\tilde{O}(\text{MATMULT}(d, c_k, d) \log \rho)$ through the UPDATE of Lemma 3.1. Bounding $\sum_k c_k$, and so turning the second part into an amortized $\tilde{O}(d^2 \log \rho)$, is what Section 3.4 does; adding the two parts gives the claimed amortized bound.

The sample of round k is not drawn afresh: row i is kept or not according to coins flipped at the round $t_i \leq k$ at which it was last refreshed, with the probability $p_i^{(t_i)}$ of that round, and the estimate that decided when to refresh was itself computed against the maintained form, hence from the earlier coins. It is worth saying why the sampling argument above still covers this. Every SOLVE of the structure, including the $O(\log n)$ internal ones behind each refresh test, is implemented by the NOISYSOLVER of Section 3.3, whose output is within total variation distance n^{-3} of the output of an ideal solver that is a function of $(A, W^{(k)}, b)$ and fresh Gaussian noise only, by Imported result 3.2, and its fixed point implementation, described after Corollary 3.3, fails to meet the hypotheses of that result with probability at most $2n^{-4}$ per call. So, at the cost of $Q(n^{-3} + 2n^{-4}) \leq 2Qn^{-3}$ in probability, we may analyze the process in which every estimate is computed by the ideal solver. In that process the refresh decisions are functions of the weights, which the oblivious adversary fixed in advance, and of fresh noise, so they are independent of the sampling coins; the sampling coins of distinct rows are independent; and the refresh rule, together with the 0.1-accuracy of the estimates, keeps $p_i^{(t_i)}$ within a constant factor of $c_s \sigma_i^{(k)} \log n$, or equal to one, for every i and every

round $k \geq t_i$ at which row i has not been refreshed again: writing $\hat{\sigma}_i^{(k)} \approx_{0.1} \sigma_i^{(k)}$ for the estimates, the rule not having fired between t_i and k means $\hat{\sigma}_i^{(k)} \in [1/2, 2] \hat{\sigma}_i^{(t_i)}$, so

$$\sigma_i^{(k)} \in \left[\frac{e^{-0.2}}{2}, 2e^{0.2} \right] \sigma_i^{(t_i)} \subseteq \left[\frac{1}{10}, 10 \right] \sigma_i^{(t_i)},$$

and therefore

$$p_i^{(t_i)} = \min \left\{ 1, c_s \hat{\sigma}_i^{(t_i)} \log n \right\} \geq \min \left\{ 1, \frac{c_s e^{-0.1}}{10} \sigma_i^{(k)} \log n \right\} \geq \min \left\{ 1, \frac{c_s}{100} \sigma_i^{(k)} \log n \right\},$$

and symmetrically $p_i^{(t_i)} \leq \min \{1, 100 c_s \sigma_i^{(k)} \log n\}$. So at round k the rows are kept independently, with probabilities fixed independently of the coins, each at least the probability of leverage score sampling at the constant $c_s/100$ and at most that at $100 c_s$. We now carry out the concentration computation. Write $N \triangleq A^\top W^{(k)} A$ and $u_i \triangleq N^{-1/2} (w_i^{(k)})^{1/2} a_i$, so that $\sum_i u_i u_i^\top = I$ and $\|u_i\|_2^2 = (w_i^{(k)})^{1/2} a_i^\top N^{-1} a_i (w_i^{(k)})^{1/2} = \sigma_i^{(k)}$, and Equation (4) is the statement

$$e^{-0.1} I \preceq \sum_i S_{ii}^{(k)} u_i u_i^\top \preceq e^{0.1} I.$$

Set

$$R \triangleq \frac{100}{c_s \log n} = \frac{1}{10^4 \log n}.$$

We write $\sum_i S_{ii}^{(k)} u_i u_i^\top$ as a sum of independent positive semidefinite matrices of norm at most R . A row with $p_i^{(t_i)} = 1$ is kept deterministically with $S_{ii}^{(k)} = 1$; its term $u_i u_i^\top$ has norm $\sigma_i^{(k)} \leq 1$, and we split it into $m_i \triangleq \lceil \sigma_i^{(k)} / R \rceil$ equal summands $u_i u_i^\top / m_i$, each of norm at most R , which are constant and hence independent of everything. For a row with $p_i^{(t_i)} < 1$, let $\xi_i \in \{0, 1\}$ be its coin and

$$X_i \triangleq \frac{\xi_i}{p_i^{(t_i)}} u_i u_i^\top,$$

so that $\mathbb{E} X_i = u_i u_i^\top$ as $\mathbb{E} \xi_i = p_i^{(t_i)}$, and, since $p_i^{(t_i)} < 1$ forces $p_i^{(t_i)} \geq (c_s/100) \sigma_i^{(k)} \log n$,

$$\lambda_{\max}(X_i) \leq \frac{\sigma_i^{(k)}}{p_i^{(t_i)}} \leq \frac{100}{c_s \log n} = R.$$

The coins of distinct rows are independent, so all summands are independent, and their expectations sum to $\sum_i u_i u_i^\top = I$, whence $\mu_{\min} = \mu_{\max} = 1$ in Imported result 2.3. Apply it with $\delta_+ \triangleq e^{0.1} - 1$ in the upper tail and $\delta_- \triangleq 1 - e^{-0.1}$ in the lower tail. Since $\ln(1 + \delta_+) = 0.1$ and $\ln(1 - \delta_-) = -0.1$, the two bases are

$$\frac{e^{\delta_+}}{(1 + \delta_+)^{1 + \delta_+}} = \exp(\delta_+ - 0.1(1 + \delta_+)) = \exp(0.9e^{0.1} - 1) \leq e^{-0.005}$$

and

$$\frac{e^{-\delta_-}}{(1 - \delta_-)^{1 - \delta_-}} = \exp(-\delta_- + 0.1(1 - \delta_-)) = \exp(1.1e^{-0.1} - 1) \leq e^{-0.004},$$

using $0.9e^{0.1} \leq 0.9947$ and $1.1e^{-0.1} \leq 0.9954$. Raised to the power $1/R = 10^4 \log n$, each is at most $e^{-40 \log n} = n^{-40}$, so

$$\Pr \left[\sum_i S_{ii}^{(k)} u_i u_i^\top \not\approx_{0.1} I \right] \leq d n^{-40} + d n^{-40} \leq n^{-9},$$

as $2d \leq n^2$. The complementary event is Equation (4) at round k . The number of kept rows has mean at most $\sum_i p_i^{(t_i)} \leq 100c_s \log n \sum_i \sigma_i^{(k)} = 10^8 d \log n$, and the scalar Chernoff bound for a sum of independent indicators, $\Pr[X \geq 2\mu'] \leq e^{-\mu'/3}$ for any $\mu' \geq \mathbb{E}X$, at $\mu' \triangleq 10^8 d \log n + 30 \log n$ gives at most $2 \cdot 10^8 d \log n + 60 \log n = \tilde{O}(d \log n)$ kept rows except with probability n^{-10} . Together with the n^{-9} of the leverage score estimate, a round fails with probability at most $n^{-9} + n^{-9} + n^{-10} \leq n^{-8}$, and a union bound over the $r_0 \leq n$ rounds and the $2Qn^{-3}$ from the coupling give the probability claimed. On the event that Equation (4) holds, the maintained inverse is a constant factor preconditioner and SOLVE meets Equation (1), whatever the coins happened to be. What obliviousness is used for is the weights and not the probabilities: $w^{(k)}$ must not be chosen in response to the coins, while the estimate of $\sigma^{(k)}$ has only to be accurate and, up to the coupling, independent of the coins.

SOLVE runs a fixed number $O(\log \rho)$ of steps of preconditioned Richardson iteration against the maintained inverse, so that in exact arithmetic it is a linear map of b . The exact operator applied at each step is $A^\top W^{(k)} A$, at the cost of one multiplication by A and one by $A^\top$, so the guarantee Equation (1) is with respect to the true matrix, and only the rate of convergence depends on the sample through Equation (4) and Imported result 2.1. The same holds of the leverage score estimates the refresh test runs against it, and of anything the caller computes from the output of SOLVE: the sample only affects how fast the solves converge, not the point they converge to, so every such quantity is a deterministic function of the inputs up to the requested accuracy, even though it is computed from a full pass over A while the preconditioner behind SOLVE is built from $\tilde{O}(d \log n)$ rows.

Since the guarantee is relative and the arithmetic is fixed point, the iteration is run on the scaled right hand side $b' \triangleq 2^{-j} b$, with j the integer for which $\|b'\|_\infty \in [1/2, 1)$, on numbers with $20 \log \rho$ bits, and the result is multiplied by 2^j ; for $b = 0$ the answer 0 is returned directly. As $b \neq 0$ has $20 \log \rho$ bits, its largest entry lies in $[\rho^{-20}, \rho^{20})$, so $-20 \log \rho < j \leq 20 \log \rho$. Write $M \triangleq A^\top W^{(k)} A$. The error $y_t - M^{-1} b'$ of the t -th iterate contracts in the M norm at rate $c \leq 1/2$ by Equation (4) and Imported result 2.1, so $\|y_t\|_M \leq 2\|M^{-1} b'\|_M$ and $\|y_t\|_2 \leq 2\|M^{-1}\| \|b'\|_2 \leq 2\rho d^{1/2}$, up to the truncation errors bounded below; the products $A y_t$, $W^{(k)} A y_t$ and $M y_t$ have magnitude at most ρ^3 times this, so every number formed has magnitude below $\rho^5 < \rho^{20}$ and fits in $20 \log \rho$ bits; the output $2^j y'$ has magnitude below ρ^{25} and at most $20 \log \rho + 20 \log \rho - 1 < 40 \log \rho$ bits after the point, so it is a number with $40 \log \rho$ bits as Item 3.1(3) requires, and costs $O(\log \rho)$ per entry to write. One Richardson step applies A , $W^{(k)}$, $A^\top$ and the maintained $d \times d$ inverse, each product being formed exactly and truncated once to $20 \log \rho$ bits; each of these $O(1)$ truncations moves a vector of dimension at most n by at most $n^{1/2} \rho^{-20}$ in ℓ_2 , and is then amplified by the operators applied after it, whose norms multiply to at most $2\rho^3$, so the step is computed to within $O(n^{1/2} \rho^{-17})$ in ℓ_2 and hence within $O(n^{1/2} \rho^{-16.5}) \leq \rho^{-15}$ in the M norm, using $\|M\| \leq \rho$ and $\rho \geq n$. Against

$$\|M^{-1} b'\|_M = \|b'\|_{M^{-1}} \geq \rho^{-1/2} \|b'\|_2 \geq \frac{\rho^{-1/2}}{2}$$

this is a relative error of at most $2\rho^{-14.5}$ per step. A contraction of rate $c \leq 1/2$ run with additive error e per step converges to within $e/(1-c) \leq 2e$ of the exact solution, so $\tilde{O}(\log \rho)$ steps reach relative accuracy ρ^{-10} , and the truncation errors never move an iterate by more than

$4\rho^{-14.5}\|M^{-1}b'\|_M$, which justifies the magnitude bound above. The same scaling is used by every solve the data structure runs; the two inner solves of NOISYSOLVER below, which are asked for accuracy $\rho^{-20}/(32n^7) \geq \rho^{-28}$, run on numbers with $60\log\rho$ bits instead, which puts the floor of the same computation below ρ^{-50} and changes the cost by a constant factor. $\square$

3.3 Hiding the randomness

Lemma 3.2 assumes an oblivious adversary: its sampling is only valid if the $w^{(k)}$ handed to round k are independent of the coin flips used in rounds $1, \dots, k-1$. This fails as stated for our application, where $w^{(k)}$ is computed by the interior point method from the approximate solutions the data structure itself returned, and where a single leaked bit about which rows were sampled could be amplified into a weight sequence that forces a refresh every round.

So we ask for two conditions on the interface of Definition 3.1, which together make the updates oblivious to the sampling:

1. the only outputs of the data structure that depend on the sample are the outputs of SOLVE, which is the only output at all; anything the caller computes from them is computed from a full pass over A with the sample entering only as a preconditioner, whose accuracy requirement is met deterministically;
2. the output of SOLVE is within total variation distance n^{-3} of a distribution that is a function of (A, W, b) alone.

The first is exactly the last assertion of Lemma 3.2. The second is achieved by injecting noise into the solver's output, following [LS15, Section 5]: SOLVE is implemented as a call to NOISYSOLVER below, and is then the only channel through which anything about the sampling could leak.

The following is Lemma 16 of [LS15], the NOISYSOLVER, stated there for $A^\top A$ and applied here to $W^{1/2}A$; a linear solver is one whose output is a linear function of its right hand side.

Imported result 3.1. *Let $\mathcal{S}$ be a $T_{\mathcal{S}}$ -time linear solver for $A^\top WA$. The algorithm that, on input b and accuracy ϵ , returns*

$$y \triangleq y_1 + \alpha_{\text{noise}} y_2, \quad y_1 \triangleq \mathcal{S}(b, \epsilon_1), \quad y_2 \triangleq \mathcal{S}(A^\top W^{1/2} \xi, \epsilon_2), \quad \alpha_{\text{noise}} \triangleq \frac{\epsilon}{8n^{1/2}} \|y_1\|_{A^\top WA},$$

with $\xi \sim N(0, I_n)$, $\epsilon_1 = \epsilon^2/(32n^7)$ and $\epsilon_2 = \epsilon^2/(12dn^6)$, is an ϵ -accurate $O(\text{nnz}(A) + T_{\mathcal{S}})$ -time solver for $A^\top WA$.

The following, Theorem 17 of [LS15], is the indistinguishability of the noisy solver from an ideal one.

Imported result 3.2. *Let $\text{IDEALSOLVER}(b) \triangleq (A^\top WA)^{-1}b + c$ where $c \sim N(0, \beta_{\text{noise}}^2 (A^\top WA)^{-1})$ and $\beta_{\text{noise}} = \frac{\epsilon}{8n^{1/2}} \|(A^\top WA)^{-1}b\|_{A^\top WA}$. Then the total variation distance between the outputs of NOISYSOLVER and IDEALSOLVER is at most n^{-3} ; consequently no algorithm making at most n^2 calls can distinguish them.*

Both statements are for a linear $\mathcal{S}$: the proof of the second uses that y_2 is a linear image of ξ and so Gaussian. The noise is a $(1 \pm \epsilon)$ perturbation in the $A^\top WA$ norm, so it is absorbed into the solve tolerance of every statement that consumes the solver, and its distribution depends on the data structure's randomness not at all. Since IDEALSOLVER is a deterministic function of (A, W, b) plus fresh noise, any weight sequence generated from its outputs is independent of the sampling, and the obliviousness hypothesis of Lemma 3.2 may be dropped.

This gives the guarantee for adaptive weight sequences.

Corollary 3.3. *Lemma 3.2 holds for weight sequences $w^{(k)}$ produced by any algorithm that sees the data structure only through the operations of Definition 3.1, provided every SOLVE is implemented by NOISYSOLVER of Imported result 3.1, at the cost of $\tilde{O}(\text{nnz}(A) \log \rho)$ extra bit operations per call; the failure probability is $n^{-7} + 2Qn^{-3}$ as in that lemma, with Q now counting the caller's SOLVE calls as well.*

Proof. By Imported result 3.2 each call is within total variation distance n^{-3} of IDEALSOLVER, so by a union bound over the Q calls the whole run may be coupled with a run against IDEALSOLVER, except with probability $2Qn^{-3}$; the internal calls of the structure are already coupled this way in the proof of Lemma 3.2, so the coupling extends to the caller's calls at no further cost than counting them in Q . Under the coupling every $w^{(k)}$ is a function of A , of $w^{(0)}, \dots, w^{(k-1)}$, and of Gaussian noise independent of the sampling used inside the data structure, so the sequence is oblivious in the sense required by Lemma 3.2, which applies verbatim. The extra cost per call is one additional solve and one multiplication by $A^\top$, and by Imported result 3.1 the accuracy degrades by a constant factor, absorbed into $\rho^{-O(1)}$. $\square$

Three points concern fixed point; throughout, $M \triangleq A^\top W A$ and NOISYSOLVER is run at accuracy $\epsilon = \rho^{-10}/2$, so that $\epsilon_1 = \epsilon^2/(32n^7) \geq \rho^{-28}$ and $\epsilon_2 = \epsilon^2/(12dn^6) \geq \rho^{-28}$ for $\rho \geq \max(n, 2^7)$. First, the output of NOISYSOLVER is formed on the scaled right hand side $b' = 2^{-j}b$ of the proof of Lemma 3.2, on numbers with $60 \log \rho$ bits, then truncated to $20 \log \rho$ bits and multiplied by 2^j , and the run it is coupled with is against $2^j \text{Trunc}(\text{IDEALSOLVER}(b'), 20 \log \rho)$; truncation and scaling are one map applied to both outputs, and total variation distance does not increase under a common map, so the bound n^{-3} of Imported result 3.2 is unaffected, whatever the size of the noise relative to ρ^{-20} , and the truncated ideal output is still a function of (A, W, b) and fresh noise. The truncation moves the scaled output by at most $d^{1/2}\rho^{-20}$ in ℓ_2 , hence by at most $d^{1/2}\rho^{-19.5} \leq \rho^{-18}\|M^{-1}b'\|_M$ in the M norm, by the lower bound $\|M^{-1}b'\|_M \geq \rho^{-1/2}/2$ displayed in that proof, so the returned vector is within Equation (1). Second, ξ is the only place the algorithm needs randomness of more than $O(\log n)$ bits per row: it is sampled with $40 \log \rho$ bits per entry, that is, the algorithm holds the truncation $\xi' = \text{Trunc}(\xi, 40 \log \rho)$ of an exact Gaussian vector ξ , a discrete distribution sampled with $\text{poly}(\log \rho)$ bit operations per entry, on the event that the entries of ξ have magnitude below ρ^{40} , which has probability $1 - ne^{-\rho^{80}/2}$. Third, the fixed point computation is not linear in its right hand side, while Imported result 3.2 is for a linear $\mathcal{S}$. Let $y'' \triangleq y_1 + \alpha_{\text{noise}} Q_2 A^\top W^{1/2} \xi$ be the output of the exact arithmetic NOISYSOLVER on b' and the exact Gaussian ξ , with the exact arithmetic Richardson iteration, a linear map Q_2 with $\|Q_2 v - M^{-1}v\|_M \leq \epsilon_2 \|M^{-1}v\|_M$ for every v , as its inner solver; Imported results 3.1 and 3.2 apply to y'' . The computed output differs from y'' in two ways. Replacing ξ by ξ' moves $\alpha_{\text{noise}} Q_2 A^\top W^{1/2} \xi$ by at most $2\alpha_{\text{noise}} \|A^\top W^{1/2}(\xi - \xi')\|_{M^{-1}} \leq 2\alpha_{\text{noise}} n^{1/2} \rho^{-40} \leq \rho^{-48}$ in the M norm, since $\|A^\top W^{1/2} z\|_{M^{-1}}^2 = z^\top W^{1/2} A M^{-1} A^\top W^{1/2} z \leq \|z\|_2^2$ and $\alpha_{\text{noise}} \leq \epsilon \|y_1\|_M / 8 \leq \rho^{-9}$ by $\|y_1\|_M \leq 2\|M^{-1}b'\|_M \leq 2\rho^{1/2} d^{1/2}$. The truncations of the inner solves, of the scale α_{noise} and of the final sum are each a relative ρ^{-50} of the quantity truncated, by the proof of Lemma 3.2 run with $60 \log \rho$ bits; by the χ^2 tail bound in the proof of Imported result 3.1, $\|\xi\|_2^2 \leq 16n$ except with probability n^{-4} , and then $\alpha_{\text{noise}} \|y_2\|_M \leq (\epsilon/8n^{1/2}) \|y_1\|_M \cdot 4n^{1/2}(1 + \epsilon_2) \leq \|M^{-1}b'\|_M$, using $\|\xi'\|_2 \leq \|\xi\|_2$, so these truncations move the output by at most $\rho^{-48} \|M^{-1}b'\|_M \leq \rho^{-47}$ in the M norm. Altogether the computed output is within $2\rho^{-47}$ of y'' in the M norm and hence within ρ^{-46} of it in every coordinate, using $\|M^{-1}\| \leq \rho$. The truncations to $20 \log \rho$ bits of the two then agree unless some coordinate of y'' lies within ρ^{-46} of a multiple of $h \triangleq \rho^{-20}$. Coordinate i of y'' is Gaussian with mean $(y_1)_i$ and standard deviation $s_i = \alpha_{\text{noise}} (Q_2 M Q_2^\top)_{ii}^{1/2} \geq \alpha_{\text{noise}} \rho^{-1/2}/2 \geq \rho^{-13}$, where $Q_2 M Q_2^\top \approx_1 M^{-1}$ by the accuracy of the linear Q_2 and $\alpha_{\text{noise}} \geq (\epsilon/8n^{1/2})(1 - \epsilon_1) \|M^{-1}b'\|_M \geq \rho^{-11}/(64n^{1/2})$. For

a unimodal density f with $\sup f \leq 1/s$ and cells of width h , the maxima of f over the cells sum to at most $1/h + 2/s$, and a δ -neighbourhood of a lattice point meets two cells, so the probability of landing within δ of the lattice $h\mathbb{Z}$ is at most $4\delta(1/h + 2/s)$; with $\delta = \rho^{-46}$ this is at most $4\rho^{-46}(\rho^{20} + 2\rho^{13}) \leq \rho^{-25}$ per coordinate and ρ^{-24} over the $d \leq \rho$ coordinates. The truncated computed output is therefore within total variation distance $n^{-3} + n^{-4} + \rho^{-24} \leq n^{-3} + 2n^{-4}$ of $2^j \text{Trunc}(\text{IDEALSOLVER}(b'), 20 \log \rho)$, and the excess $2n^{-4}$ over the imported n^{-3} is what is charged per call in the proof of Lemma 3.2; the coupled ideal output remains a function of (A, W, b) and fresh noise.

3.4 Overall accounting and adaptive solver interface

Condition 3 of Definition 3.2 converts into a bound on the leverage score mass of the coordinates that a lazy refresh rule touches over a window of rounds. This bound is the only way the stability of the sequence enters the cost.

The following corollary bounds the refreshed mass. A refresh schedule is, for each coordinate i , an increasing sequence of rounds $t_{i,1} < t_{i,2} < \dots$ in the window, the rounds at which the coordinate is refreshed. It is (η, η') -triggered, for $0 < \eta \leq \eta'$, if for every consecutive pair $t = t_{i,j} < t' = t_{i,j+1}$ of refresh rounds of a coordinate i the refresh at t' was caused by a movement of at least η and not by a smaller one:

$$w_i^{(t')} \not\approx_{\eta} w_i^{(t)} \quad \text{or} \quad \sigma_i^{(t')} \not\approx_{\eta} \sigma_i^{(t)}, \quad \text{while} \quad w_i^{(k)} \approx_{\eta'} w_i^{(t)} \quad \text{and} \quad \sigma_i^{(k)} \approx_{\eta'} \sigma_i^{(t)} \quad \text{for } t \leq k < t'.$$

This is what a lazy rule produces when it refreshes a coordinate at the first round its weight or leverage score is seen to have moved by about a constant factor, the gap between η and η' being the accuracy of the estimate it acts on.

Corollary 3.4. *Let $w^{(k_0)}, \dots, w^{(k_0+r_0)}$ be ζ -leverage stable and let a refresh schedule on this window be (η, η') -triggered. Then the leverage score mass refreshed over the window, not counting the first refresh of each coordinate, satisfies*

$$\sum_i \sum_{j \geq 1} \sigma_i^{(t_{i,j+1})} \leq \frac{30e^{3\eta'} r_0^2 \zeta^2}{\eta^2}.$$

The same bound holds, up to a constant factor, for a sequence $\hat{w}^{(k_0)}, \dots, \hat{w}^{(k_0+r_0)}$ that is ζ -leverage stable up to noise, with $\hat{w}$ and its own leverage scores in the trigger conditions, provided $\eta \geq \zeta$.

Proof. Write $\delta^{(m)} \triangleq \log w^{(m+1)} - \log w^{(m)}$ for the single round increments, so that condition 3 reads $\|\delta^{(m)}\|_{\sigma^{(m+1)}} \leq \zeta$, and condition 2 with the fact recorded after Definition 3.2 gives $\sigma^{(m)} \approx_{2\zeta} \sigma^{(m+1)}$. Fix a consecutive pair $t < t'$ of refresh rounds of a coordinate i . The second half of the trigger condition, together with $\sigma_i^{(t')} \approx_{2\zeta} \sigma_i^{(t'-1)}$, gives

$$\sigma_i^{(k)} \approx_{2\eta'+2\zeta} \sigma_i^{(t')} \quad \text{for every } t \leq k \leq t', \tag{5}$$

so the leverage score of a coordinate is comparable to its value at the refresh throughout the interval that ends at that refresh. This is the only comparison between different rounds that the proof uses; the weights themselves may drift by large factors over the window.

Weight triggers. If the pair is triggered by the weight, then $|\sum_{m=t}^{t'-1} \delta_i^{(m)}| \geq \eta$, and by Cauchy-Schwarz over the $t' - t \leq r_0$ terms,

$$\eta^2 \sigma_i^{(t')} \leq r_0 \sum_{m=t}^{t'-1} \sigma_i^{(t')} \left(\delta_i^{(m)} \right)^2 \leq e^{2\eta'+2\zeta} r_0 \sum_{m=t}^{t'-1} \sigma_i^{(m+1)} \left(\delta_i^{(m)} \right)^2,$$

by Equation (5) at $k = m + 1$.

Leverage score triggers. Interpolate between consecutive weight vectors along $w(s) \triangleq w^{(m)}e^{(s-m)\delta^{(m)}}$ for $s \in [m, m+1]$, a continuous path through all the weights of the window, with $W(s) = \text{diag}(w(s))$, $M(s) \triangleq A^\top W(s)A$, $\sigma(s) \triangleq \tau(W(s)^{1/2}A)$ and $P(s) \triangleq W(s)^{1/2}AM(s)^{-1}A^\top W(s)^{1/2}$ the projection, so that $\sigma_i(s) = w_i(s)a_i^\top M(s)^{-1}a_i$ and $P(s)_{ij}^2 = w_i(s)w_j(s)(a_i^\top M(s)^{-1}a_j)^2$. On $[m, m+1]$, writing $u \triangleq \delta^{(m)}$, we have $\frac{d}{ds}M(s)^{-1} = -M(s)^{-1}A^\top W(s)\text{diag}(u)AM(s)^{-1}$, so

$$\frac{d}{ds} \log \sigma_i(s) = u_i - \frac{\sum_j w_j(s)u_j(a_i^\top M(s)^{-1}a_j)^2}{a_i^\top M(s)^{-1}a_i} = u_i - \frac{1}{\sigma_i(s)} \sum_j P(s)_{ij}^2 u_j.$$

By Cauchy–Schwarz and $\sum_j P(s)_{ij}^2 = \sigma_i(s)$,

$$\begin{aligned} \sum_i \sigma_i(s) \left(\frac{d}{ds} \log \sigma_i(s) \right)^2 &\leq 2 \sum_i \sigma_i(s) u_i^2 + 2 \sum_i \frac{1}{\sigma_i(s)} \left(\sum_j P(s)_{ij}^2 u_j \right)^2 \\ &\leq 2 \|u\|_{\sigma(s)}^2 + 2 \sum_j u_j^2 \sum_i P(s)_{ij}^2 = 4 \|u\|_{\sigma(s)}^2, \end{aligned}$$

the last step by symmetry of $P(s)$. On $[m, m+1]$ the weights $w(s)$ are within $e^{\pm\zeta}$ of $w^{(m+1)}$ coordinatewise, so $\sigma(s) \approx_{2\zeta} \sigma^{(m+1)}$ and, with $D_i(s) \triangleq \sigma_i(s) \left(\frac{d}{ds} \log \sigma_i(s) \right)^2$,

$$\int_m^{m+1} \sum_i D_i(s) ds \leq 4e^{2\zeta} \|\delta^{(m)}\|_{\sigma^{(m+1)}}^2 \leq 4e^{2\zeta} \zeta^2. \quad (6)$$

If the pair is triggered by the leverage score, then $|\int_t^{t'} \frac{d}{ds} \log \sigma_i(s) ds| \geq \eta$, and by Cauchy–Schwarz in s over an interval of length at most r_0 ,

$$\eta^2 \sigma_i^{(t')} \leq r_0 \int_t^{t'} \sigma_i^{(t')} \left(\frac{d}{ds} \log \sigma_i(s) \right)^2 ds \leq e^{2\eta' + 4\zeta} r_0 \int_t^{t'} D_i(s) ds,$$

since $\sigma_i^{(t')} \leq e^{2\eta' + 2\zeta} \sigma_i^{(m+1)} \leq e^{2\eta' + 4\zeta} \sigma_i(s)$ for $s \in [m, m+1] \subseteq [t, t']$.

Summing. For a fixed i the intervals $[t_{i,j}, t_{i,j+1})$ are disjoint, so the sums over m and the integrals over s belonging to distinct pairs of the same coordinate do not overlap. Summing the two trigger bounds over all consecutive pairs of all coordinates, and using $\zeta \leq 0.1$,

$$\begin{aligned} \eta^2 \sum_i \sum_{j \geq 1} \sigma_i^{(t_{i,j+1})} &\leq e^{2\eta' + 0.4} r_0 \sum_{m=k_0}^{k_0+r_0-1} \left(\|\delta^{(m)}\|_{\sigma^{(m+1)}}^2 + \int_m^{m+1} \sum_i D_i(s) ds \right) \\ &\leq e^{2\eta' + 0.4} r_0 \cdot r_0 \cdot 5e^{0.2} \zeta^2 \leq 30e^{3\eta'} r_0^2 \zeta^2. \end{aligned}$$

Noisy version. A trigger $\hat{w}_i^{(t')} \not\approx_\eta \hat{w}_i^{(t)}$ together with $\hat{w}^{(m)} \approx_{\zeta/10} w^{(m)}$ at both ends forces $w_i^{(t')} \not\approx_{\eta-\zeta/5} w_i^{(t)}$, and $\eta - \zeta/5 \geq 0.8\eta$ when $\eta \geq \zeta$; conversely the second half of the trigger condition for $\hat{w}$ gives it for w with $\eta' + \zeta/5 \leq 1.2\eta'$. The leverage scores of $\hat{w}^{(m)}$ are within $e^{\pm\zeta/5}$ of $\sigma^{(m)}$, so the same two implications hold for the leverage score triggers, and the exact bound for a $(0.8\eta, 1.2\eta')$ -triggered schedule gives the noisy one up to a constant factor. $\square$

This bounds the quantity left open in Lemma 3.2, and with a restart schedule it proves the overall guarantee.

Proof of Theorem 2. By Corollary 3.3 it suffices to prove the statement for an oblivious sequence, at the cost of the $2Qn^{-3}$ in the failure probability and $\tilde{O}(\text{nnz}(A) \log \rho)$ per SOLVE already included in the claimed bounds, so we may apply Lemma 3.2.

Fix a window length r_0 and restart the data structure, resampling from scratch, every r_0 rounds. Inside a window, a coordinate is refreshed when its weight or its estimated leverage score has moved by more than a factor of 2 since its last refresh, which, the estimates being 0.1-accurate, forces the exact weight or leverage score to have moved by more than a factor of $2e^{-0.2} > e^{0.1}$, while until that round the exact quantities had moved by less than a factor of $2e^{0.2} < e^1$: the refresh schedule is $(0.1, 1)$ -triggered in the sense of Corollary 3.4, and the sequence handed to the structure is ζ -leverage stable, or stable up to noise with $\zeta < 0.1 = \eta$. So the corollary bounds the leverage score mass of all coordinates refreshed in the window, beyond the first refresh of each at the restart, by $\tilde{O}(r_0^2 \zeta^2)$, where the mass of a refreshed coordinate is measured at the round of its refresh. A refresh of coordinate i at round t' , its previous refresh being at t , changes the sampled matrix only if the row is kept before or after the refresh, that is, only if the coin flipped at t or the coin flipped at t' came up kept; so $\sum_k c_k$ is at most twice the number of kept coins among the coins flipped at the refreshes of the window, plus the number of kept coins flipped at the restart for coordinates refreshed later in the window. Under the coupling of Lemma 3.2 the refresh schedule and the probabilities $p_i^{(t)} = \min\{1, c_s \hat{\sigma}_i^{(t)} \log n\}$ are functions of the oblivious weights and of the ideal solver's noise, hence independent of the coins, and conditioned on them the coins are independent. The means of the coins counted, each at most twice, sum to at most twice

$$\begin{aligned} \sum_{\text{refreshes } (i, t')} p_i^{(t')} + \sum_{i \in I} p_i^{(k_0)} &\leq c_s e^{0.1} \log n \left(\sum_{\text{refreshes } (i, t')} \sigma_i^{(t')} + \sum_{i \in I} \sigma_i^{(k_0)} \right) \\ &\leq 2c_s e^{2.3} \log n \sum_{\text{refreshes } (i, t')} \sigma_i^{(t')} = \tilde{O}(r_0^2 \zeta^2), \end{aligned}$$

where I is the set of coordinates refreshed at least once after the restart, the refreshes summed over are those after the restart, and $\sigma_i^{(k_0)} \leq e^{2.2} \sigma_i^{(t_{i,2})}$ for $i \in I$ by Equation (5) applied to the pair $(k_0, t_{i,2})$ with $2\eta' + 2\zeta \leq 2.2$, the restart counting as the first refresh $t_{i,1} = k_0$ of every coordinate; the last step is the mass bound of the corollary. This is the expected count of [LS15, Lemma 15]. A Chernoff bound for independent indicators with this sum of means then gives

$$\sum_{k \leq r_0} c_k = \tilde{O}(r_0^2 \zeta^2 + \log n) \quad (7)$$

with probability at least $1 - n^{-10}$, and the additive $\log n$ is absorbed below, since $\text{MATMULT}(d, \log n, d) = \tilde{O}(d^{\sqrt{2}} \log n)$ is inside the per window budget. By superadditivity of $\text{MATMULT}(d, \cdot, d)$ in its middle dimension the UPDATE cost over the window is therefore

$$\sum_{k \leq r_0} \text{MATMULT}(d, c_k, d) \leq \text{MATMULT}\left(d, \tilde{O}(r_0^2 \zeta^2 + \log n), d\right) = \tilde{O}\left(\frac{r_0^2}{d} \text{MATMULT}(d, d, d)\right),$$

on top of the one $\tilde{O}(\text{MATMULT}(d, d, d) \log \rho)$ paid at the restart, with probability at least $1 - n^{-10}$ by Equation (7), and with probability at least $1 - n^{-8}$ in every one of the at most n^2 windows by a union bound. The cost bound is therefore a high probability one, as the theorem states, and not merely a bound in expectation. Balancing $\text{MATMULT}(d, d, d)/r_0$ against $r_0 \text{MATMULT}(d, d, d)/d$ gives $r_0 = d^{1/2}$ and an amortized

$$\tilde{O}\left(\frac{\text{MATMULT}(d, d, d)}{d^{1/2}} \log \rho\right) = \tilde{O}\left(d^{1/2 + \sqrt{2}} \log \rho\right) = \tilde{O}(d^2 \log \rho)$$

per round, the last two steps by Equation (2), on top of the $\tilde{O}((\text{nnz}(A) + d^2) \log^2 \rho)$ per round spent reading the update vector and re-estimating the leverage scores for the refresh test. The two together are the amortized

$$\tilde{O}((\text{nnz}(A) + d^2) \log^2 \rho)$$

of Theorem 2, the second $\log \rho$ being the iteration count of the solves inside the estimate, the first their bit length. Resampling at a restart re-estimates the leverage scores against the solver the structure already has, at $\tilde{O}((\text{nnz}(A) + d^2) \log^2 \rho)$, the sample of the previous window being a constant factor preconditioner still, by Item 3.2(2), so no later window repeats the dense inverse of the first round, and the $\text{nnz}(A) + \text{MATMULT}(d, d, d)$ term is paid once. The failure probability is n^{-8} per round from the proof of Lemma 3.2, whose union bound was over rounds, so n^{-6} over the $r \leq n^2$ rounds, plus the $2Qn^{-3}$ of its coupling and the n^{-8} of Equation (7), at most the $n^{-5} + 2Qn^{-3}$ of the theorem. It remains to account for SOLVE, which neither Lemma 3.1 nor Lemma 3.2 charges, and which Item 2(3) of the theorem does.

Correctness of SOLVE. Let $N^{(k)} \triangleq A^\top S^{(k)} W^{(k)} A$ be the matrix whose inverse the structure holds exactly, and $A^\top W^{(k)} A$ the matrix Equation (1) is stated for. These are within a factor $e^{0.1}$ of each other by Equation (4), so $(N^{(k)})^{-1}$ is a constant factor preconditioner for $A^\top W^{(k)} A$, and preconditioned Richardson iteration on $A^\top W^{(k)} A y = b$ contracts the error in the $A^\top W^{(k)} A$ norm by a constant factor per step, by Imported result 2.1. The operator applied at each step is the exact $A^\top W^{(k)} A$, one multiplication by A and one by $A^\top$, so the sample enters only through the contraction rate: what the iteration converges to is the true solution, and Equation (1) is a guarantee against $A^\top W^{(k)} A$ and not against the sampled matrix. Reaching ρ^{-10} takes $\tilde{O}(\log \rho)$ steps.

Cost of SOLVE. A step costs one multiplication by A , one by $A^\top$, and one by the dense $d \times d$ inverse, so $\tilde{O}(\text{nnz}(A) + d^2)$ arithmetic operations, each on numbers of $\tilde{O}(\log \rho)$ bits by Item 3.2(1) and Equation (4). Over $\tilde{O}(\log \rho)$ steps this is

$$\tilde{O}((\text{nnz}(A) + d^2) \log^2 \rho)$$

bit operations, which is Item 2(3); the two logarithms are the iteration count and the bit length, and it is also what the refresh test inside UPDATE pays for its own estimate, already counted above.

Adaptivity of SOLVE. Every SOLVE is served by NOISYSOLVER of Imported result 3.1, which is one extra solve, one extra multiplication by $A^\top$, and the $\tilde{O}(\text{nnz}(A) \log \rho)$ of sampling and rounding the Gaussian, so a constant factor on the above, within Item 2(3). This is what Corollary 3.3 charges for dropping the obliviousness hypothesis, and it is why the theorem may be stated for an adaptive sequence. $\square$

The restart balance is the only place Equation (2) is used, and it is used as [LS15] use it, to make the square recomputation at a restart fit under the per round budget. Keeping the Woodbury updates rectangular does not remove it. Suppose the structure held its inverse in factored form, as the inverse at the last restart plus the accumulated rank one corrections, applying it to a vector in $\tilde{O}(d^2 + Cd)$ after C corrections and never forming a square product between restarts. Every window still begins by inverting a $d \times d$ matrix, at $\text{MATMULT}(d, d, d)$, and $\tilde{O}(d^2 + Cd)$ stays inside the per round budget only while $C = \tilde{O}(d)$, which by Equation (7) caps the window at $r_0 = \tilde{O}(d^{1/2}/\zeta)$. The amortized restart cost is then $\text{MATMULT}(d, d, d)\zeta/d^{1/2}$, the same balance up to the constant ζ , so Equation (2) is needed either way.

4 Lewis Weights and Potential

Symbol	Meaning
q, α	Lewis weight parameter $q = 1 - \alpha \in [1/2, 2)$, $\alpha = 1/(10 \log(10n/d))$; Lemma 4.2, Definition 4.2
v	regularizer, $d/n \leq v_i \leq 1$ and $\ v\ _1 \leq 4d$; Section 4.1, Lemma 4.2, Definition 4.2
$\tau_q(A, v)$	v -regularized ℓ_q -Lewis weights of A ; Section 4.1, Lemma 4.1
$\tilde{\tau}$	an estimate of τ_q , supplied to Lemma 4.2; maintained along the path by Definition 4.6
$\varsigma(w), M_w$	Lewis map $\tau(\text{diag}(w)^{1/2-1/q}A) + v$ and the quadratic form $A^\top \text{diag}(w)^{1-2/q}A$ behind it; Equation (8)
ε	target accuracy of the estimation routines, distinct from the centrality ϵ of Equation (11); Lemma 4.1, Lemma 4.2
$\mathcal{S}, T_{\mathcal{S}}$	solver supplied to Lemma 4.2, and the cost of one call; Item 4.2(6)
$T(x), \tilde{T}$	$\text{diag}(\tau(x))$ and $\text{diag}(\tilde{\tau})$; Definition 4.6
$\phi_i(t)$	log barrier $-\log(t - l_i) - \log(u_i - t)$; $\Phi''(x) = \text{diag}(\phi_i''(x_i))$; Lemma 4.3
$\tau(x)$	$\tau_q(\Phi''(x)^{-1/2}A, v)$; Definition 4.2
μ	path parameter, from μ_0 down to μ_f ; Definition 4.3, Lemma 5.10
λ	potential scale, $\Theta(\log(n/\epsilon)/\epsilon)$; Equation (11), Definition 4.4
$y(x, s, \mu), \bar{y}$	centrality vector of Definition 4.4, and its computed value in a step, Lemma 5.2
$\Psi(x, s, \mu)$	centrality potential; Definition 4.4, Lemma 5.8
C, C_{start}	universal constant of Equation (11), and $10C$; Lemma 4.4
γ, C_{norm}	step size and norm scale of Equation (11); Definition 4.3, Definition 4.5
$\ g\ _{\tau+\infty}$	$\ g\ _\infty + C_{\text{norm}}\ g\ _\tau$, where $\ g\ _\tau^2 = \sum_i \tau_i g_i^2$; Definition 4.5
$g^{b(\tau)}$	the dual unit vector of g in $\ \cdot\ _{\tau+\infty}$; Definition 4.5, Definition 5.1

Figure 3: Variables local to the Lewis weights and potential of Section 4.

This section collects everything about the weights of the central path: the regularized Lewis weights and the two routines that estimate them, the barrier and the parameters of the method, and the centrality potential that Section 5 drives down one step at a time. All of it is the fixed point analogue of the regularized Lewis weight central path of [vdBLL⁺21], transposed to our dimensions (n rows, d columns, so $m \rightarrow n$ and $n \rightarrow d$ there), and with the ℓ_p exponent there written as q here. The data structure of Section 3 knows nothing of these weights: it sees only the vector handed to UPDATE. The estimation routines below read A only through matrix vector products and through calls of a supplied solver, and the interior point method supplies SOLVE of Definition 3.1 as that solver.

4.1 Regularized Lewis weights

Definition 4.1. For $q \in [1/2, 2)$, $n \times d$ matrix A with $n > d$ and rank d , and length n positive vector v with $d/n \leq v_i \leq 1$ and $\|v\|_1 \leq 4d$, the v -regularized ℓ_q -Lewis weights of A , $\tau_q(A, v)$ is the unique vector such that

$$\tau_q(A, v) = \tau \left(\text{diag}(\tau_q(A, v))^{1/2-1/q} A \right) + v$$

The fixed point that defines the weights is reached by a damped iteration that touches A only through matrix vector products and through linear systems in a quadratic form spectrally close to the Lewis one. We state it twice. Once from scratch, where the starting point is the all ones vector and the solver has to be built as well, which is the only place in the paper a dense $d \times d$

inverse is formed. And once from a coarse estimate $\tilde{\tau}$ of the answer together with a solver for the quadratic form that estimate induces, which is the form the interior point method calls it in, the estimate being the previous round's and the solver being the SOLVE of Definition 3.1. Throughout this subsection write

$$\varsigma(w) \triangleq \tau \left(\text{diag}(w)^{1/2-1/q} A \right) + v \quad (8)$$

for the map whose fixed point is $\tau_q(A, v)$, and $M_w \triangleq A^\top \text{diag}(w)^{1-2/q} A$ for the quadratic form behind it, so that $\varsigma_i(w) - v_i = w_i^{1-2/q} a_i^\top M_w^{-1} a_i$.

The two facts about Equation (8) that both routines rest on are Lemma 5.3 and Lemma 5.5 of [vdBLL⁺21], which transfer the contraction argument of [CP15] to the regularized weights; the regularizer v is their z .

Imported result 4.1. *Let $q \in (0, 2)$ and $w, \bar{\varsigma} \in \mathbb{R}_{>0}^n$ with $w \approx_\delta \varsigma(w)$ and $\bar{\varsigma} \approx_\eta \varsigma(w)$, and let $w' \triangleq (w^{2/q-1} \bar{\varsigma})^{q/2}$ coordinatewise. Then $w' \approx_{(\delta+\eta)|1-q/2|+\eta} \varsigma(w')$ and $w' \approx_{(\delta+\eta)q/2} w$. Moreover any $w \approx_\delta \varsigma(w)$ satisfies $w \approx_\delta \tau_q(A, v)$.*

The static Lewis estimation is the following.

Lemma 4.1. *For $n > d$, given:*

1. *an $n \times d$ matrix A with column rank d and $A^\top A$ has magnitude at most ρ ,*
2. *parameter $1/2 \leq q < 2$,*
3. *vector v with $d/n \leq v_i \leq 1$ and $\|v\|_1 \leq 4d$,*
4. *accuracy $0.1 > \varepsilon > \rho^{-1}$,*

we can return an ε -approximation to the v -regularized ℓ_q -Lewis weights of A , with probability at least $1 - n^{-7}$, in bit complexity

$$\tilde{O} \left((\varepsilon^{-2} (\text{nnz}(A) + d^2) + \text{MATMULT}(d, d, d)) \log \rho \log^2(1/\varepsilon) \right).$$

Proof. Set $\eta \triangleq \varepsilon/20$ and run

$$w^{(t+1)} \triangleq \text{Trunc} \left(\left(\left(w^{(t)} \right)^{2/q-1} \bar{\varsigma}^{(t)} \right)^{q/2}, 20 \log \rho \right), \quad \bar{\varsigma}^{(t)} \approx_\eta \varsigma \left(w^{(t)} \right), \quad (9)$$

coordinatewise from $w^{(0)} \triangleq \mathbf{1}$ for $L \triangleq \lceil 4 \log(2 \log n / \varepsilon) \rceil$ steps, where $\bar{\varsigma}^{(t)}$ is the estimate of $\varsigma(w^{(t)})$ computed below, and return $w^{(L)}$.

Convergence. Let δ_t be such that $w^{(t)} \approx_{\delta_t} \varsigma(w^{(t)})$. Since leverage scores lie in $[0, 1]$ and $v_i \in [d/n, 1]$, $\varsigma(\mathbf{1}) \in [d/n, 2]^n$, so $\delta_0 \leq \log n$. As long as $\delta_t \leq \log n$ the iterate satisfies $w^{(t)} \approx_{\log n} \tau_q(A, v) \in [d/n, 2]^n$ by the last part of Imported result 4.1, so $w^{(t)} \in [\rho^{-2}, \rho]^n$, the untruncated update has magnitude $\rho^{O(1)}$, and the truncation in Equation (9) changes each coordinate by a factor $e^{\pm \rho^{-17}}$, which we absorb into η (it is below $\eta/100$ as $\varepsilon > \rho^{-1}$). The first part of Imported result 4.1 with $|1 - q/2| \leq 3/4$ gives

$$\delta_{t+1} \leq \frac{3}{4} (\delta_t + \eta) + \eta \leq \frac{3}{4} \delta_t + 2\eta, \quad \text{so} \quad \delta_t \leq \left(\frac{3}{4} \right)^t \log n + 8\eta \leq \left(\frac{3}{4} \right)^t \log n + \frac{\varepsilon}{2}$$

for all t , which in particular keeps $\delta_t \leq \log n$ throughout. At $t = L$ the first term is at most $\varepsilon/2$, so $w^{(L)} \approx_\varepsilon \varsigma(w^{(L)})$, and the last part of Imported result 4.1 turns this into $w^{(L)} \approx_\varepsilon \tau_q(A, v)$. Since $\varepsilon < 0.1$, $L = O(\log(1/\varepsilon) + \log \log n) = \tilde{O}(\log(1/\varepsilon))$.

Estimating $\varsigma(w^{(t)})$. Write $D \triangleq \text{diag}(w^{(t)})^{1/2-1/q}$ and $P \triangleq DAM_{w^{(t)}}^{-1}A^\top D$ for the projection onto the column span of DA , so $\varsigma_i(w^{(t)}) - v_i = \|Pe_i\|_2^2$. Let $G \in \mathbb{R}^{k \times n}$ have independent $N(0, 1/k)$ entries with $k \triangleq \lceil 10^4 \eta^{-2} \log n \rceil$, and set $\bar{\varsigma}_i^{(t)} \triangleq v_i + \|GPe_i\|_2^2$. By the Johnson–Lindenstrauss lemma in its Gaussian form, for each fixed i the event $\|GPe_i\|_2^2 \approx_{\eta/2} \|Pe_i\|_2^2$ fails with probability at most $2e^{-k\eta^2/100} \leq n^{-10}$, so all n coordinates satisfy it with probability at least $1 - n^{-9}$, and then $\bar{\varsigma}^{(t)} \approx_{\eta/2} \varsigma(w^{(t)})$ since adding $v_i \geq 0$ to both sides can only shrink a ratio. We also use that $\|G\|_F^2 \leq 2n$ except with probability $e^{-kn/8} \leq n^{-9}$: $k\|G\|_F^2$ is a chi-squared variable with kn degrees of freedom, and $\Pr[\chi_m^2 \geq 2m] \leq e^{-m/8}$. The k rows of GP are $(Pg_j)^\top$ for the rows g_j of G , and $Pg_j = DA \left(M_{w^{(t)}}^{-1} (A^\top Dg_j) \right)$ is two matrix vector products with DA at $O(\text{nnz}(A))$ bit operations and one linear solve in $M_{w^{(t)}}$. Summing squares of the k rows costs $O(kn)$, so the estimate costs $O(k \text{nnz}(A))$ bit operations plus k solves.

The solves and their accuracy. Each solve is run by preconditioned Richardson iteration to relative accuracy

$$\delta \triangleq \frac{\eta}{100n^2}$$

in the $M_{w^{(t)}}$ norm. The preconditioner is rebuilt at every iterate: apply a sparse embedding Π with $\tilde{O}(d)$ rows and constant factor spectral guarantee to DA at $\tilde{O}(\text{nnz}(A))$ bit operations, form the Gram matrix of ΠDA at $\tilde{O}(\text{MATMULT}(d, d, d))$ and invert it with Imported result 2.2 at $\tilde{O}(\text{MATMULT}(d, d, d) \log \rho)$; the inverse is a constant factor spectral approximation of $M_{w^{(t)}}^{-1}$, so $O(\log(1/\delta)) = O(\log(n/\eta))$ Richardson steps, each one application of $M_{w^{(t)}}$ at $O(\text{nnz}(A) \log \rho)$ bit operations and of the dense inverse at $O(d^2 \log \rho)$, reach accuracy δ . In fixed point the iteration is run exactly as the SOLVE in the proof of Lemma 3.2: the right hand side is scaled by a power of two into $[1/2, 1)$, the numbers have $20 \log \rho$ bits, and every matrix vector product is formed exactly and truncated once, so the argument there bounds the relative error of one step by $2\rho^{-14.5}$, which is below $\delta \geq \rho^{-1}/(2000\rho^2)$ by a factor of more than ρ^{10} and is absorbed into the rate. The residual error is negligible: the computed j -th row of GP is $(DA\tilde{y}_j)^\top$ with $\|\tilde{y}_j - M_{w^{(t)}}^{-1}A^\top Dg_j\|_{M_{w^{(t)}}} \leq \delta\|Pg_j\|_2 \leq \delta\|g_j\|_2$, since $\|y\|_{M_{w^{(t)}}} = \|DAy\|_2$ and P is a projection; so the row is $(Pg_j)^\top + f_j^\top$ with $\|f_j\|_2 \leq \delta\|g_j\|_2$. Writing $a_j \triangleq (Pg_j)_i$ and using $\sum_j a_j^2 \leq e^{\eta/2}\|Pe_i\|_2^2 \leq 2$ on the Johnson–Lindenstrauss event and $\sum_j (f_j)_i^2 \leq \delta^2\|G\|_F^2 \leq 2n\delta^2$,

$$\begin{aligned} \left| \sum_j (a_j + (f_j)_i)^2 - \sum_j a_j^2 \right| &\leq 2 \left(\sum_j a_j^2 \right)^{1/2} \left(\sum_j (f_j)_i^2 \right)^{1/2} + \sum_j (f_j)_i^2 \\ &\leq 2 \cdot 2^{1/2} \cdot (2n)^{1/2} \delta + 2n\delta^2 \\ &\leq 6n^{1/2} \delta = \frac{6\eta}{100n^{3/2}} \leq \frac{\eta}{4} \cdot \frac{d}{n} \leq \frac{\eta}{4} \varsigma_i(w^{(t)}), \end{aligned}$$

using $\delta \leq 1/n$ in the third step and $\varsigma_i(w^{(t)}) \geq v_i \geq d/n$ in the last. Together with the Johnson–Lindenstrauss error this gives $\bar{\varsigma}^{(t)} \approx_\eta \varsigma(w^{(t)})$, as Equation (9) requires.

Total. Per iterate: one preconditioner at $\tilde{O}(\text{MATMULT}(d, d, d) \log \rho)$, and $k = \tilde{O}(\varepsilon^{-2})$ solves at $O((\text{nnz}(A) + d^2) \log \rho \log(n/\varepsilon))$ each, plus $O(k \text{nnz}(A) \log \rho)$ around them. Over the $L = O(\log(1/\varepsilon))$ iterates this is

$$\tilde{O}((\varepsilon^{-2} (\text{nnz}(A) + d^2) \log(n/\varepsilon) + \text{MATMULT}(d, d, d)) \log \rho \log(1/\varepsilon)),$$

which is the stated bound since $\log(n/\varepsilon) \leq \log n \cdot \log(1/\varepsilon)$ for $\varepsilon < 0.1$ and $\tilde{O}$ hides the $\log n$. The failure probability is at most $L(n^{-9} + n^{-9}) \leq n^{-7}$, one Johnson–Lindenstrauss event and one Frobenius norm event per iterate. $\square$

Inside the interior point method the estimate of the previous round is already at hand, and with it a solver, so neither the all ones start nor the dense inverse is needed.

The fixed-point Lewis estimation from a solver is the following.

Lemma 4.2. *For $n > d$, given:*

1. *an $n \times d$ matrix A with column rank d and $A^\top A$ has magnitude at most ρ ,*
2. *parameter $1/2 \leq q < 2$,*
3. *vector v with $d/n \leq v_i \leq 1$ and $\|v\|_1 \leq 4d$,*
4. *accuracy $0.1 > \varepsilon > \rho^{-1}$,*
5. *a vector $\tilde{\tau}$ that 0.5-approximates the weights being estimated, $\tilde{\tau} \approx_{0.5} \tau_q(A, v)$,*
6. *a solver $\mathcal{S}$ for a fixed $d \times d$ matrix M with $M \approx_{0.5} A^\top \text{diag}(\tilde{\tau})^{1-2/q} A$, which on input $b \in \mathbb{R}^d$ returns y with $\|y - M^{-1}b\|_M \leq \rho^{-10} \|M^{-1}b\|_M$ in $T_{\mathcal{S}}$ bit operations,*

we can return an ε -approximation to the v -regularized ℓ_q -Lewis weights of A , with probability at least $1 - n^{-7}$, in bit complexity

$$\tilde{O}(\varepsilon^{-2} (\text{nnz}(A) \log \rho + T_{\mathcal{S}}) \log^2(1/\varepsilon)).$$

Proof. Run the iteration of Equation (9) from $w^{(0)} \triangleq \tilde{\tau}$ for $L \triangleq \lceil 4 \log(4/\varepsilon) \rceil = O(\log(1/\varepsilon))$ steps, with the solves in $M_{w^{(t)}}$ preconditioned by $\mathcal{S}$ instead of by a rebuilt dense inverse.

Convergence. By Item 4.2(5), $\varsigma(\tilde{\tau}) \approx_{0.5|1-2/q|} \varsigma(\tau_q(A, v)) = \tau_q(A, v) \approx_{0.5} \tilde{\tau}$, where the first step raises the coordinatewise ratio $\tilde{\tau}/\tau_q(A, v)$ to the power $1 - 2/q$ inside the quadratic form and the leverage scores of DA change by at most the factor D changes. As $|1 - 2/q| \leq 3$ on $[1/2, 2]$ this gives $\delta_0 \leq 2$, and the recursion of the proof of Lemma 4.1 gives $\delta_t \leq (3/4)^t \cdot 2 + \varepsilon/2 \leq 2$ for all t and $\delta_L \leq \varepsilon$, so $w^{(L)} \approx_\varepsilon \tau_q(A, v)$ by Imported result 4.1. The magnitude bookkeeping is as there, now with $w^{(t)} \approx_2 \tau_q(A, v)$.

The one fixed preconditioner suffices. Since every iterate satisfies $w^{(t)} \approx_2 \tau_q(A, v) \approx_{0.5} \tilde{\tau}$,

$$M_{w^{(t)}} \approx_{2.5|1-2/q|} A^\top \text{diag}(\tilde{\tau})^{1-2/q} A \approx_{0.5} M,$$

the second step being Item 4.2(6). So M is an e^8 spectral approximation of $M_{w^{(t)}}$ at every iterate, and a constant factor preconditioner is all Richardson iteration needs: $O(\log(n/\eta))$ steps, each one application of $M_{w^{(t)}}$ at $O(\text{nnz}(A) \log \rho)$ bit operations, its result truncated to the $20 \log \rho$ bits the solver takes, and one call to $\mathcal{S}$ at $T_{\mathcal{S}}$, reach the relative accuracy $\delta = \eta/(100n^2)$ of the proof of Lemma 4.1, the fixed point bookkeeping being the one there: the residual is scaled into $[1/2, 1)$ before it is handed to $\mathcal{S}$, whose relative error ρ^{-10} and the truncation of the residual, of relative size at most $2\rho^{-14.5}$ as in the proof of Lemma 3.2, are both far below $\delta \geq \rho^{-3}/2000$ and are absorbed into the rate. At the value $q = 1 - \alpha$ of Definition 4.2 the exponent is $|1 - 2/q| \leq 1.1$ and the constant is below $e^{3.3}$. Only the rate at which Richardson contracts depends on M , not the point it converges to, so M enters as a preconditioner only, and both approximations are asked at the loose accuracy 0.5; stating Item 4.2(6) at the supplied $\tilde{\tau}$ rather than at the answer is what lets the caller check it.

Total. The estimate of $\varsigma(w^{(t)})$, its accuracy, and its failure probability are exactly as in the proof of Lemma 4.1: $k = \tilde{O}(\varepsilon^{-2})$ solves and $O(k \text{nnz}(A) \log \rho)$ bit operations per iterate, each solve $O(\log(n/\varepsilon))$ Richardson steps at $O(\text{nnz}(A) \log \rho + T_S)$, over $L = O(\log(1/\varepsilon))$ iterates, which is $\tilde{O}(\varepsilon^{-2}(\text{nnz}(A) \log \rho + T_S) \log^2(1/\varepsilon))$ as $\log(n/\varepsilon) \leq \log n \cdot \log(1/\varepsilon)$; the failure probability is $L(n^{-9} + n^{-9}) \leq n^{-7}$. $\square$

4.2 Barrier, weights, and parameters

The barrier of the i -th box constraint $l_i \leq x_i \leq u_i$ is

$$\phi_i(x_i) \triangleq -\log(x_i - l_i) - \log(u_i - x_i),$$

and $\Phi''(x) \triangleq \text{diag}(\phi_i''(x_i))$.

The barrier is highly self-concordant in the sense of Definition 4.1 of [vdBLL⁺21]; the following is Lemma 4.2 there.

Lemma 4.3. *Each ϕ_i is highly 1-self-concordant on (l_i, u_i) : for all $t \in (l_i, u_i)$,*

$$|\phi_i'(t)| \leq \phi_i''(t)^{1/2}, \quad |\phi_i'''(t)| \leq 2\phi_i''(t)^{3/2}, \quad |\phi_i''''(t)| \leq 6\phi_i''(t)^2.$$

Moreover $\phi_i'((l_i + u_i)/2) = 0$ and $\phi_i''(t) \geq (u_i - l_i)^{-2}$.

The fourth derivative bound is what makes the second order expansion of the weights in Lemma 5.6 valid, and is the reason we cannot work with an arbitrary 1-self-concordant barrier. One consequence is used repeatedly: if $|\phi_i''(t)^{1/2}(t' - t)| \leq \beta \leq 1/100$ then, since the two distances to the bounds change by factors in $[1 - \beta, 1 + \beta]$,

$$\phi_i''(t') \approx_{3\beta} \phi_i''(t), \quad |\phi_i'(t') - \phi_i'(t)| \leq 2\beta \phi_i''(t)^{1/2}. \quad (10)$$

The central path weights are defined as follows.

Definition 4.2. *Fix the Lewis exponent and the regularizer*

$$q \triangleq 1 - \alpha, \quad \alpha \triangleq \frac{1}{10 \log(10n/d)}, \quad v \in \mathbb{R}_{>0}^n \text{ with } \frac{d}{n} \leq v_i \leq 1 \text{ and } \|v\|_1 \leq 4d,$$

and define the central path weights at a point x with $l < x < u$ by

$$\tau(x) \triangleq \tau_q \left(\Phi''(x)^{-1/2} A, v \right), \quad T(x) \triangleq \text{diag}(\tau(x)).$$

Since $\log(10n/d) \geq \log 10$ we have $\alpha \leq 1/23$, so $q \in [1/2, 2)$ as required by Lemma 4.2, and the weights are bounded on both sides: $d/n \leq v_i \leq \tau_i(x) \leq 1 + v_i \leq 2$ for every i , and $\|\tau(x)\|_1 = d + \|v\|_1 \leq 5d$.

The weights are stable under rescaling; the following is Lemma 4.22 of [vdBLL⁺21].

Lemma 4.4. *For $q \in (0, 4)$ and positive diagonal $C \approx_\epsilon \tilde{C}$ we have $\tau_q(CA, v) \approx_{4\epsilon} \tau_q(\tilde{C}A, v)$.*

Throughout, C denotes a sufficiently large universal constant and the algorithm's parameters are

$$\begin{aligned}
\epsilon &\triangleq \frac{\alpha}{C}, \\
\lambda &\triangleq \frac{C \log(Cn/\epsilon^2)}{\epsilon}, \\
\gamma &\triangleq \frac{\epsilon}{C\lambda}, \\
C_{\text{norm}} &\triangleq \frac{C}{1-q} = \frac{C}{\alpha}, \\
\theta &\triangleq \frac{\epsilon\gamma}{C_{\text{norm}}d^{1/2}}, \\
C_{\text{start}} &\triangleq 10C,
\end{aligned} \tag{11}$$

so that $\theta^{-1} = \tilde{O}(d^{1/2})$, as used in Figure 5. The explicit requirements on C that this section uses are the following. From $\alpha \leq 1/23$, $\epsilon \leq 1/80$ once $C \geq 4$, as Definition 4.3 requires. Next $e^{\lambda\epsilon} = (Cn/\epsilon^2)^C \geq n^{10}$ once $C \geq 10$, which is what converts a polynomial bound on the potential into ϵ -centrality in Lemma 5.10. And $\lambda\epsilon \leq 3C \log n$ for n larger than a constant depending on C , which with $C_{\text{start}} = 10C$ keeps the initial potential below n^2 in Lemma 5.10. Both $\gamma^{-1} = C^4\alpha^{-2} \log(Cn/\epsilon^2) = O(\log^3 n)$ and $C_{\text{norm}} = O(\log n)$ are $\tilde{O}(1)$. The remaining constraints on C are the hidden constants of the imported Lemmas 5.6 to 5.8, which [vdBLL⁺21] asserts for C sufficiently large without a value.

The algorithm uses, for each of the parameters of Equation (11) and for α , a number with $40 \log \rho$ bits within a factor $1 \pm \rho^{-30}$ of the value displayed; each requirement above is an inequality with room for such a factor, and we do not distinguish the two in the notation.

The magnitude bound ρ is, throughout this section, a common bound for the magnitudes of $A^\top A$ and of every $A^\top W A$ the data structure holds, and also for $\phi_i''(x_i)^{\pm 1}$, $\mu^{\pm 1}$, $\|A\|$, $\|c\|_2$, $\|l\|_\infty$ and $\|u\|_\infty$ along the path; we take $\rho \geq n$ as well, which together with $\rho = 2^{O(B)}$ is the standing assumption $\log n = O(B)$ on the bit length of the input. Lemma 6.1 gives $\rho = 2^{O(B)}$, and enlarging ρ affects only the $\log \rho$ factors of the costs. All arithmetic inside a step is carried out on numbers with $40 \log \rho = O(B)$ bits, every intermediate result being truncated by $\text{Trunc}(\cdot, 40 \log \rho)$, except that a right hand side handed to SOLVE is truncated to the $20 \log \rho$ bits of Definition 3.1. Every quantity a step forms, other than the calls to SOLVE, is a formula in the sense of Lemma 2.2 whose inputs are stored numbers or outputs of SOLVE, whose inputs and intermediate results have magnitude at most ρ , and whose divisors and radicands are at least ρ^{-1} , by Lemma 6.1 with ρ enlarged by a constant power; so $M = \rho$, and the requirements $M \leq 2^{\ell-1}$ and on the bits before the decimal point are met. The formulas are of two kinds: an entrywise formula with at most 30 operations and multiplicative depth at most 6, and a sum over the rows of at most $3n$ products of two stored numbers, of multiplicative depth 1 and with at most $6n \leq 6\rho$ operations, such as an entry of $A^\top v$ or of Ah ; an entrywise formula may take sums of the second kind as inputs. Lemma 2.2 bounds the error of a sum of the second kind by $6\rho \cdot 2\rho^2 \cdot \rho^{-40} \leq \rho^{-36}$, and then that of an entrywise formula with at most 30 such inputs by

$$(30\rho^{-40} + 30\rho^{-36}) (2\rho^2)^6 \leq \rho^{-20} \tag{12}$$

for $\rho \geq 4000$, which is below the $1/(2\rho)$ that the lemma requires. A sum of the second kind whose factors are themselves entrywise formulas, such as an entry of $A^\top \Phi''(x)^{-1/2}g$, has inputs off by at most ρ^{-20} each, and Lemma 2.2 bounds its error by

$$(6\rho \cdot \rho^{-40} + 6n\rho^{-20}) 2\rho^2 \leq \rho^{-16}. \tag{13}$$

Every truncation error quoted in Sections 5 and 6 is an instance of Equation (12) or (13), or a direct application of Lemma 2.2 with $M = \rho$, and the proof states the operation count and the depth of the formula concerned.

4.3 Centrality and the potential

An ϵ -centered point is defined as follows.

Definition 4.3. A triple (x, s, μ) with $l < x < u$ and $\mu > 0$ is ϵ -centered for $\epsilon \in (0, 1/80]$ if

1. (Approximate centrality) $\| \frac{s + \mu T(x) \phi'(x)}{\mu T(x) \Phi''(x)^{1/2}} \|_\infty \leq \epsilon$,
2. (Dual feasibility) $s = Az + c$ for some $z \in \mathbb{R}^d$,
3. (Approximate primal feasibility) $\|A^\top x - b\|_{(A^\top (T(x) \Phi''(x))^{-1} A)^{-1}} \leq \frac{\epsilon \gamma}{C_{\text{norm}}}$.

The algorithm stores the multiplier z of Item 4.3(2) rather than s ; wherever s appears below it stands for $Az + c$, evaluated in $O(\text{nnz}(A))$ operations when needed. This makes dual feasibility hold by construction, truncation included.

The centrality potential is the following.

Definition 4.4. For $\psi(t) \triangleq \cosh(\lambda t)$ with λ as in Equation (11), and the centrality vector

$$y_i(x, s, \mu) \triangleq \frac{s_i + \mu \tau_i(x) \phi'_i(x_i)}{\mu \tau_i(x) \phi''_i(x_i)^{1/2}}, \quad \Psi(x, s, \mu) \triangleq \sum_{i=1}^n \psi(y_i(x, s, \mu)).$$

Exact centrality is $s + \mu T(x) \phi'(x) = 0$; the denominators rescale each coordinate by the local Hessian. Since $\cosh(\lambda y_i) \leq \Psi$, a bound $\Psi \leq n^{10}$ gives $|y_i| \leq \log(2n^{10})/\lambda \leq \epsilon$ for every i by the choice $e^{\lambda \epsilon} \geq n^{10}$ in Equation (11), which is Item 4.3(1).

The $\tau + \infty$ norm is the following.

Definition 4.5. For $g \in \mathbb{R}^n$ and a weight vector τ ,

$$\|g\|_{\tau+\infty} \triangleq \|g\|_\infty + C_{\text{norm}} \|g\|_\tau, \quad \|g\|_\tau \triangleq \left(\sum_i \tau_i g_i^2 \right)^{1/2},$$

with dual norm $\|g\|_{\tau+\infty}^*$ and $g^{b(\tau)} \triangleq \arg \max_{\|h\|_{\tau+\infty}=1} h^\top g$.

The step direction is measured in $\|\cdot\|_{\tau+\infty}$ because the ℓ_∞ part keeps every coordinate inside the region where the barrier Hessian is stable, while the τ part is what controls the movement of the Lewis weights, Lemma 5.7, and hence the work of the data structure. Two elementary comparisons between the norms are used below: for every g ,

$$\|g\|_\tau \leq \|\tau\|_1^{1/2} \|g\|_\infty \leq (5d)^{1/2} \|g\|_\infty, \quad \|g\|_{\tau+\infty} \leq \left(1 + C_{\text{norm}} (5d)^{1/2}\right) \|g\|_\infty, \quad (14)$$

and if $\tau \approx_\delta \tau'$ then $\|g\|_{\tau+\infty} \leq e^{\delta/2} \|g\|_{\tau'+\infty}$.

The algorithm stores x exactly, as a vector of numbers with $40 \log \rho$ bits, but never $\tau(x)$ exactly; it stores the fixed point approximation of Lemma 4.2, run against the SOLVE call of Definition 3.1.

The approximations maintained across steps are the following.

Definition 4.6. *At the start of every call to STEP we hold a point $\bar{x}$ and a vector $\tilde{\tau}$ with*

$$\left\| \Phi''(x)^{1/2}(\bar{x} - x) \right\|_{\infty} \leq \epsilon, \quad \tilde{\tau} \approx_{\epsilon} \tau(\bar{x}),$$

and write $\tilde{T} \triangleq \text{diag}(\tilde{\tau})$.

This is Invariant 4.10 of [vdBLL⁺21]. There the proxy $\bar{x}$ is needed because the iterate lives implicitly and only $\tilde{O}(d)$ of its coordinates may be read per step; here every coordinate of x is explicit, so we take $\bar{x} \triangleq x$ throughout, the first condition holds with slack zero, and $\bar{x}$ is kept in the statements only so that they read as in the source. The estimate $\tilde{\tau}$ is produced by Lemma 4.2 at accuracy $\gamma/100 \leq \epsilon$ (step 2 of Figure 5); the finer accuracy is what step 4 needs, see Lemma 5.2.

5 Effect of One Step

Symbol	Meaning
$x^{(0)}$	initial point given with the LP data, $l+2^{-B} < x^{(0)} < u-2^{-B}$; Theorem 1, Lemma 6.4
$\bar{x}$	proxy for x inside a step, Definition 4.6; equal to x here, Figure 5
x, sl, su	primal iterate and dual slacks, $sl - su - c$ in the range of A ; Definition 4.3, Lemma 6.4
z, s	stored dual multiplier and the slack it defines, $s = Az + c$; Item 4.3(2), Figure 5
θ	path step size, $\theta^{-1} = \tilde{O}(d^{1/2})$; Figure 5, Lemma 5.8, Lemma 5.10
$\bar{y}, g$	centrality proxy of step 4 of Figure 5, Lemma 5.2, and the admissible direction, Definition 5.1
$\hat{g}, \pi, s_k, h^{(k)}, \nu$	inside DIRECTION, Figure 6: computed gradient, its normalization, scale, candidate, certified norm
$\tilde{q}_k, \tilde{\nu}_k$	inside DIRECTION: values computed on numbers with $40 \log \rho$ bits; proof of Lemma 5.3
$\bar{A}$	rescaled matrix $\tilde{T}^{-1/2} \Phi''(\bar{x})^{-1/2} A$, so $\bar{A}^{\top} \bar{A} = A^{\top} W A$; Figure 5, Lemma 5.4
$\delta_1, \delta_2, \delta_r$	the two solve residuals of a step and their sum; Figure 5, Lemma 5.4
R, C_{valid}	step-sampling diagonal of [vdBLL ⁺ 21] and its parameter; $R = I$ here, Section 5
$\hat{x}, \hat{z}, \hat{s}, \hat{\mu}$	the iterate, multiplier, slack and path parameter a step returns; Figure 5, Lemma 5.9
$\hat{x}^*, \hat{z}^*, h_j^*$	the exact step of Equation (24) and its exact solves; Lemmas 5.4, 5.6, 5.8
β	size of a perturbation in Lemma 5.5; proof of Lemma 5.9
D_c	$\text{diag}(c) = \Phi''(x)^{-1/2}$, the barrier scaling of Lemma 5.6; Lemma 5.1, Lemma 5.7
$\tau^{(k)}, T^{(k)}$	the path Lewis weights $\tau(x^{(k)})$ at round k and their diagonal; Lemma 5.11, Lemma 6.2

Figure 4: Variables local to the interior point method, its step (Section 5) and initialization (Section 6).

This section states the top level guarantees of one step of the interior point method, in the setting of Section 4, and chains them into the path following bound. Two simplifications are specific to this version. A step reads and writes all n coordinates explicitly, which the budget $\text{nnz}(A) \geq n$ per step allows, so the step is never subsampled: the random diagonal R of [vdBLL⁺21] is the identity here, and the imported per-step statements, which are in expectation over R there, are deterministic. And the only errors beyond the analysis there are the ρ^{-10} accuracy of SOLVE and fixed point truncation; both are handled by one perturbation lemma, Lemma 5.5, which compares the computed step with the exact step on the same inputs. The section ends with the movement bounds on the path weights that Section 6 converts into the leverage stability of Definition 3.2.

A step touches A only through the calls of Definition 3.1 and through $O(1)$ matrix vector products per solver call, so it costs

$$\tilde{O}(\gamma^{-2}(\text{nnz}(A) + d^2) \log \rho) + (\text{cost of one UPDATE})$$

bit operations, the d^2 being the applications of the maintained inverse inside the SOLVE calls, including those the Lewis iteration makes, and $\gamma^{-2} = \tilde{O}(1)$ the number of Johnson–Lindenstrauss directions that iteration uses at accuracy $\gamma/100$; the coordinatewise work of the step, including the computation of the direction in Lemma 5.3, is $\tilde{O}(n \log \rho) \leq \tilde{O}(\text{nnz}(A) \log \rho)$ and is absorbed. No matrix produced inside the data structure is visible here.

No subsampling of the step. Algorithm 1 of [vdBLL⁺21] multiplies the residual direction δ_r of Figure 5 by a random diagonal R drawn from a C_{valid} -valid distribution, [vdBLL⁺21, Definition 4.13], because there a step may only touch $\tilde{O}(d)$ coordinates. Here a step already pays $\text{nnz}(A) \geq n$, so we take $R \triangleq I$. The identity is C_{valid} -valid for every $C_{\text{valid}} \geq C_{\text{norm}}$: its expectation is I ; the variances of its entries and the deviation $\|R\delta_r - \delta_r\|_\infty$ are zero; $\mathbb{E}[R_{ii}^2] = 1 \leq 2\tau(\bar{A})_i^{-1}$ and $\mathbb{E}[R_{ii}R_{jj}] = 1 \leq 2$ since leverage scores are at most one; and $\bar{A}^\top R \bar{A} = \bar{A}^\top \bar{A}$ exactly. Consequently every statement of [vdBLL⁺21] that is in expectation over R , or holds with probability $1 - d^{-10}$ over R , holds deterministically here, and the imported lemmas below are stated in that form. The only randomness left in a step is inside SOLVE and inside the Lewis estimate.

5.1 The step

STEP(x, z, μ), with $s \triangleq Az + c$:

1. $\hat{\mu} \triangleq (1 - \theta)\mu$.

2. Set $\bar{x} \triangleq x$ and

$$\tilde{\tau} \triangleq \text{LEWIS_ESTIMATE}\left(\Phi''(x)^{-1/2}A, q, v, \gamma/100, \text{SOLVE}\right),$$

the routine of Lemma 4.2 run with the data structure's SOLVE as its solver and the previous round's $\tilde{\tau}$ as its starting estimate, giving $\tilde{\tau} \approx_{\gamma/100} \tau(x)$.

3. Call

$$\text{UPDATE}(\hat{w}), \quad \hat{w}_i \triangleq \text{Trunc}\left((\tilde{\tau}_i \phi_i''(x_i))^{-1}, 40 \log \rho\right) \text{ for every } i \in [n],$$

so that the data structure now holds $\widehat{W} \approx_{\rho^{-18}} W \triangleq \tilde{T}^{-1} \Phi''(x)^{-1}$ by Equation (12), as $w_i \geq \rho^{-1}$ by Lemma 6.1.

4. Let

$$\bar{y}_i \triangleq \frac{s_i + \mu \tilde{\tau}_i \phi_i'(x_i)}{\mu \tilde{\tau}_i \phi_i''(x_i)^{1/2}}, \quad g \triangleq \text{DIRECTION}(\bar{y}, \tilde{\tau}),$$

the routine of Figure 6, which returns a fixed point approximation of $-\gamma \nabla \Psi(\bar{y})^{b(\tilde{\tau})}$ that is admissible in the sense of Definition 5.1; in particular $\|g\|_{\tilde{\tau}+\infty} \leq \gamma$.

5. With $\bar{A} \triangleq \tilde{T}^{-1/2} \Phi''(x)^{-1/2} A$, so that $\bar{A}^\top \bar{A} = A^\top W A$, solve the two Newton systems by

$$h_1 \triangleq \text{SOLVE}\left(\text{Trunc}\left(A^\top \Phi''(x)^{-1/2} g, 20 \log \rho\right)\right), \quad h_2 \triangleq \text{SOLVE}\left(\text{Trunc}\left(A^\top x - b, 20 \log \rho\right)\right),$$

and set

$$\delta_1 \triangleq \tilde{T}^{-1} \Phi''(x)^{-1/2} A h_1, \quad \delta_2 \triangleq \tilde{T}^{-1} \Phi''(x)^{-1/2} A h_2, \quad \delta_r \triangleq \delta_1 + \delta_2.$$

6. Set

$$\hat{x} \triangleq \text{Trunc}\left(x + \Phi''(x)^{-1/2}(g - \delta_r), 40 \log \rho\right), \quad \hat{z} \triangleq \text{Trunc}\left(z + \mu h_1, 40 \log \rho\right),$$

so that $\hat{s} \triangleq A\hat{z} + c$ equals $s + \mu \tilde{T} \Phi''(x)^{1/2} \delta_1$ up to the solve error and the truncation.

7. **return** $(\hat{x}, \hat{z})$.

Figure 5: One step of the path-following algorithm for the primal. Given an ϵ -centered pair $(x, s = Az + c)$ at path parameter μ , STEP decreases the path parameter by a factor $1 - \theta$, refreshes the Lewis-weight estimate and the maintained inverse, and takes one Newton step on the penalized objective; Lemma 5.9 shows the output is ϵ -centered at the new parameter with contracted potential.

Three things about the call order in Figure 5 are worth recording. First, the Lewis estimate of step 2 is made before UPDATE in step 3, so the solver it is handed is the one belonging to the previous round's weights; that this solver satisfies Item 4.2(6) is checked in Lemma 5.1 below. Second, SOLVE is accurate to ρ^{-10} in the norm of the held matrix $A^\top \widehat{W} A \approx_{\rho^{-18}} A^\top W A$, hence to $3\rho^{-10}$ in the $A^\top W A$ norm, and the two places its output enters the step, the direction δ_r and the multiplier update μh_1 , are both linear in it; Lemma 5.4 converts the guarantee into coordinatewise bounds on these two quantities, and Lemma 5.5 shows that perturbations of that size do not disturb centrality or feasibility. Third, step 3 hands UPDATE the whole weight vector rather than the coordinates that moved, which costs $O(n) \leq O(\text{nnz}(A))$ per round and leaves the data structure to decide which of its rows to refresh, which is the decision Corollary 3.4 counts.

The following lemma shows that the maintained form serves the Lewis estimate.

Lemma 5.1. *Let $\tilde{\tau}'$ be the estimate of the previous round, x' the previous iterate, and $W' = (\tilde{T}')^{-1} \Phi''(x')^{-1}$, so that the data structure holds $\widehat{W}' \approx_{\rho^{-18}} W'$ when step 2 of Figure 5 runs at the current iterate x . If $\tilde{\tau}' \approx_{\gamma/100} \tau(x')$ and $\|\Phi''(x')^{1/2}(x - x')\|_\infty \leq 1/100$, then $\tilde{\tau}'$ and $M \triangleq A^\top \widehat{W}' A$ satisfy Items 4.2(5) and 4.2(6) for the matrix $D_c A$, $D_c = \Phi''(x)^{-1/2}$, whose weights step 2 estimates.*

Proof. Item 4.2(5) asks $\tilde{\tau}' \approx_{0.5} \tau(x)$. By Equation (10) $\Phi''(x)^{-1/2} \approx_{3/200} \Phi''(x')^{-1/2}$, so Lemma 4.4 gives $\tau(x) \approx_{3/50} \tau(x') \approx_{\gamma/100} \tilde{\tau}'$. Item 4.2(6) asks $M \approx_{0.5} (D_c A)^\top \text{diag}(\tilde{\tau}')^{1-2/q} D_c A$. At $q = 1 - \alpha$,

$$1 - \frac{2}{q} = -\frac{1 + \alpha}{1 - \alpha} = -1 - \frac{2\alpha}{1 - \alpha},$$

so

$$\text{diag}(\tilde{\tau}')^{1-2/q} D_c^2 = \text{diag}(\tilde{\tau}')^{-2\alpha/(1-\alpha)} \cdot \frac{\Phi''(x')}{\Phi''(x)} \cdot W', \quad (15)$$

the quotient of diagonal matrices being entrywise. The second factor is within $e^{\pm 3/100}$ of the identity by Equation (10). For the first, $d/n \leq \tau_i(x') \leq 5$ gives $e^{-1/100} d/n \leq \tilde{\tau}'_i \leq 6$, so $|\log \tilde{\tau}'_i| \leq \log(10n/d)$, and with $\alpha = 1/(10 \log(10n/d)) \leq 1/23$,

$$\frac{2\alpha}{1 - \alpha} |\log \tilde{\tau}'_i| \leq \frac{2}{10(1 - 1/23)} = \frac{23}{110} < \frac{1}{4}.$$

Hence $A^\top W' A$ and the form of Item 4.2(6) are within $e^{\pm(1/4+3/100)}$, and $M \approx_{\rho^{-18}} A^\top W' A$ adds ρ^{-18} to the exponent, still inside the 0.5 of Item 4.2(6). $\square$

Along the path the hypotheses of Lemma 5.1 are supplied by the previous round: the accuracy of $\tilde{\tau}'$ by its step 2, and the movement bound by Item 5.6(2) together with Lemma 5.9, as $\gamma \leq 1/100$.

The following lemma controls the computed centrality vector.

Lemma 5.2. *If (x, s, μ) is ϵ -centered and $\tilde{\tau} \approx_{\gamma/100} \tau(x)$, then the vector $\bar{y}$ of step 4 satisfies $\|\bar{y} - y(x, s, \mu)\|_\infty \leq \gamma/20$, truncation included.*

Proof. Write $a_i \triangleq s_i/(\mu \tau_i(x) \phi_i''(x_i)^{1/2})$ and $b_i \triangleq \phi_i'(x_i)/\phi_i''(x_i)^{1/2}$, so $y_i = a_i + b_i$, $|b_i| \leq 1$ by Lemma 4.3, and $|a_i| \leq |y_i| + 1 \leq 1 + \epsilon$ by centrality. Only the first term depends on τ , and

$$|\bar{y}_i - y_i| = |a_i| \left| \frac{\tau_i(x)}{\tilde{\tau}_i} - 1 \right| \leq (1 + \epsilon) \left(e^{\gamma/100} - 1 \right) \leq \frac{1.03\gamma}{100}.$$

Each entry of $\bar{y}$ is a formula in the stored numbers $x_i, l_i, u_i, \mu, \tilde{\tau}_i, c_i$ and the sum $(Az)_i$, with at most 20 operations, namely the two distances to the bounds, their reciprocals and squared reciprocals for ϕ_i' and ϕ_i'' , one square root, four products and one quotient, of multiplicative depth 5; so Equation (12) bounds the truncation error by $\rho^{-20} \leq \gamma/100$ per entry. $\square$

5.2 Direction finding

The exact direction $-\gamma \nabla \Psi(\bar{y})^{b(\tilde{\tau})}$ of [vdBLL⁺21] is the maximizer of a linear functional over the unit ball of $\|\cdot\|_{\tilde{\tau}+\infty}$; it is not a rational function of $\bar{y}$ and $\tilde{\tau}$, and in fixed point only a near maximizer can be formed. The progress analysis uses the direction through the three properties below only, and tolerates a constant fraction of the margin $\alpha/4$ that the imported proof has, so a near maximizer suffices.

An admissible direction is defined as follows.

Definition 5.1. *Given $\bar{y} \in \mathbb{R}^n$ and a positive $\tilde{\tau} \in \mathbb{R}^n$, a vector $g \in \mathbb{R}^n$ is an admissible direction at $(\bar{y}, \tilde{\tau})$ if*

1. $\|g\|_{\tilde{\tau}+\infty} \leq \gamma$;
2. $g_i \nabla \Psi(\bar{y})_i \leq 0$ for every $i \in [n]$;
3. $-\nabla \Psi(\bar{y})^\top g \geq (1 - \alpha/16) \gamma \|\nabla \Psi(\bar{y})\|_{\tilde{\tau}+\infty}^*$.

The exact direction is admissible with 0 in place of $\alpha/16$: Item 5.1(1) holds with equality by Definition 4.5, Item 5.1(3) is the definition of the dual norm, and Item 5.1(2) holds for the maximizer because flipping the sign of a coordinate of h that disagrees in sign with $\nabla \Psi(\bar{y})_i$ leaves $\|h\|_{\tilde{\tau}+\infty}$ unchanged and does not decrease $h^\top \nabla \Psi(\bar{y})$.

$\text{DIRECTION}(\bar{y}, \tilde{\tau})$, all arithmetic on numbers with $40 \log \rho$ bits, every intermediate result truncated by $\text{Trunc}(\cdot, 40 \log \rho)$; a tilde marks a quantity computed this way, so $\tilde{q}_k$ is the computed value of $\pi^\top h^{(k)}$ and $\tilde{\nu}_k$ that of $\|h^{(k)}\|_{\tilde{\tau}+\infty}$.

1. If $\bar{y} = 0$, **return** 0. Otherwise let j be the integer with $2^{-j-1} < \|\bar{y}\|_\infty \leq 2^{-j}$, and for every i compute a number $\hat{g}_i$ with $40 \log \rho$ bits such that $|\hat{g}_i - 2^j \lambda \sinh(\lambda \bar{y}_i)| \leq \rho^{-20}$, as $\hat{g}_i \triangleq \text{Trunc}(\text{Trunc}(\lambda z_i) \tilde{S}_i, 40 \log \rho)$ with $z_i \triangleq 2^j \bar{y}_i$ and $\tilde{S}_i$ the partial sum of the Taylor series of $\sinh(t)/t$ at $t = \text{Trunc}(\lambda \bar{y}_i, 40 \log \rho)$ with $\lceil 6\lambda\epsilon + 40 \log \rho \rceil$ terms.

2. Normalize and threshold: $\pi_i \triangleq \text{Trunc}(|\hat{g}_i|/\|\hat{g}\|_\infty, 40 \log \rho)$, then $\pi_i \triangleq 0$ wherever $\pi_i < \rho^{-10}$.

3. Let η be a number with $40 \log \rho$ bits in $[\alpha/128, \alpha/64]$ and $K \triangleq \lceil 512\alpha^{-1} \log(12\rho^{12}) \rceil$. Set $s_0 \triangleq \text{Trunc}(\rho^{-2}, 40 \log \rho)$, $s_{k+1} \triangleq \text{Trunc}((1+\eta)s_k, 40 \log \rho)$ for $k < K$, and for every $k \leq K$ form

$$h_i^{(k)} \triangleq \min\left(1, \frac{s_k \pi_i}{\tilde{\tau}_i}\right), \quad \nu_k \triangleq \tilde{\nu}_k + \rho^{-17},$$

the computed norm shifted up so that $\nu_k \geq \|h^{(k)}\|_{\tilde{\tau}+\infty}$.

4. Let k^* maximize $\tilde{q}_k/\nu_k$ over k , comparing two fractions by cross multiplication, and write $h \triangleq h^{(k^*)}$, $\nu \triangleq \nu_{k^*}$.

5. **return** g with $g_i \triangleq -\text{sgn}(\hat{g}_i) \text{Trunc}(\gamma h_i/\nu, 40 \log \rho)$.

Figure 6: Computing the step direction in fixed point. Up to scale, the maximizer of $\pi^\top h$ over the unit ball of $\|\cdot\|_{\tilde{\tau}+\infty}$ has the form $h_i = \min(1, s\pi_i/\tilde{\tau}_i)$ for a single scalar s , and the ratio $\pi^\top h/\|h\|_{\tilde{\tau}+\infty}$ is scale free, so no constraint has to be solved: the geometric sequence s_k replaces the exact s by one within a factor $1 + \eta$, and the final normalization by ν makes Item 5.1(1) hold exactly.

The following lemma shows that DIRECTION computes an admissible direction.

Lemma 5.3. *Let (x, s, μ) be ϵ -centered with $\Psi(x, s, \mu) \leq n^{10}$, let $\tilde{\tau} \approx_{\gamma/100} \tau(x)$, and let $\bar{y}$ be the vector of step 4 of Figure 5. Then $\text{DIRECTION}(\bar{y}, \tilde{\tau})$ returns an admissible direction at $(\bar{y}, \tilde{\tau})$, using $\tilde{O}(n \log \rho)$ bit operations.*

Proof. Write $p \triangleq \nabla \Psi(\bar{y})$, so $p_i = \lambda \sinh(\lambda \bar{y}_i)$, let $U \triangleq \{h : \|h\|_{\tilde{\tau}+\infty} \leq 1\}$, and abbreviate

$$\|\cdot\|^* \triangleq \|\cdot\|_{\tilde{\tau}+\infty}^* = \max_{h \in U} \langle \cdot, h \rangle.$$

As noted after Definition 4.4, $\Psi \leq n^{10}$ gives $|y_i| \leq \epsilon$, so $|\bar{y}_i| \leq \epsilon + \gamma/20 \leq 2\epsilon$ by Lemma 5.2 and, by Equation (11),

$$|p_i| \leq \lambda e^{2\lambda\epsilon} = \lambda \left(\frac{Cn}{\epsilon^2} \right)^{2C} = n^{O(1)};$$

and $d/(2n) \leq \tilde{\tau}_i \leq 6$ as in the proof of Lemma 5.1. So every quantity below has magnitude $\rho^{O(1)}$ and occupies $O(\log \rho)$ bits. Two facts about the dual norm are used repeatedly. Since U is symmetric under sign changes of coordinates,

$$\|p\|_{\tilde{\tau}+\infty}^* = \max_{h \in U} \sum_i |p_i| |h_i|, \tag{16}$$

so the dual norm is monotone in $|p|$ entrywise, and $\|p - p'\|_{\tilde{\tau}+\infty}^* \leq \|p - p'\|_1$ because $\|h\|_\infty \leq 1$ on U . And testing $h = e_i / \|e_i\|_{\tilde{\tau}+\infty}$ for the i maximizing $|p_i|$, where $\|e_i\|_{\tilde{\tau}+\infty} = 1 + C_{\text{norm}} \tilde{\tau}_i^{1/2} \leq 4C_{\text{norm}}$,

$$\|p\|_{\tilde{\tau}+\infty}^* \geq \frac{\|p\|_\infty}{4C_{\text{norm}}} \geq \frac{\|p\|_1}{4nC_{\text{norm}}}. \quad (17)$$

The gradient, step 1. If $\bar{y} = 0$ then $p = 0$, and $g = 0$ is admissible. Otherwise Items 5.1(2) and 5.1(3) are unchanged when p is scaled by a positive constant, so it suffices to prove them for $\hat{p} \triangleq 2^j p$. Write $z_i \triangleq 2^j \bar{y}_i$, which is an exact shift of a stored number since $j \geq 0$, with $|z_i| \leq 1$ and $\|z\|_\infty > 1/2$, and $S(t) \triangleq \sinh(t)/t = \sum_{k \geq 0} t^{2k}/(2k+1)!$, so that $\hat{p}_i = \lambda z_i S(t_i)$ for $t_i \triangleq \lambda \bar{y}_i$, and $1 \leq S(t) \leq e^{|t|}$. Here $|t_i| \leq 2\lambda\epsilon$ and $e^{2\lambda\epsilon} = n^{O(1)} \leq \rho$ once ρ is enlarged by a constant power as agreed after Equation (11), so $\|\hat{p}\|_\infty \geq \lambda/2$ and every quantity below is at most ρ in magnitude. The computed $\tilde{t}_i = \text{Trunc}(\lambda \bar{y}_i, 40 \log \rho)$ is within ρ^{-40} of t_i , and $c_i \triangleq \text{Trunc}(\tilde{t}_i^2, 40 \log \rho)$ within $3\rho^{-39}$ of t_i^2 . The partial sum $\tilde{S}_i$ is formed as $b_0 \triangleq 1$, $b_k \triangleq \text{Trunc}(\text{Trunc}(b_{k-1} c_i)/(2k(2k+1)), 40 \log \rho)$ and $\tilde{S}_i \triangleq \text{Trunc}(\sum_{k \leq K} b_k)$ with $K \triangleq \lceil 6\lambda\epsilon + 40 \log \rho \rceil$, the sum accumulated left to right. This recurrence has multiplicative depth K , for which the bound of Lemma 2.2 is too coarse, so we bound its error directly. Let $\beta_k \triangleq c_i^k/(2k+1)!$ be the exact terms at c_i and $e_k \triangleq |b_k - \beta_k|$; then $e_0 = 0$ and $e_k \leq r_k e_{k-1} + 2\rho^{-40}$ with $r_k \triangleq c_i/(2k(2k+1))$, since the product is off by $c_i e_{k-1} + \rho^{-40}$ and the division by an integer at least 2 does not increase this before its own truncation. Unrolling,

$$e_k \leq 2\rho^{-40} \sum_{m=0}^k \prod_{i'=m+1}^k r_{i'} = 2\rho^{-40} \sum_{m=0}^k \frac{\beta_k}{\beta_m} \leq 2\rho^{-40} (k+1) \max_{m \leq k} \frac{c_i^{k-m}}{(2(k-m))!} \leq 2e\rho^{-39} (k+1),$$

using $(2k+1)!/(2m+1)! \geq (2(k-m))!$ and $\sum_{j' \geq 0} c^{j'}/(2j')! = \cosh(c^{1/2}) \leq e^{|t_i|+1} \leq e\rho$, as $c_i \leq t_i^2 + 1$. Hence $|\tilde{S}_i - \sum_{k \leq K} \beta_k| \leq \sum_{k \leq K} e_k + K\rho^{-40} \leq 2e(K+1)^2 \rho^{-39} \leq \rho^{-36}$, as $K+1 \leq \rho$. Next, $\sum_{k \leq K} \beta_k$ differs from $\sum_{k \leq K} t_i^{2k}/(2k+1)!$ by at most $|c_i - t_i^2| \sum_{k \leq K} k t_i^{2k-1}/(2k+1)! \leq 3\rho^{-39} \cosh(u^{1/2}) \leq 3e\rho^{-38}$ for some u between c_i and t_i^2 , and the tail of the series of $S(t_i)$ after K terms is at most ρ^{-42} : for $k \geq K/2 \geq 3\lambda\epsilon$ the ratio of consecutive terms is $t_i^2/(2k(2k+1)) \leq 1/9$, so the term at K is at most $3^{-K} \cosh(t_i) \leq \rho^{-43.9} \cdot \rho$ and the tail at most $9/8$ of it. Altogether $|\tilde{S}_i - S(t_i)| \leq 2\rho^{-36}$, and $\hat{g}_i = \text{Trunc}(\text{Trunc}(\lambda z_i) \tilde{S}_i)$ is within

$$\rho^{-40} (\rho + 1) + \rho \cdot 2\rho^{-36} + \rho^{-40} \leq \rho^{-20}$$

of $\lambda z_i S(t_i) = \hat{p}_i$, as step 1 requires. In particular $\|\hat{g}\|_\infty \geq \lambda/2 - \rho^{-20} \geq 1/4$, so $\hat{g} \neq 0$. We work with $\hat{g}$ in place of $\hat{p}$ and charge the difference at the end: wherever $|\hat{g}_i| \geq \rho^{-10} \|\hat{g}\|_\infty \geq \rho^{-10}/4 > \rho^{-20}$, which is the case wherever step 2 leaves $\pi_i > 0$ since the truncation there is toward zero, the sign of $\hat{g}_i$ is that of $\hat{p}_i$; and since $\|\hat{g} - \hat{p}\|^* \leq \|\hat{g} - \hat{p}\|_1 \leq n\rho^{-20}$ while $\|\hat{p}\|^* \geq \|\hat{p}\|_\infty/(4C_{\text{norm}}) \geq 1/(8C_{\text{norm}})$ by Equation (17),

$$\|\hat{g}\|^* \geq (1 - 8nC_{\text{norm}}\rho^{-20}) \|\hat{p}\|^* \geq (1 - \rho^{-17}) \|\hat{p}\|^*,$$

using $\rho \geq n \geq 8C_{\text{norm}}$.

Normalization and thresholding, step 2. Let $\bar{\pi} \triangleq \|\hat{g}\|/\|\hat{g}\|_\infty \in [0, 1]^n$, a vector with a unit entry, so $\|\bar{\pi}\|^* \geq 1/(4C_{\text{norm}})$ by Equation (17). Each π_i differs from $\bar{\pi}_i$ by at most ρ^{-20} from the truncation to $40 \log \rho$ bits, one operation on stored numbers in Equation (12), and by at most ρ^{-10} from the thresholding, so, using $\rho \geq n \geq C_{\text{norm}}$,

$$\|\pi - \bar{\pi}\|_1 \leq 2n\rho^{-10} \leq 2\rho^{-9} \leq 8C_{\text{norm}}\rho^{-9} \|\bar{\pi}\|^* \leq \rho^{-8} \|\bar{\pi}\|^*.$$

Hence, for every $h \in U$, $\bar{\pi}^\top h \geq \pi^\top h - \rho^{-8} \|\bar{\pi}\|^*$, and $\|\pi\|^* \geq (1 - \rho^{-8}) \|\bar{\pi}\|^*$: a vector $h \in U$ with $\pi^\top h \geq (1 - \eta') \|\pi\|^*$ has $\bar{\pi}^\top h \geq (1 - \eta' - 2\rho^{-8}) \|\bar{\pi}\|^*$. The nonzero entries of π lie in $[\rho^{-10}, 1]$, and $\pi_i > 0$ implies $\hat{g}_i \neq 0$. Since $-\hat{g}^\top g = \gamma \|\hat{g}\|_\infty \bar{\pi}^\top h / \nu$ up to the final truncation and $\|\hat{g}\|^* = \|\hat{g}\|_\infty \|\bar{\pi}\|^*$, it remains to find $h \geq 0$ with $h_i = 0$ wherever $\pi_i = 0$ and

$$\frac{\pi^\top h}{\|h\|_{\tilde{\tau}+\infty}} \geq \left(1 - \frac{\alpha}{32} - \rho^{-9}\right) \|\pi\|^*. \quad (18)$$

Structure of the maximizer, in exact arithmetic. By Equation (16) the maximizer defining $\|\pi\|^*$ may be taken nonnegative, and setting to 0 its coordinates with $\pi_i = 0$ leaves $\pi^\top h$ unchanged and does not increase the norm. As $\pi^\top h$ and $\|h\|_{\tilde{\tau}+\infty}$ are both homogeneous of degree one,

$$\|\pi\|^* = \max \left\{ \frac{\pi^\top h}{\|h\|_{\tilde{\tau}+\infty}} : h \geq 0, h \neq 0, h_i = 0 \text{ wherever } \pi_i = 0 \right\}, \quad (19)$$

and only the direction of h matters. Let h^* attain the maximum, scaled so that $\|h^*\|_\infty = 1$, and put $a^* \triangleq C_{\text{norm}} \|h^*\|_{\tilde{\tau}}$. Then h^* maximizes $\pi^\top h$ over $\{h \geq 0 : \|h\|_\infty \leq 1, C_{\text{norm}} \|h\|_{\tilde{\tau}} \leq a^*\}$, since every h in this set has

$$\pi^\top h \leq \|\pi\|^* \|h\|_{\tilde{\tau}+\infty} \leq \|\pi\|^* (1 + a^*) = \pi^\top h^*.$$

This is a linear objective over the intersection of a box with an ellipsoid, a convex program with a strictly feasible point, so the Karush–Kuhn–Tucker conditions hold at h^* : for some multiplier $\xi \geq 0$ of the ellipsoid constraint, $h_i^* = \min(1, \pi_i / (2\xi \tilde{\tau}_i))$ for every i with $\pi_i > 0$ when $\xi > 0$, and $h_i^* = 1$ for every such i when $\xi = 0$. For $s \in (0, \infty]$ write

$$h_i^{(s)} \triangleq \min \left(1, \frac{s \pi_i}{\tilde{\tau}_i} \right), \quad r(s) \triangleq \frac{\pi^\top h^{(s)}}{\|h^{(s)}\|_{\tilde{\tau}+\infty}},$$

so that $h^* = h^{(s^*)}$ with $s^* = 1/(2\xi)$, or $s^* = \infty$, and $r(s^*) = \|\pi\|^*$.

The range of s . Let $s_{\min} \triangleq \min_{i:\pi_i>0} \tilde{\tau}_i/\pi_i$ and $s_{\max} \triangleq \max_{i:\pi_i>0} \tilde{\tau}_i/\pi_i$. For $s \leq s_{\min}$ no coordinate is capped and $h^{(s)} = s \pi / \tilde{\tau}$, so $r(s) = r(s_{\min})$; for $s \geq s_{\max}$ every coordinate with $\pi_i > 0$ is capped and $h^{(s)} = h^{(\infty)}$, so $r(s) = r(s_{\max})$. Hence s^* may be taken in $[s_{\min}, s_{\max}]$. As the nonzero π_i lie in $[\rho^{-10}, 1]$, $d/(2n) \leq \tilde{\tau}_i \leq 6$ and $\rho \geq n \geq 2$,

$$s_{\min} \geq \frac{d}{2n} \geq \rho^{-2}, \quad s_{\max} \leq 6\rho^{10}.$$

Monotonicity in s . Since $\min(1, cz) \leq c \min(1, z)$ for $c \geq 1$ and $z \geq 0$, for $c \geq 1$

$$h^{(s)} \leq h^{(cs)} \leq c h^{(s)} \quad \text{entrywise}, \quad (20)$$

and as $\pi^\top h$ and $\|h\|_{\tilde{\tau}+\infty}$ are monotone and homogeneous on $h \geq 0$,

$$r(cs) \geq \frac{\pi^\top h^{(s)}}{c \|h^{(s)}\|_{\tilde{\tau}+\infty}} = \frac{r(s)}{c}, \quad r(s) \geq \frac{r(cs)}{c} \quad \text{for } c \geq 1 : \quad (21)$$

changing s by a factor c changes r by at most the factor c .

The sequence s_k . The computed s_{k+1} differs from $(1 + \eta)s_k$ by less than ρ^{-20} , two operations of depth one in Equation (12), and by induction

$$s_k \geq s_0 \geq \rho^{-2} - \rho^{-20} \geq \frac{\rho^{-2}}{2},$$

so with $\alpha = 1/(10 \log(10n/d)) \geq \rho^{-1}$,

$$\left(1 + \frac{\alpha}{256}\right) s_k \leq (1 + \eta) (1 - 2\rho^{-18}) s_k \leq s_{k+1} \leq \left(1 + \frac{\alpha}{64}\right) s_k.$$

Hence $s_0 \leq \rho^{-2} \leq s_{\min} \leq s^*$, and, by $\log(1+t) \geq t/2$ for $t \leq 1$ and the choice of K ,

$$s_K \geq \left(1 + \frac{\alpha}{256}\right)^K \frac{\rho^{-2}}{2} \geq e^{\alpha K/512} \frac{\rho^{-2}}{2} \geq 12\rho^{12} \cdot \frac{\rho^{-2}}{2} = 6\rho^{10} \geq s_{\max} \geq s^*.$$

So the first $\bar{k}$ with $s_{\bar{k}} \geq s^*$ has

$$s_{\bar{k}} \leq \left(1 + \frac{\alpha}{64}\right) s_{\bar{k}-1} < \left(1 + \frac{\alpha}{64}\right) s^*,$$

and Equation (21) gives

$$r(s_{\bar{k}}) \geq \frac{\|\pi\|^*}{1 + \alpha/64} \geq \left(1 - \frac{\alpha}{64}\right) \|\pi\|^*. \quad (22)$$

Truncation of the candidates, step 3. Fix k and write $\bar{h} \triangleq h^{(s_k)}$. The computed $h_i^{(k)}$ is $\min(1, \cdot)$ of the truncated quotient of the truncated product $s_k \pi_i$ by $\tilde{\tau}_i$; both truncations are toward zero on nonnegative quantities, each by less than ρ^{-40} , and the division by $\tilde{\tau}_i \geq \rho^{-2}$ amplifies the error of the product by at most ρ^2 , so

$$0 \leq \bar{h}_i - h_i^{(k)} \leq \rho^{-40} (1 + \rho^2) \leq \rho^{-17},$$

and $h_i^{(k)} = 0$ wherever $\pi_i = 0$. Changing h by at most ρ^{-17} per entry changes $\pi^\top h$ by at most $n\rho^{-17}$ and $\|h\|_{\tilde{\tau}+\infty}$ by at most $\rho^{-17}(1 + C_{\text{norm}}(6d)^{1/2})$, both at most ρ^{-15} . On the other side, for the i_0 with $\pi_{i_0} = 1$,

$$\bar{h}_{i_0} = \min\left(1, \frac{s_k}{\tilde{\tau}_{i_0}}\right) \geq \frac{\rho^{-2}}{12} \geq \rho^{-3}, \quad \text{so} \quad \pi^\top \bar{h} \geq \bar{h}_{i_0} \geq \rho^{-3}, \quad \|\bar{h}\|_{\tilde{\tau}+\infty} \geq \|\bar{h}\|_\infty \geq \rho^{-3}.$$

Hence

$$(1 - \rho^{-12}) \pi^\top \bar{h} \leq \pi^\top h^{(k)} \leq \pi^\top \bar{h}, \quad (1 - \rho^{-12}) \|\bar{h}\|_{\tilde{\tau}+\infty} \leq \|h^{(k)}\|_{\tilde{\tau}+\infty} \leq \|\bar{h}\|_{\tilde{\tau}+\infty}.$$

The computed $\tilde{q}_k$ is the sum of the n products $\pi_i h_i^{(k)}$ of stored numbers, a formula with $2n \leq 2\rho$ operations of multiplicative depth 1 on numbers of magnitude at most ρ , so Lemma 2.2 with $M = \rho$ and $\ell = 40 \log \rho$ puts it within $2\rho \cdot 2\rho^2 \cdot \rho^{-40} = 4\rho^{-37}$ of $\pi^\top h^{(k)}$. For $\tilde{\nu}_k$, the radicand $\sum_i \tilde{\tau}_i (h_i^{(k)})^2$ is a formula with $3n \leq 3\rho$ operations of depth 2 on stored numbers, so it is off by at most $3\rho \cdot (2\rho^2)^2 \cdot \rho^{-40} = 12\rho^{-35}$ by the same lemma; it is at least $\tilde{\tau}_{i_0} (h_{i_0}^{(k)})^2 \geq \rho^{-2} \cdot \rho^{-6}/4 \geq \rho^{-9}$, so the square root, whose derivative is at most $\rho^{4.5}/2$ there, is off by at most $6\rho^{-30.5} + \rho^{-40}$, and the product with $C_{\text{norm}} \leq \rho$ and the sum with the exact maximum $\|h^{(k)}\|_\infty$ leave $\tilde{\nu}_k$ within $\rho^{-40} + \rho(6\rho^{-30.5} + \rho^{-40}) + 2\rho^{-40} \leq \rho^{-17}$ of $\|h^{(k)}\|_{\tilde{\tau}+\infty}$; hence

$$\|h^{(k)}\|_{\tilde{\tau}+\infty} \leq \nu_k \leq \|h^{(k)}\|_{\tilde{\tau}+\infty} + 2\rho^{-17},$$

and against the lower bounds $\rho^{-3}/2$ on $\pi^\top h^{(k)}$ and $\|h^{(k)}\|_{\tilde{\tau}+\infty}$ both

$$\frac{\pi^\top h^{(k)}}{\nu_k} \quad \text{and} \quad \frac{\tilde{q}_k}{\nu_k} \quad \text{lie in} \quad \left[(1 - \rho^{-11}) r(s_k), (1 + \rho^{-11}) r(s_k)\right]. \quad (23)$$

The selection, steps 4 and 5. The cross multiplication compares $\tilde{q}_k \nu_{k'}$ with $\tilde{q}_{k'} \nu_k$, products of numbers with $40 \log \rho$ bits that are exact on $80 \log \rho$ bits, so k^* maximizes $\tilde{q}_k / \nu_k$ exactly, and by Equation (23) at k^* and at $\bar{k}$ and Equation (22),

$$\frac{\pi^\top h}{\nu} \geq (1 - \rho^{-11}) r(s_{k^*}) \geq (1 - \rho^{-11})^2 \frac{\tilde{q}_{k^*}}{\nu_{k^*}} \geq (1 - \rho^{-11})^2 \frac{\tilde{q}_{\bar{k}}}{\nu_{\bar{k}}}$$

and

$$(1 - \rho^{-11})^2 \frac{\tilde{q}_{\bar{k}}}{\nu_{\bar{k}}} \geq (1 - \rho^{-11})^3 r(s_{\bar{k}}) \geq \left(1 - \frac{\alpha}{64} - \rho^{-10}\right) \|\pi\|^*,$$

which is Equation (18) for h/ν . Finally

$$|g_i| = \text{Trunc} \left(\frac{\gamma h_i}{\nu}, 40 \log \rho \right) \leq \frac{\gamma h_i}{\nu}, \quad \text{so} \quad \|g\|_{\tilde{\tau}+\infty} \leq \frac{\gamma \|h\|_{\tilde{\tau}+\infty}}{\nu} \leq \gamma,$$

which is Item 5.1(1), and $h_i = 0$ where $\pi_i = 0$ while $\text{sgn}(g_i) = -\text{sgn}(\hat{g}_i) = -\text{sgn}(\hat{p}_i) = -\text{sgn}(p_i)$ elsewhere, which is Item 5.1(2). The truncation toward zero, two operations of depth two in Equation (12), lowers each $|g_i|$ by at most ρ^{-20} and so $\sum_i |\hat{g}_i| |g_i|$ by at most

$$n\rho^{-20} \|\hat{g}\|_\infty \leq 4C_{\text{norm}} \rho^{-19} \|\hat{g}\|^* \leq \rho^{-17} \gamma \|\hat{g}\|^*,$$

so that

$$-\hat{g}^\top g \geq \left(1 - \frac{\alpha}{64} - \rho^{-10} - 2\rho^{-8} - \rho^{-17}\right) \gamma \|\hat{g}\|^*.$$

Passing from $\hat{g}$ to $\hat{p}$: on the support of g the signs agree and $|\hat{p}_i| \geq |\hat{g}_i| - \rho^{-20}$, so

$$-\hat{p}^\top g \geq -\hat{g}^\top g - n\rho^{-20} \gamma \geq -\hat{g}^\top g - 16nC_{\text{norm}} \rho^{-20} \gamma \|\hat{g}\|^* \geq -\hat{g}^\top g - \rho^{-17} \gamma \|\hat{g}\|^*,$$

using $\|\hat{g}\|^* \geq \|\hat{g}\|_\infty / (4C_{\text{norm}}) \geq 1/(16C_{\text{norm}})$ and $\rho \geq n \geq 16C_{\text{norm}}$, and $\|\hat{g}\|^* \geq (1 - \rho^{-17}) \|\hat{p}\|^*$ from step 1. Collecting the losses of the normalization, the gradient and the truncation,

$$-\hat{p}^\top g \geq (1 - \rho^{-17}) \left(1 - \frac{\alpha}{64} - \rho^{-10} - 2\rho^{-8} - 2\rho^{-17}\right) \gamma \|\hat{p}\|^*$$

and

$$(1 - \rho^{-17}) \left(1 - \frac{\alpha}{64} - \rho^{-10} - 2\rho^{-8} - 2\rho^{-17}\right) \geq 1 - \frac{\alpha}{64} - \rho^{-7} \geq 1 - \frac{\alpha}{16},$$

so

$$-\hat{p}^\top g \geq \left(1 - \frac{\alpha}{16}\right) \gamma \|\hat{p}\|^*,$$

which is Item 5.1(3) for $\hat{p}$, hence for $p = 2^{-j} \hat{p}$.

Cost. Step 1 is $O(nK) = O(n(\lambda\epsilon + \log \rho)) = O(n \log \rho)$ operations, as $\lambda\epsilon \leq 3C \log n$ and $\rho \geq n$; step 2 is $O(n)$, and step 3 is $K + 1 = O(\alpha^{-1} \log \rho) = O(\log n \log \rho)$ candidates at $O(n + \log \rho)$ operations each, all on numbers with $40 \log \rho$ bits: $\tilde{O}(n \log \rho)$ bit operations in total, within the budget of a step. $\square$

5.3 Centrality and feasibility

The following lemma bounds the solve error in the step.

Lemma 5.4. *In step 5 of Figure 5 let $h_j^* \triangleq (A^\top W A)^{-1} b_j$ be the exact solutions for the right hand sides $b_1 = A^\top \Phi''(x)^{-1/2} g$ and $b_2 = A^\top x - b$, and let δ_j^*, δ_r^* be formed from them as δ_j, δ_r are from h_j . If (x, s, μ) is ϵ -centered, $\tilde{\tau} \approx_{\gamma/100} \tau(x)$ and $\|g\|_{\tilde{\tau}+\infty} \leq \gamma$, then*

$$\|h_1^*\|_{A^\top W A} \leq \gamma, \quad \|h_2^*\|_{A^\top W A} \leq \gamma,$$

and, for the outputs of SOLVE,

$$\|\delta_r - \delta_r^*\|_\infty \leq 12 \left(\frac{n}{d}\right)^{1/2} \rho^{-10} \gamma, \quad \left\| \left(\mu \tilde{T} \Phi''(x)^{1/2} \right)^{-1} A (\mu h_1 - \mu h_1^*) \right\|_\infty \leq 6 \left(\frac{n}{d}\right)^{1/2} \rho^{-10} \gamma.$$

Proof. For any $e \in \mathbb{R}^d$, $\tilde{T}^{-1} \Phi''(x)^{-1/2} A e = \tilde{T}^{-1/2} \bar{A} e$ and

$$\|\bar{A} e\|_2^2 = e^\top \bar{A}^\top \bar{A} e = \|e\|_{A^\top W A}^2,$$

so every coordinate of $\tilde{T}^{-1} \Phi''(x)^{-1/2} A e$ is at most $\tilde{\tau}_i^{-1/2} \|e\|_{A^\top W A} \leq 2(n/d)^{1/2} \|e\|_{A^\top W A}$, using $\tilde{\tau}_i \geq e^{-\gamma/100} d/n$. Both displayed bounds follow by taking $e = h_j - h_j^*$ once $\|e\|_{A^\top W A} \leq 3\rho^{-10} \|h_j^*\|_{A^\top W A}$ is shown and the two norms $\|h_j^*\|_{A^\top W A}$ are bounded by γ . For the first, write $H \triangleq A^\top W A$ and $\hat{H} \triangleq A^\top \hat{W} A$ for the matrix the data structure holds after step 3, so that $\hat{H} \approx_{\rho^{-18}} H$ and Equation (1) reads $\|h_j - \hat{H}^{-1} b_j\|_{\hat{H}} \leq \rho^{-10} \|b_j\|_{\hat{H}^{-1}}$. For symmetric positive definite $\hat{H} \approx_\delta H$ and any b , $\|b\|_{\hat{H}^{-1}} \leq e^{\delta/2} \|b\|_{H^{-1}}$ and

$$\left\| \hat{H}^{-1} b - H^{-1} b \right\|_H = \left\| \left(H^{1/2} \hat{H}^{-1} H^{1/2} - I \right) H^{-1/2} b \right\|_2 \leq (e^\delta - 1) \|b\|_{H^{-1}},$$

so with $\delta = \rho^{-18}$ and $\|b_j\|_{H^{-1}} = \|h_j^*\|_H$,

$$\|h_j - h_j^*\|_H \leq e^{\delta/2} \left\| h_j - \hat{H}^{-1} b_j \right\|_{\hat{H}} + \left\| \hat{H}^{-1} b_j - H^{-1} b_j \right\|_H \leq (e^\delta \rho^{-10} + e^\delta - 1) \|h_j^*\|_H \leq 3\rho^{-10} \|h_j^*\|_H.$$

For h_1^* : $b_1 = \bar{A}^\top \tilde{T}^{1/2} g$, so $\bar{A} h_1^* = P \tilde{T}^{1/2} g$ with P the orthogonal projection onto the range of $\bar{A}$, and

$$\|h_1^*\|_{A^\top W A} = \left\| P \tilde{T}^{1/2} g \right\|_2 \leq \left\| \tilde{T}^{1/2} g \right\|_2 = \|g\|_{\tilde{\tau}} \leq \frac{\gamma}{C_{\text{norm}}}.$$

For h_2^* : $\|h_2^*\|_{A^\top W A}^2 = (A^\top x - b)^\top (A^\top W A)^{-1} (A^\top x - b)$, and

$$W = \left(\tilde{T} \Phi''(x) \right)^{-1} \approx_{\gamma/100} (T(x) \Phi''(x))^{-1},$$

so

$$\|h_2^*\|_{A^\top W A} \leq e^{\gamma/200} \left\| A^\top x - b \right\|_{(A^\top (T(x) \Phi''(x))^{-1} A)^{-1}} \leq e^{\gamma/200} \frac{\epsilon \gamma}{C_{\text{norm}}} \leq \gamma$$

by Item 4.3(3). □

The following lemma bounds the effect of perturbations on centrality and feasibility.

Lemma 5.5. *Let $s = Az + c$ with $\Psi(x, s, \mu) \leq n^{10}$, and let x', z' satisfy*

$$\left\| \Phi''(x)^{1/2} (x' - x) \right\|_\infty \leq \beta, \quad \left\| \left(\mu T(x) \Phi''(x)^{1/2} \right)^{-1} A (z' - z) \right\|_\infty \leq \beta$$

for some $\beta \leq 1/100$, and put $s' \triangleq Az' + c$. Then

$$1. \Psi(x', s', \mu) \leq e^{40\lambda\beta} \Psi(x, s, \mu);$$

$$2. \left\| A^\top x' - b \right\|_{(A^\top (T(x') \Phi''(x'))^{-1} A)^{-1}} \leq e^{5\beta} \left(\left\| A^\top x - b \right\|_{(A^\top (T(x) \Phi''(x))^{-1} A)^{-1}} + (5d)^{1/2} \beta \right);$$

3. s' is dual feasible.

Proof. By Equation (10), $\Phi''(x') \approx_{3\beta} \Phi''(x)$ and $|\phi'_i(x'_i) - \phi'_i(x_i)| \leq 2\beta\phi''_i(x_i)^{1/2}$; by Lemma 4.4 applied to $\Phi''(x')^{-1/2} \approx_{3\beta/2} \Phi''(x)^{-1/2}$, $\tau(x') \approx_{6\beta} \tau(x)$.

For Item 5.5(1), write $y_i = a_i + b_i$ with $a_i = s_i/(\mu\tau_i(x)\phi''_i(x_i)^{1/2})$ and $b_i = \phi'_i(x_i)/\phi''_i(x_i)^{1/2}$, and $y'_i = a'_i + b'_i$ likewise at (x', s') . From $\cosh(\lambda y_i) \leq n^{10}$, $|y_i| \leq \log(2n^{10})/\lambda \leq 1$, so $|a_i| \leq 2$ and $|b_i| \leq 1$. The numerator of a'_i is $s_i + (A(z' - z))_i$, and the second summand is at most β times the denominator of a_i ; the denominator of a'_i is within $e^{\pm 7.5\beta}$ of that of a_i . Hence

$$|a'_i - a_i| \leq (2 + \beta) (e^{7.5\beta} - 1) + \beta \leq 18\beta, \quad |b'_i - b_i| \leq (e^{1.5\beta} - 1) + 2\beta e^{1.5\beta} \leq 4\beta,$$

so $|y'_i - y_i| \leq 22\beta$. Finally

$$\cosh(a + b) = \cosh a \cosh b + \sinh a \sinh b \leq \cosh(a) e^{|b|},$$

so $\psi(y'_i) \leq \psi(y_i) e^{22\lambda\beta}$ for every i , and summing gives Item 5.5(1).

For Item 5.5(2) put $D \triangleq T(x)\Phi''(x)$ and $D' \triangleq T(x')\Phi''(x')$, so $D' \approx_{9\beta} D$, and the two norms in the statement are within $e^{\pm 4.5\beta}$ of each other. By the triangle inequality it remains to bound $\|A^\top(x' - x)\|_{(A^\top D^{-1}A)^{-1}}$, and with $r \triangleq x' - x$,

$$\left\| A^\top r \right\|_{(A^\top D^{-1}A)^{-1}}^2 = \left(D^{1/2} r \right)^\top D^{-1/2} A \left(A^\top D^{-1} A \right)^{-1} A^\top D^{-1/2} \left(D^{1/2} r \right) \leq \left\| D^{1/2} r \right\|_2^2,$$

because the middle matrix is an orthogonal projection, and

$$\left\| D^{1/2} r \right\|_2^2 = \sum_i \tau_i(x) \phi''_i(x_i) r_i^2 \leq 5d\beta^2$$

by $\|\tau(x)\|_1 \leq 5d$. Item 5.5(3) is the definition of s' . □

5.4 Per-step bounds and progress

We now import the per-step analysis. To separate what is imported from what is new, define the *exact step* on the inputs of STEP and the direction g of its step 4 as the pair

$$\hat{x}^* \triangleq x + \Phi''(x)^{-1/2} (g - \delta_r^*), \quad \hat{z}^* \triangleq z + \mu h_1^*, \quad \hat{s}^* \triangleq A\hat{z}^* + c = s + \mu \tilde{T} \Phi''(x)^{1/2} \delta_1^*, \quad (24)$$

that is, steps 5 and 6 with exact solves and no truncation. The exact step is Algorithm 1 of [vdBLL⁺21] with $R = I$, $\bar{x} = x$ and $\bar{\tau} = \tilde{\tau}$, and the three lemmas below are its analysis there, stated for $R = I$. Their hypotheses are those of Definitions 4.3 and 4.6 together with $\|g\|_{\tilde{\tau}+\infty} \leq \gamma$; the source states the norm bound on g in $\|\cdot\|_{\tau(x)+\infty}$, which differs from $\|\cdot\|_{\tilde{\tau}+\infty}$ by the factor $e^{\gamma/200}$ of Equation (14), absorbed by the $O(\gamma)$ slack of the statements.

The bounds on the exact step are Lemmas 4.29, 4.31 and 4.33 of [vdBLL⁺21].

Lemma 5.6. *Let (x, s, μ) be ϵ -centered, let $\bar{x} = x, \tilde{\tau}$ satisfy Definition 4.6, let $\|g\|_{\tilde{\tau}+\infty} \leq \gamma$, and let $\delta_x \triangleq \hat{x}^* - x$, $c \triangleq \Phi''(x)^{-1/2}$, $D_c \triangleq \text{diag}(c)$, $\delta_c \triangleq \Phi''(\hat{x}^*)^{-1/2} - \Phi''(x)^{-1/2}$, and $\tau \triangleq \tau(x)$. Then*

1. $\|D_c^{-1}\delta_x\|_{\tau+\infty} \leq \gamma + O((\epsilon + C_{\text{norm}}^{-1})\gamma);$
2. $\|D_c^{-1}\delta_x\|_\infty \lesssim \gamma$ and $\|D_c^{-1}\delta_c\|_\infty \lesssim \gamma;$
3. $\|D_c^{-1}\delta_c\|_{\tau+\infty} \leq \gamma + O((\epsilon + C_{\text{norm}}^{-1})\gamma);$
4. $\|D_c^{-2}\delta_c^2\|_{\tau+\infty} \lesssim \gamma^2;$
5. $\|D_c^{-1}|\delta_x|\|_{P^{(2)}} \lesssim \gamma/C_{\text{norm}},$ where $P = \overline{A}(\overline{A}^\top \overline{A})^{-1} \overline{A}^\top$ and $P^{(2)}$ is its entrywise square.

Because δ_c is γ -bounded in the sense of [vdBLL⁺21, Definition 4.23], the Jacobian bounds for the regularized Lewis weight map transfer these bounds to the weights themselves. This is the τ_q statement referred to in Section 3, at the value $q = 1 - \alpha$ of Definition 4.2.

The movement of τ_q in one exact step is Lemmas 4.20 and 4.35 of [vdBLL⁺21].

Lemma 5.7. *In the setting of Lemma 5.6, let $\delta_\tau \triangleq \tau(\hat{x}^*) - \tau(x)$, $T \triangleq T(x)$, and let J be the Jacobian of $c \mapsto \tau_q(\text{diag}(c)A, v)$ at c . Then*

1. $\|T^{-1}\delta_\tau\|_\infty \lesssim \gamma;$
2. $\|T^{-1}(\delta_\tau - J\delta_c)\|_{\tau+\infty} \lesssim \gamma^2;$
3. $\|T^{-1}Jh\|_{\tau+\infty} \lesssim \|D_c^{-1}h\|_{\tau+\infty}$ for every $h \in \mathbb{R}^n$.

Consequently $\|T^{-1}\delta_\tau\|_{\tau+\infty} \lesssim \gamma$.

Item 5.7(3) is the decomposition $T^{-1}JD_c = \Lambda + K$ of [vdBLL⁺21, Lemma 4.20] into a diagonal $0 \preceq \Lambda \preceq I$ and a matrix K with $\|Kh\|_\infty \lesssim \|h\|_\infty$ and $\|Kh\|_\tau \lesssim \|h\|_{P^{(2)}} \leq \|h\|_\tau$, the last step by Cauchy–Schwarz and $\sum_j P_{ij}^2 = P_{ii} \leq \tau_i$. The consequence is Items 5.7(2) and 5.7(3) applied to $h = \delta_c$, with Item 5.6(3).

The progress of the exact step with an admissible direction follows Lemmas 4.38, 4.39 and Corollary 4.40 of [vdBLL⁺21], with the hypothesis on the direction relaxed.

Lemma 5.8. *In the setting of Lemma 5.6, suppose moreover that $\|\bar{y} - y(x, s, \mu)\|_\infty \leq \gamma/20$, that g is an admissible direction at $(\bar{y}, \tilde{\tau})$ in the sense of Definition 5.1, and let $\hat{\mu} = (1 - \theta)\mu$. Then $\hat{s}^*$ is dual feasible,*

$$\left\| A^\top \hat{x}^* - b \right\|_{(A^\top (T(\hat{x}^*) \Phi''(\hat{x}^*))^{-1} A)^{-1}} \leq \frac{\epsilon \gamma}{2C_{\text{norm}}},$$

and

$$\Psi(\hat{x}^*, \hat{s}^*, \hat{\mu}) \leq (1 - \kappa) \Psi(x, s, \mu) + n.$$

Proof. Write $y \triangleq y(x, s, \mu)$, $\psi'(y)_i \triangleq \lambda \sinh(\lambda y_i)$ and $\psi''(y)_i \triangleq \lambda^2 \cosh(\lambda y_i)$, so that $\nabla \Psi(y) = \psi'(y)$, and abbreviate $\|\cdot\|^* \triangleq \|\cdot\|_{\tilde{\tau}+\infty}^*$ and $U \triangleq \{h : \|h\|_{\tilde{\tau}+\infty} \leq 1\}$. Dual feasibility and the feasibility bound are [vdBLL⁺21, Lemma 4.38], and [vdBLL⁺21, Lemma 4.39] gives, for a universal constant c_2 ,

$$\Psi(\hat{x}^*, \hat{s}^*, \hat{\mu}) \leq \Psi(x, s, \mu) + \psi'(y)^\top g + \left(1 - \frac{\alpha}{4}\right) \gamma \|\psi'(y)\|^* + c_2 \gamma^2 \|\psi''(y)\|^*. \quad (25)$$

Both statements use g only through $\|g\|_{\tilde{\tau}+\infty} \leq \gamma$, which is Item 5.1(1); the direction enters Lemma 4.39 there only through the term $\psi'(y)^\top g$, which is kept explicit in Equation (25). The source then bounds this term for the exact maximizer by [vdBLN⁺20, Lemma 4.36] (the argument is [vdBLL⁺21, Corollary 4.40]); we rerun that argument with Items 5.1(2) and 5.1(3) in place of the maximizer.

The elementary inequality. For $a \geq 0$ and $|t| \leq \delta' \leq 1/5$, expanding $\sinh(a+t) = \sinh(a) \cosh(t) + \cosh(a) \sinh(t)$ with $\cosh(t) \geq 1$, $\sinh(t) \geq -\sinh(\delta') \geq -2\delta'$ and $\cosh(a) \leq \sinh(a) + 1$ gives

$$\sinh(a+t) \geq (1-2\delta') \sinh(a) - 2\delta'. \quad (26)$$

Let $\delta \triangleq \gamma/20$, so that $\lambda\delta \leq 1/5$ by Equation (11), and write $p \triangleq \nabla \Psi(\bar{y})$.

The correlation with $\psi'(y)$. Fix i with $g_i \neq 0$; by Item 5.1(2), $p_i \neq 0$ and g_i has the sign opposite to $\bar{y}_i$, so with $a \triangleq \lambda|\bar{y}_i|$ and $t \triangleq \lambda \operatorname{sgn}(\bar{y}_i)(y_i - \bar{y}_i)$, $|t| \leq \lambda\delta$, Equation (26) gives

$$-\psi'(y)_i g_i = |g_i| \lambda \sinh(a+t) \geq (1-2\lambda\delta) |g_i| |p_i| - 2\lambda^2 \delta |g_i|.$$

Summing over i and using Item 5.1(3) together with $\|g\|_1 \leq n\|g\|_\infty \leq n\gamma$,

$$-\psi'(y)^\top g \geq (1-2\lambda\delta) \left(1 - \frac{\alpha}{16}\right) \gamma \|p\|^* - 2\lambda^2 \delta \gamma n. \quad (27)$$

Next, $\|p\|^*$ is compared with $\|\psi'(y)\|^*$. By Equation (16) the dual norm is $\max_{h \in U} \sum_i |p_i| |h_i|$; for the maximizer h of $\psi'(y)$, Equation (26) with $a \triangleq \lambda|y_i|$ and $t \triangleq \lambda(|\bar{y}_i| - |y_i|)$ gives $|p_i| \geq (1-2\lambda\delta)|\psi'(y)_i| - 2\lambda^2 \delta$, and since $\|h\|_\infty \leq 1$ on U ,

$$\|p\|^* \geq \sum_i |p_i| |h_i| \geq (1-2\lambda\delta) \|\psi'(y)\|^* - 2\lambda^2 \delta n.$$

Substituting into Equation (27) and using $4\lambda\delta = \lambda\gamma/5 \leq \alpha/(5C^2)$,

$$\psi'(y)^\top g \leq -\left(1 - \frac{\alpha}{16} - \frac{\lambda\gamma}{5}\right) \gamma \|\psi'(y)\|^* + \frac{\lambda^2 \gamma^2 n}{5}. \quad (28)$$

The second order term. Entrywise

$$\psi''(y)_i = \lambda^2 \cosh(\lambda y_i) \leq \lambda |\psi'(y)_i| + \lambda^2,$$

so by the monotonicity of the dual norm in the absolute values and $\|1\|^* \leq n$,

$$c_2 \gamma^2 \|\psi''(y)\|^* \leq c_2 \lambda \gamma \cdot \gamma \|\psi'(y)\|^* + c_2 \lambda^2 \gamma^2 n \leq \frac{\alpha}{32} \gamma \|\psi'(y)\|^* + c_2 \lambda^2 \gamma^2 n,$$

using $c_2 \lambda \gamma = c_2 \alpha / C^2 \leq \alpha / 32$ for $C \geq 32c_2$.

Collecting. Equations (25), (28) and the last display give

$$\begin{aligned} \Psi(\hat{x}^*, \hat{s}^*, \hat{\mu}) &\leq \Psi(x, s, \mu) - \left(\frac{\alpha}{4} - \frac{\alpha}{16} - \frac{\alpha}{32} - \frac{\lambda\gamma}{5}\right) \gamma \|\psi'(y)\|^* + \left(c_2 + \frac{1}{5}\right) \lambda^2 \gamma^2 n \\ &\leq \Psi(x, s, \mu) - \frac{\alpha}{8} \gamma \|\psi'(y)\|^* + \left(c_2 + \frac{1}{5}\right) \lambda^2 \gamma^2 n. \end{aligned}$$

Finally $\|\psi'(y)\|^*$ is converted into the potential. Since $\|h\|_{\tilde{\tau}+\infty} \leq \|h\|_\infty \|1\|_{\tilde{\tau}+\infty}$ and

$$\|1\|_{\tilde{\tau}+\infty} = 1 + C_{\text{norm}} \|\tilde{\tau}\|_1^{1/2} \leq 1 + C_{\text{norm}} (6d)^{1/2} \leq 4C_{\text{norm}} d^{1/2} = \frac{4Cd^{1/2}}{\alpha},$$

the vector $h \triangleq \alpha \operatorname{sgn}(\psi'(y)) / (4Cd^{1/2})$ lies in U and gives $\|\psi'(y)\|^* \geq \alpha \|\psi'(y)\|_1 / (4Cd^{1/2})$, while $|\lambda \sinh(\lambda y_i)| \geq \lambda (\cosh(\lambda y_i) - 1)$ gives $\|\psi'(y)\|_1 \geq \lambda \Psi(x, s, \mu) - \lambda n$. Hence, with $\kappa = \alpha^2 \lambda \gamma / (32Cd^{1/2})$,

$$\Psi(\hat{x}^*, \hat{s}^*, \hat{\mu}) \leq (1 - \kappa) \Psi(x, s, \mu) + \kappa n + \left(c_2 + \frac{1}{5}\right) \lambda^2 \gamma^2 n \leq (1 - \kappa) \Psi(x, s, \mu) + n,$$

since $\kappa \leq \lambda\gamma/32 \leq 1/256$ and $(c_2 + 1/5)\lambda^2 \gamma^2 \leq (1/32 + 1/40)/8$ by $\lambda\gamma \leq 1/8$ and $c_2 \lambda \gamma \leq 1/32$. $\square$

The following lemma is the guarantee for the computed step.

Lemma 5.9. *Let $(x, s = Az + c, \mu)$ be ϵ -centered with $\Psi(x, s, \mu) \leq n^2$, and let the estimate $\tilde{\tau}$ of step 2 and the SOLVE calls of step 5 meet their guarantees. Then the output $(\hat{x}, \hat{z})$ of STEP(x, z, μ) satisfies*

$$\left\| \Phi''(x)^{1/2} (\hat{x} - \hat{x}^*) \right\|_{\infty} \leq \rho^{-8} \gamma, \quad \left\| \left(\hat{\mu} T(\hat{x}^*) \Phi''(\hat{x}^*)^{1/2} \right)^{-1} A (\hat{z} - \hat{z}^*) \right\|_{\infty} \leq \rho^{-8} \gamma,$$

and, with $\hat{s} = A\hat{z} + c$, the triple $(\hat{x}, \hat{s}, \hat{\mu})$ is ϵ -centered and

$$\Psi(\hat{x}, \hat{s}, \hat{\mu}) \leq \left(1 - \frac{\kappa}{2}\right) \Psi(x, s, \mu) + 2n.$$

Proof. Step 2 gives $\tilde{\tau} \approx_{\gamma/100} \tau(x)$, so Definition 4.6 holds, Lemma 5.2 supplies the hypothesis on $\bar{y}$ of Lemma 5.8, and Lemma 5.3, applicable as $\Psi \leq n^2 \leq n^{10}$, shows that the g of step 4 is an admissible direction, which is the remaining hypothesis of Lemma 5.8; in particular $\|g\|_{\tilde{\tau}+\infty} \leq \gamma$, as Lemmas 5.6 and 5.4 require.

The two displayed bounds. By Equation (24), $\hat{x} - \hat{x}^* = \Phi''(x)^{-1/2}(\delta_r^* - \delta_r) + r_x$ and $\hat{z} - \hat{z}^* = \mu(h_1 - h_1^*) + r_z$, where r_x, r_z collect the truncation. The solve error terms are bounded by Lemma 5.4: the first by $12(n/d)^{1/2} \rho^{-10} \gamma \leq 12\rho^{-9.5} \gamma$ as $\rho \geq n$, and the second by $6(n/d)^{1/2} \rho^{-10} \gamma$ in the norm with $\mu \tilde{T} \Phi''(x)^{1/2}$ in place of $\hat{\mu} T(\hat{x}^*) \Phi''(\hat{x}^*)^{1/2}$; the two diagonals are within a factor 3 since $\hat{\mu} \geq \mu/2$, $\tau(\hat{x}^*) \approx_{O(\gamma)} \tau(x) \approx_{\gamma/100} \tilde{\tau}$ and $\Phi''(\hat{x}^*) \approx_{O(\gamma)} \Phi''(x)$ by Item 5.6(2) and Equation (10). For the truncation, every entry of $\hat{x}$ and $\hat{z}$ is a formula with at most 12 operations and multiplicative depth at most 6 on the stored numbers, the outputs h_1, h_2 of SOLVE and the sums $(Ah_j)_i$: two distances to the bounds, their squared reciprocals, a sum, a square root and a reciprocal for $\phi_i''(x_i)^{-1/2}$, then the difference $g_i - (\delta_r)_i$, one product and one sum for $\hat{x}_i$, and one product and one sum for $\hat{z}_i$; so Equation (12) gives $\|r_x\|_{\infty}, \|r_z\|_{\infty} \leq \rho^{-20}$. The right hand sides passed to SOLVE are sums of the second kind over entrywise formulas, cut to $20 \log \rho$ bits, so by Equation (13) each entry of b_j is off by at most $\rho^{-16} + \rho^{-20} \leq 2\rho^{-16}$, which moves $h_j^* = (A^\top W A)^{-1} b_j$ by at most $\|(A^\top W A)^{-1/2}\| \cdot 2d^{1/2} \rho^{-16} \leq 2\rho^{1/2} d^{1/2} \rho^{-16}$ in the $A^\top W A$ norm, and this is handled like the solve error. Then $\|\Phi''(x)^{1/2} r_x\|_{\infty} \leq \rho^{1/2} \rho^{-20}$, and

$$|(Ar_z)_i| \leq \|A\| \|r_z\|_2 \leq \rho^{1/2} d^{1/2} \rho^{-20}$$

against a denominator $\hat{\mu} \tau_i \phi_i''^{1/2} \geq \rho^{-1} (d/2n) \rho^{-1/2}$. All of these are below $\rho^{-8} \gamma/2$ for $\rho \geq n$ large enough, and the two bounds follow.

The potential and feasibility. Lemma 5.8 gives $\Psi(\hat{x}^*, \hat{s}^*, \hat{\mu}) \leq n^2 + n \leq n^{10}$, so Lemma 5.5 applies at $(\hat{x}^*, \hat{z}^*, \hat{\mu})$ with $\beta = \rho^{-8} \gamma$:

$$\begin{aligned} \Psi(\hat{x}, \hat{s}, \hat{\mu}) &\leq e^{40\lambda\gamma\rho^{-8}} ((1 - \kappa)\Psi(x, s, \mu) + n) \\ &\leq \left(1 + \frac{\kappa}{4}\right) ((1 - \kappa)\Psi(x, s, \mu) + n) \\ &\leq \left(1 - \frac{\kappa}{2}\right) \Psi(x, s, \mu) + 2n, \end{aligned}$$

where the middle step uses

$$40\lambda\gamma\rho^{-8} \leq \frac{\kappa}{8} = \frac{\alpha^2 \lambda \gamma}{256 C d^{1/2}}, \quad \text{i.e.} \quad \rho^8 \geq 10240 C d^{1/2} \alpha^{-2},$$

true for $\rho \geq n$ and n larger than a constant. Item 5.5(2) with the feasibility bound of Lemma 5.8 gives

$$\left\| A^\top \hat{x} - b \right\|_{(A^\top (T(\hat{x}) \Phi''(\hat{x}))^{-1} A)^{-1}} \leq e^{5\beta} \left(\frac{\epsilon\gamma}{2C_{\text{norm}}} + (5d)^{1/2} \rho^{-8} \gamma \right) \leq \frac{\epsilon\gamma}{C_{\text{norm}}},$$

Item 5.5(3) gives dual feasibility, and centrality follows from $\Psi(\hat{x}, \hat{s}, \hat{\mu}) \leq n^2 + 2n \leq n^{10}$ as noted after Definition 4.4. $\square$

The path following lemma is Lemma 4.12 of [vdBLL⁺21].

Lemma 5.10. *Start from an $\epsilon/C_{\text{start}}$ -centered triple $(x, s = Az + c, \mu_0)$ with an estimate $\tilde{\tau} \approx_{\gamma/100} \tau(x)$, and iterate STEP until $\mu \leq \mu_f$. This makes*

$$r = \lceil \theta^{-1} \log(\mu_0/\mu_f) \rceil = \tilde{O}\left(d^{1/2} \log(\mu_0/\mu_f)\right)$$

calls, and on the event that every SOLVE call along the way meets the guarantee of Definition 3.1, with probability at least $1 - rn^{-7}$ over the randomness of the Lewis estimates, the triple (x, s, μ) at the start and end of every call is ϵ -centered and satisfies $\Psi(x, s, \mu) \leq n^2$.

Proof. Induction on the round, using Lemma 5.9. The inductive hypothesis at the start of round k is threefold: the triple is ϵ -centered with $\Psi \leq n^2$, the estimate $\tilde{\tau}$ carried from round $k-1$ satisfies $\tilde{\tau} \approx_{\gamma/100} \tau(x^{(k-1)})$, and $\|\Phi''(x^{(k-1)})^{1/2}(x^{(k)} - x^{(k-1)})\|_\infty \leq 1/100$; at $k=1$ the last two hold with $x^{(0)} \triangleq x^{(1)}$ by the hypothesis on $\tilde{\tau}$. Given them, Lemma 5.1 supplies Items 4.2(5) and 4.2(6) to the call of Lemma 4.2 in step 2, whose output is $\tilde{\tau} \approx_{\gamma/100} \tau(x^{(k)})$, so Definition 4.6 holds for the round and Lemma 5.9 applies; Item 5.6(2) of Lemma 5.6 for the exact step, plus the $\rho^{-8}\gamma$ by which Lemma 5.9 lets the computed step differ from it, bounds the movement to $x^{(k+1)}$ by $O(\gamma) \leq 1/100$ in the local norm, which is the third hypothesis for round $k+1$, and the new $\tilde{\tau}$ is the second. The base case is the initial potential: an $\epsilon/C_{\text{start}}$ -centered point has $|y_i| \leq \epsilon/(10C)$ for every i , so

$$\Psi(x, s, \mu_0) \leq n \cosh\left(\frac{\lambda\epsilon}{10C}\right) \leq n e^{3 \log n/10} \leq n^2$$

by $\lambda\epsilon \leq 3C \log n$ from Equation (11). For the inductive step, if $\Psi \leq n^2$ at the start of a round then Lemma 5.9 gives, at its end, $\Psi \leq (1 - \kappa/2)n^2 + 2n \leq n^2$, because $\kappa = \alpha^3/(32C^3 d^{1/2}) \geq 4/n$ for n larger than a constant depending on C , using $n \geq d$; the same lemma gives ϵ -centrality at the end of the round, and step 2 of the next round restores the hypothesis on $\tilde{\tau}$. The count of calls is $\mu_r = (1 - \theta)^r \mu_0 \leq \mu_f$. The failure probability is a union bound over the r rounds, each making one call to Lemma 4.2, which fails with probability at most n^{-7} once its solver meets Item 4.2(6); that every SOLVE call, inside and outside the estimate, does so is the event conditioned on in the statement, and its probability is the subject of Theorem 2, not of this lemma. $\square$

Hence the potential stays at most n^2 throughout and μ decreases from μ_0 to μ_f in $\tilde{O}(d^{1/2} \log(\mu_0/\mu_f))$ steps; Section 6 turns this into a bit complexity.

5.5 Stability of the weights handed to the data structure

With $R = I$ the iterates themselves move stably, so unlike [LPT⁺25, Section 4.4] and [vdBLL⁺21, Lemmas 4.44 and 4.45] no nearby auxiliary sequence is needed: the stability below is a statement about the actual iterates, which are vectors of numbers with $40 \log \rho$ bits by construction.

The following lemma is the stability of the path weights.

Lemma 5.11. *Let $(x^{(k)}, z^{(k)})$, $1 \leq k \leq r+1$, be the iterates of Lemma 5.10, on the event of that lemma, the k -th call being made from $(x^{(k)}, z^{(k)})$, and write $\tau^{(k)} \triangleq \tau(x^{(k)})$, $T^{(k)} \triangleq \text{diag}(\tau^{(k)})$, $w_i^{(k)} \triangleq (\tau_i^{(k)} \phi_i''(x_i^{(k)}))^{-1}$ for the exact path weights with $W^{(k)} \triangleq \text{diag}(w^{(k)})$, and $\hat{w}^{(k)}$ for the vector handed to UPDATE in round k . Then for every $k \leq r$:*

1. (ℓ_2 stability of the iterate) $\|\Phi''(x^{(k)})^{1/2}(x^{(k+1)} - x^{(k)})\|_{\tau^{(k)}+\infty} \lesssim \gamma$;
2. (stability of the barrier) $\|\Phi''(x^{(k)})^{1/2}(\Phi''(x^{(k+1)})^{-1/2} - \Phi''(x^{(k)})^{-1/2})\|_{\tau^{(k)}+\infty} \lesssim \gamma$;
3. (τ_q stability) $\|(T^{(k)})^{-1}(\tau^{(k+1)} - \tau^{(k)})\|_{\tau^{(k)}+\infty} \lesssim \gamma$ and $\|(W^{(k)})^{-1}(w^{(k+1)} - w^{(k)})\|_{\tau^{(k)}+\infty} \lesssim \gamma$;
4. (the fed weights) $\hat{w}^{(k)} \approx_{\gamma/50} w^{(k)}$.

Proof. Fix k and write $x = x^{(k)}$, $\hat{x} = x^{(k+1)}$, and $\hat{x}^*$ for the exact step of Equation (24) from x . By Lemma 5.9, $\|\Phi''(x)^{1/2}(\hat{x} - \hat{x}^*)\|_\infty \leq \rho^{-8}\gamma$, and by Equation (14) this is at most $(1 + C_{\text{norm}}(5d)^{1/2})\rho^{-8}\gamma \leq \gamma$ in $\|\cdot\|_{\tau^{(k)}+\infty}$; through Equation (10) and Lemma 4.4 the same $O(\rho^{-8}\gamma)$ bound holds for the induced changes of $\Phi''^{-1/2}$ and of τ between $\hat{x}$ and $\hat{x}^*$. So it suffices to prove Items 5.11(1) to 5.11(3) with $\hat{x}^*$ in place of $\hat{x}$, and there they are, respectively, Item 5.6(1), Item 5.6(3), and the consequence of Lemma 5.7; for the weights, $w = \tau^{-1}c^2$ with $c = \Phi''^{-1/2}$, and with $u \triangleq \tau^{(k+1)}/\tau^{(k)} - 1$ and $v \triangleq c^{(k+1)}/c^{(k)} - 1$, both of ℓ_∞ norm $O(\gamma) \leq 1/10$ by Item 5.6(2) and Item 5.7(1),

$$\left| \frac{w_i^{(k+1)}}{w_i^{(k)}} - 1 \right| = \left| \frac{(1 + v_i)^2}{1 + u_i} - 1 \right| \leq 2|u_i| + 3|v_i|,$$

so the weight bound is 2 times the τ bound plus 3 times Item 5.11(2). Item 5.11(4) is $\hat{w}^{(k)} \approx_{\rho^{-18}} (\tilde{T}\Phi''(x))^{-1}$ from the truncation in step 3 together with $\tilde{\tau} \approx_{\gamma/100} \tau(x)$ from step 2, and $\rho^{-18} \leq \gamma/100$. $\square$

Item 5.11(3) in $\|\cdot\|_{\tau^{(k)}+\infty}$ contains both an ℓ_∞ bound of $O(\gamma)$ and a τ -norm bound of $O(\gamma/C_{\text{norm}})$, which are the shapes of conditions 2 and 3 of Definition 3.2, except that the definition weighs condition 3 by the leverage scores $\sigma^{(k)}$ of the reweighted matrix $(W^{(k)})^{1/2}A$ rather than by the Lewis weights $\tau^{(k)} = \tau(x^{(k)})$ of $\Phi''(x^{(k)})^{-1/2}A$. These are different vectors, but the reweighting is by $\tau^{(k)}$ itself, and at $q = 1 - \alpha$ that makes them comparable.

The following lemma compares the leverage scores of the reweighted matrix with the path weights.

Lemma 5.12. *Let x be a point with $\tau = \tau(x)$, $T = \text{diag}(\tau)$, $W = T^{-1}\Phi''(x)^{-1}$ the exact path weights at x , and $\sigma \triangleq \tau(W^{1/2}A)$ the leverage scores of the reweighted matrix. Then*

$$v + \sigma \approx_{1/2} \tau, \quad \text{in particular} \quad \sigma_i \leq e^{1/2}\tau_i \quad \text{for every } i.$$

Proof. By Definition 4.2, $\tau = v + \tau(T^{1/2-1/q}\Phi''(x)^{-1/2}A)$, and at $q = 1 - \alpha$

$$\frac{1}{2} - \frac{1}{q} = -\frac{1}{2} - \frac{\alpha}{1 - \alpha}, \quad \text{so} \quad T^{1/2-1/q}\Phi''(x)^{-1/2} = T^{-\alpha/(1-\alpha)}W^{1/2} = (D^2W)^{1/2}, \quad D \triangleq T^{-\alpha/(1-\alpha)}.$$

The computation of Lemma 5.1, with τ in place of $\tilde{\tau}'$ and $d/n \leq \tau_i \leq 5$, gives $D^2 = T^{-2\alpha/(1-\alpha)} \approx_{1/4} I$, hence $D^2W \approx_{1/4} W$, and leverage scores move by at most twice the movement of the weights, as recorded after Definition 3.2: $\tau((D^2W)^{1/2}A) \approx_{1/2} \tau(W^{1/2}A) = \sigma$. The second claim is $\sigma_i \leq e^{1/2}(\tau_i - v_i) \leq e^{1/2}\tau_i$. $\square$

Lemma 6.2 combines this with Lemma 5.11 to verify Definition 3.2, and Corollary 3.4 and Theorem 2 then bound the total work spent on inverse maintenance over the whole path. Condition 3 of Definition 3.2 is proved for the exact $w^{(k)}$ and not for the fed $\hat{w}^{(k)}$, whose deviation $e^{\pm\gamma/50}$ from $w^{(k)}$ has leverage score norm up to $d^{1/2}\gamma/25$; this is why Section 3 states its guarantee for sequences that are stable up to noise, and why the fed sequence qualifies is the last step of Lemma 6.2.

6 Initialization and Overall Guarantees

This section collects the three things still missing: that every matrix the algorithm stores has magnitude $2^{O(B)}$, so that $O(B)$ bits per entry suffice throughout; that the weights fed to UPDATE meet Definition 3.2, so that Theorem 2 and Corollary 3.3 apply; and an initial point from which $\tilde{O}(d^{1/2})$ path steps suffice.

6.1 Aspect ratio, magnitudes, and bit lengths

Everything rests on the path aspect ratio, which is where the assumption that the midpoint is feasible is used.

The following is Lemma 4.46 of [vdBLL⁺21].

Imported result 6.1. *Suppose the midpoint $(l + u)/2$ is feasible, and let W_0 be the ratio of the largest to the smallest entry of $\Phi''((l + u)/2)^{1/2}$. Then, over all points x encountered by the path following of Lemma 5.10, the ratio W' of the largest to the smallest entry of $\Phi''(x)^{1/2}$ satisfies*

$$\log W' = \tilde{O}(\log(2 + W_0) + \log(1/\mu_f) + \log(2 + \|c\|_\infty)).$$

The following lemma bounds the magnitudes along the path.

Lemma 6.1. *Run the algorithm on the augmented instance of Lemma 6.4, whose data has $O(B)$ bits and whose midpoint is feasible, with $\mu_f = 2^{-O(B)}$. Then at every point x of the path, and for the maintained weights $w_i = (\tau_i(x)\phi_i''(x_i))^{-1}$,*

$$\begin{aligned} 2^{-O(B)} &\leq \phi_i''(x_i) \leq 2^{O(B)}, \\ \frac{d}{n} &\leq \tau_i(x) \leq 1 + v_i \leq 5, \\ 2^{-O(B)} &\leq w_i \leq 2^{O(B)}, \end{aligned}$$

and consequently $A^\top W A$, $A^\top \Phi''(x)^{-1} A$ and every normal matrix formed by STEP has magnitude at most $\rho = 2^{O(B)}$. In particular $O(\log \rho) = O(B)$ bits per entry suffice in Lemma 3.1 and Theorem 2.

Proof. Imported result 6.1 bounds the ratio between the largest and smallest $\phi_i''(x_i)^{1/2}$ along the path by $2^{O(B)}$, since the augmented widths, $\|c\|_\infty$ and $1/\mu_f$ are all $2^{O(B)}$ by Lemma 6.4. An absolute bound follows from centrality. Let $m \triangleq (l + u)/2$, so $A^\top m = b$, and let (x, s, μ) be ϵ -centered with $s = Az + c$, so that $s_i = \mu\tau_i(x)(-\phi_i'(x_i) + \nu_i\phi_i''(x_i)^{1/2})$ with $\|\nu\|_\infty \leq \epsilon \leq 1/80$ by Item 4.3(1). Pairing s with $x - m$ gives

$$z^\top (A^\top x - b) + c^\top (x - m) + \mu \sum_i \tau_i(x) g_i = 0,$$

where

$$g_i \triangleq (\phi_i'(x_i) - \nu_i\phi_i''(x_i)^{1/2})(x_i - m_i),$$

since $z^\top A^\top (x - m) = z^\top (A^\top x - b)$. Write $N \triangleq A^\top (T(x)\Phi''(x))^{-1}A$. The first term is at most $\|z\|_N \|A^\top x - b\|_{N^{-1}} \leq \|z\|_N \epsilon \gamma / C_{\text{norm}}$ by Item 4.3(3), and

$$\begin{aligned} \|z\|_N^2 &= \sum_i \frac{(s_i - c_i)^2}{\tau_i(x) \phi_i''(x_i)} \\ &\leq 2\mu^2(1 + \epsilon)^2 \sum_i \tau_i(x) + 2\|c\|_\infty^2 \frac{n}{d} \sum_i \frac{(u_i - l_i)^2}{8} \\ &= 2^{O(B)}, \end{aligned}$$

using $|s_i| \leq \mu \tau_i(1 + \epsilon) \phi_i''(x_i)^{1/2}$ from Lemma 4.3, $d/n \leq \tau_i \leq 5$, $\phi_i''(t) \geq 8/(u_i - l_i)^2$, and $\mu \leq \mu_0 = 2^{O(B)}$. The second term is at most $\|c\|_\infty \sum_i (u_i - l_i) = 2^{O(B)}$. Each $g_i \geq -1$: $\phi_i'(x_i)$ has the sign of $x_i - m_i$, and in the middle half of $[l_i, u_i]$ one has $\phi_i''(x_i)^{1/2} |x_i - m_i| \leq 2^{1/2}$, so $|\nu_i| \phi_i''(x_i)^{1/2} |x_i - m_i| \leq 1$. For a coordinate at distance $\delta_i \leq (u_i - l_i)/4$ from its nearer bound, $|\phi_i'(x_i)| \geq 1/\delta_i - 1/(3\delta_i) \geq 1/(2\delta_i)$ and $|\nu_i| \phi_i''(x_i)^{1/2} \leq 1.1/(80\delta_i) \leq 1/(4\delta_i)$, so $g_i \geq |x_i - m_i|/(4\delta_i) \geq (u_i - l_i)/(16\delta_i)$. Isolating such a coordinate, with $g_j \geq -1$ for the others,

$$\begin{aligned} \mu \frac{d}{n} \cdot \frac{u_i - l_i}{16\delta_i} &\leq \mu \tau_i(x) g_i \\ &\leq \left| z^\top (A^\top x - b) \right| + \left| c^\top (x - m) \right| + 5d\mu \\ &= 2^{O(B)}, \end{aligned}$$

so $1/\delta_i \leq 2^{O(B)} n / (\mu_f d(u_i - l_i)) = 2^{O(B)}$ and $\phi_i''(x_i) = \Theta(\delta_i^{-2})$ is $2^{O(B)}$, while the lower bound on ϕ_i'' is $8/(u_i - l_i)^2 = 2^{-O(B)}$. The bounds on τ are Definition 4.2, and the bound on w follows. Finally $A^\top W A \preceq \max_i w_i \cdot A^\top A$ and $A^\top W A \succeq \min_i w_i \cdot A^\top A$, and $A^\top A$ has magnitude $2^{O(B)}$ by assumption, so $A^\top W A$ has magnitude $2^{O(B)}$. $\square$

6.2 The maintained weights are leverage stable

The following lemma shows that the path weights are leverage stable.

Lemma 6.2. *Let $w_i^{(k)} = (\tau_i(x^{(k)}) \phi_i''(x_i^{(k)}))^{-1}$ be the exact path weights of Lemma 5.11, over the $r = \tilde{O}(d^{1/2} \log(\mu_0/\mu_f))$ rounds of Lemma 5.10. Then $w^{(1)}, \dots, w^{(r+1)}$ is ζ -leverage stable in the sense of Definition 3.2, with*

$$\zeta = O(\gamma) \quad \text{and} \quad \rho = 2^{O(B)},$$

and condition 3 of that definition holds with the smaller value $O(\gamma/C_{\text{norm}})$ in place of ζ . Consequently the sequence $\hat{w}^{(1)}, \dots, \hat{w}^{(r+1)}$ actually handed to UPDATE in step 3 of Figure 5 is ζ -leverage stable up to noise, in the sense recorded after Definition 3.2.

Proof. Condition 1 is Lemma 6.1. For conditions 2 and 3, Item 5.11(3) gives

$$\left\| (W^{(k)})^{-1} \left(w^{(k+1)} - w^{(k)} \right) \right\|_{\tau^{(k)} + \infty} \lesssim \gamma,$$

and by Definition 4.5 this single bound contains both $\|\cdot\|_\infty \lesssim \gamma$, which is condition 2 since relative and logarithmic movement agree up to a factor of two once the movement is at most 0.1 coordinatewise, and $\|\cdot\|_{\tau^{(k)}} \lesssim \gamma/C_{\text{norm}}$. Condition 3 at round $k+1$ weighs the same logarithmic

movement by $\sigma^{(k+1)} = \tau((W^{(k+1)})^{1/2}A)$. By the fact recorded after Definition 3.2 and condition 2, $\sigma^{(k+1)} \approx_{2\zeta} \sigma^{(k)}$, and by Lemma 5.12 at $x^{(k)}$, $\sigma_i^{(k)} \leq e^{1/2} \tau_i^{(k)}$, so

$$\begin{aligned} \sum_i \sigma_i^{(k+1)} \log^2 \left(\frac{w_i^{(k+1)}}{w_i^{(k)}} \right) &\leq 4e^{1/2+2\zeta} \sum_i \tau_i^{(k)} \left(\frac{w_i^{(k+1)} - w_i^{(k)}}{w_i^{(k)}} \right)^2 \\ &\lesssim \frac{\gamma^2}{C_{\text{norm}}^2}, \end{aligned}$$

which is condition 3 with $O(\gamma/C_{\text{norm}})$, and a fortiori with $\zeta = O(\gamma)$. Finally $\gamma < 0.1$ by Equation (11). For the fed sequence, Item 5.11(4) gives $\hat{w}^{(k)} \approx_{\gamma/50} w^{(k)}$, and taking the constant in $\zeta = O(\gamma)$ to be at least $1/5$ makes this the $\approx_{\zeta/10}$ that stability up to noise asks for. $\square$

So the interior point method supplies the stability conditions with room to spare: the ℓ_∞ condition holds with $O(\gamma)$ where 0.1 is asked, and the leverage score condition, which is the one that drives the update counts of Corollary 3.4, is C_{norm} times inside even that. The weights are computed from the solver's outputs, so it is the adaptive half of Theorem 2, that is Corollary 3.3, that is invoked; the number of rounds, $\tilde{O}(d^{1/2} \log(\mu_0/\mu_f)) = \tilde{O}(d^{1/2} B) \leq d \log^{O(1)} n \leq n^2$ by the standing assumption $B \leq d^{1/2}$, is within the hypothesis $r \leq n^2$ of that corollary, and the number Q of solver calls, a polylogarithmic factor more, enters its failure probability as $2Qn^{-3}$.

6.3 Initialization

The final point is the projection of Lemma 4.11 of [vdBLL⁺21], which is proved there from the matrix norm bounds of its Lemma 4.28 and the self-concordance inequalities of Nesterov. We state and prove a version adapted to the bounds of this paper. Three things change relative to the source. The projection is taken in the estimate $\tilde{\tau}$ the algorithm holds, not in the exact $\tau(x)$, since the exact weights are never available; the movement bound is the explicit $2\epsilon\gamma/C_{\text{norm}}$, from Item 4.3(3) through the leverage scores of the reweighted matrix and Lemma 5.12, in place of the source's $O(\epsilon)$, so that no lower bound on C is needed for the point to stay inside the box; and the objective gap carries the explicit constant 2, obtained from the one sided bound $(-\phi'_i(x_i) + \nu_i \phi''_i(x_i)^{1/2})(x_i - x'_i) \leq 1 + \epsilon$ on the log barrier, in place of an unspecified absolute constant.

Lemma 6.3. *Let (x, s, μ) be ϵ -centered with $\epsilon \leq 1/80$, let $\tilde{\tau} \approx_\epsilon \tau(x)$, and let $\overline{W} \triangleq (\tilde{T}\Phi''(x))^{-1}$. Then*

$$x^{\text{fin}} \triangleq x + \overline{W}A \left(A^\top \overline{W}A \right)^{-1} \left(b - A^\top x \right)$$

satisfies $A^\top x^{\text{fin}} = b$,

$$\left\| \Phi''(x)^{1/2} (x^{\text{fin}} - x) \right\|_\infty \leq \frac{2\epsilon\gamma}{C_{\text{norm}}},$$

so in particular $l < x^{\text{fin}} < u$, and

$$c^\top x^{\text{fin}} - \min \left\{ c^\top x' : A^\top x' = b, l \leq x' \leq u \right\} \leq 2\mu \|\tau(x)\|_1.$$

Proof. Write $r \triangleq b - A^\top x$, $\overline{H} \triangleq A^\top \overline{W}A$, $W \triangleq (T(x)\Phi''(x))^{-1}$ the exact path weights and $N \triangleq A^\top W A$ the matrix of Item 4.3(3). Since $\tilde{\tau} \approx_\epsilon \tau(x)$ we have $\overline{W} \approx_\epsilon W$ and $\overline{H} \approx_\epsilon N$. Feasibility is $A^\top x^{\text{fin}} = A^\top x + \overline{H} \overline{H}^{-1} r = b$.

Movement. Coordinate i of the correction is $\bar{w}_i a_i^\top \bar{H}^{-1} r$, and by Cauchy–Schwarz in the $\bar{H}^{-1}$ inner product

$$\begin{aligned} \left| \bar{w}_i a_i^\top \bar{H}^{-1} r \right| &\leq \bar{w}_i \|a_i\|_{\bar{H}^{-1}} \|r\|_{\bar{H}^{-1}} \\ &= (\bar{w}_i \bar{\sigma}_i)^{1/2} \|r\|_{\bar{H}^{-1}}, \end{aligned}$$

where

$$\bar{\sigma}_i \triangleq \bar{w}_i a_i^\top \bar{H}^{-1} a_i = \tau_i \left(\bar{W}^{1/2} A \right).$$

By the fact recorded after Definition 3.2, $\bar{W} \approx_\epsilon W$ gives $\bar{\sigma} \approx_{2\epsilon} \tau(W^{1/2} A)$, and Lemma 5.12 gives $\tau_i(W^{1/2} A) \leq e^{1/2} \tau_i(x) \leq e^{1/2+\epsilon} \tilde{\tau}_i$, so $\bar{\sigma}_i \leq e^{1/2+3\epsilon} \tilde{\tau}_i \leq 2\tilde{\tau}_i$ and $\bar{w}_i \bar{\sigma}_i \leq 2\bar{w}_i \tilde{\tau}_i = 2/\phi_i''(x_i)$. Hence, with $\beta \triangleq 2\epsilon\gamma/C_{\text{norm}}$,

$$\begin{aligned} \phi_i''(x_i)^{1/2} \left| x_i^{\text{fin}} - x_i \right| &\leq 2^{1/2} \|r\|_{\bar{H}^{-1}} \\ &\leq 2^{1/2} e^{\epsilon/2} \|r\|_{N^{-1}} \\ &\leq 2^{1/2} e^{\epsilon/2} \frac{\epsilon\gamma}{C_{\text{norm}}} \\ &\leq \frac{2\epsilon\gamma}{C_{\text{norm}}} \\ &= \beta, \end{aligned}$$

by $\bar{H} \approx_\epsilon N$ and Item 4.3(3). Since $\phi_i''(x_i)^{1/2} \geq 1/\min(x_i - l_i, u_i - x_i)$ the correction moves coordinate i by at most β times its distance to the nearer bound, and $\beta < 1$, so $l < x^{\text{fin}} < u$.

Objective. Let x' be feasible, $A^\top x' = b$ and $l \leq x' \leq u$. By Item 4.3(2), $c = s - Az$, and $A^\top x^{\text{fin}} = A^\top x'$, so $c^\top (x^{\text{fin}} - x') = s^\top (x^{\text{fin}} - x')$. By Item 4.3(1), $s_i = \mu \tau_i(x) (-\phi_i'(x_i) + \nu_i \phi_i''(x_i)^{1/2})$ with $|\nu_i| \leq \epsilon$. For the log barrier, $-\phi_i'(t) = 1/(t-l_i) - 1/(u_i-t)$ and $\phi_i''(t)^{1/2} \leq 1/(t-l_i) + 1/(u_i-t)$, so for $x'_i \in [l_i, u_i]$ with $x'_i \leq x_i$, using $0 \leq x_i - x'_i \leq x_i - l_i$,

$$\begin{aligned} \left(-\phi_i'(x_i) + \nu_i \phi_i''(x_i)^{1/2} \right) (x_i - x'_i) &\leq (1 + \epsilon) \frac{x_i - x'_i}{x_i - l_i} - (1 - \epsilon) \frac{x_i - x'_i}{u_i - x_i} \\ &\leq 1 + \epsilon, \end{aligned}$$

and symmetrically when $x'_i \geq x_i$. For the remaining piece, $|\phi_i'(x_i) + \nu_i \phi_i''(x_i)^{1/2}| \leq (1 + \epsilon) \phi_i''(x_i)^{1/2}$ by Lemma 4.3, and $\phi_i''(x_i)^{1/2} |x_i^{\text{fin}} - x_i| \leq \beta$ by the movement bound. Adding,

$$\begin{aligned} c^\top (x^{\text{fin}} - x') &= \mu \sum_i \tau_i(x) \left(-\phi_i'(x_i) + \nu_i \phi_i''(x_i)^{1/2} \right) \left((x_i - x'_i) + (x_i^{\text{fin}} - x_i) \right) \\ &\leq \mu (1 + \epsilon) (1 + \beta) \|\tau(x)\|_1 \\ &\leq 2\mu \|\tau(x)\|_1, \end{aligned}$$

since $(1 + \epsilon)(1 + \beta) \leq (81/80)(1 + 1/40) < 2$. Taking x' optimal gives the claim. $\square$

The statement is for the exact diagonal $\bar{W}$; the algorithm holds a truncated copy of it, and the difference is handled by a perturbation bound in the proof of Lemma 6.4, not by weakening the hypothesis.

The augmented instance is the following.

Lemma 6.4. *Given A, b, c, l, u with B bits and $A^\top A$ of magnitude at most 2^B , one can construct in $\tilde{O}(\text{nnz}(A)B)$ bit operations an augmented instance $(\tilde{A}, b, \tilde{c}, \tilde{l}, \tilde{u})$ with*

$$\tilde{A} = \begin{bmatrix} A \\ E \end{bmatrix} \in \mathbb{R}^{(n+k) \times d}, \quad k \leq d, \quad \text{nnz}(\tilde{A}) \leq \text{nnz}(A) + d,$$

all of whose data has $O(B)$ bits, such that

1. the midpoint $\tilde{x} = (\tilde{l} + \tilde{u})/2$ is exactly feasible, $\tilde{A}^\top \tilde{x} = b$, and $\tilde{A}$ has full column rank d ;
2. with $s_{\text{init}} = \tilde{c}$ and $\mu_0 = 2^{O(B)}$, the triple $(\tilde{x}, s_{\text{init}}, \mu_0)$ is $\epsilon/C_{\text{start}}$ -centered in the sense of Definition 4.3;
3. if the primal is feasible, running the path to $\mu_f = 2^{-O(B)}$ and rounding the output inwards to $[l, u]$ gives a point $x \in [l, u]$ with $\|A^\top x - b\|_2 \leq 2^{-B}$ and $c^\top x \leq 2^{-B} + \min\{c^\top x' : A^\top x' = b, l \leq x' \leq u\}$;
4. $\log(\mu_0/\mu_f) = O(B)$.

Proof. Let $r_{\text{mid}} = b - A^\top(l+u)/2$ be the residual of the midpoint, $J = \{j : (r_{\text{mid}})_j \neq 0\}$, $k = |J| \leq d$, and Δ the least power of two above $\max_i(u_i - l_i)$. For each $j \in J$ add an artificial coordinate with box $[0, 4\Delta]$, midpoint 2Δ , and row $E_{j,:} = (r_{\text{mid}})_j(2\Delta)^{-1}e_j^\top$, so that $E^\top(2\Delta\mathbf{1}) = r_{\text{mid}}$ and the augmented midpoint $\tilde{x} = ((l+u)/2, 2\Delta\mathbf{1})$ satisfies $\tilde{A}^\top \tilde{x} = A^\top(l+u)/2 + r_{\text{mid}} = b$, giving the feasibility of Item 6.4(1); adding rows preserves column rank. Give the artificial coordinates objective coefficient Λ , the least power of two above $2^{B+2}\|E\|_2(1+M)$ with $M \triangleq \sum_i |c_i|(u_i - l_i)$, where $\|E\|_2 = \max_j |(r_{\text{mid}})_j|/(2\Delta)$ since the rows of E are supported on distinct columns. All the new data, Δ , the entries of E , and Λ , is formed from the input by $O(1)$ arithmetic operations per entry on numbers with $O(B)$ bits, using $\log n = O(B)$ as recorded after Equation (11), which is the cost claimed. The augmented instance has $n+k \leq 2n$ rows, so Section 4 applies to it with n replaced by $n+k$, which changes no $\tilde{O}(\cdot)$.

For Item 6.4(2), at the midpoint every barrier gradient vanishes, so with $s_{\text{init}} = \tilde{c}$ and $z = 0$ Item 4.3(2) of Definition 4.3 holds, Item 4.3(3) holds with left hand side zero by exact feasibility, and the centrality vector is $y_i = \tilde{c}_i/(\mu_0\tau_i\phi_i''(\tilde{x}_i)^{1/2})$ with $\phi_i''(\tilde{x}_i)^{1/2} = 2^{3/2}/(\tilde{u}_i - \tilde{l}_i)$ and $\tau_i \geq d/(n+k) \geq d/(2n)$, so

$$|y_i| \leq \frac{2n \|\tilde{c}\|_\infty \max_i(\tilde{u}_i - \tilde{l}_i)}{2^{3/2}d\mu_0} \leq \frac{\epsilon}{C_{\text{start}}}$$

for μ_0 any power of two above $2C_{\text{start}}\epsilon^{-1}(n/d)\max_i(\tilde{u}_i - \tilde{l}_i)\|\tilde{c}\|_\infty$, which is $2^{O(B)}$ because $\|\tilde{c}\|_\infty \leq \Lambda = 2^{O(B)}$. We take $\mu_f \triangleq 2^{-B-7}/d$, also $2^{-O(B)}$, and Item 6.4(4) follows.

For Item 6.4(3), let $(\tilde{x}, \tilde{s}, \mu_f)$ be the ϵ -centered triple for the augmented instance at the end of the path, with the estimate $\tilde{r}$ of Definition 4.6, and let $\hat{w}$ be the weights of step 3 of Figure 5 at this point, $\hat{w}_i = \text{Trunc}((\tilde{\tau}_i\phi_i''(\tilde{x}_i))^{-1}, 40\log\rho)$, so that SOLVE is against $H \triangleq \tilde{A}^\top \hat{W} \tilde{A}$. Let $\bar{W} \triangleq (\tilde{T}\Phi''(\tilde{x}))^{-1}$ be the exact diagonal of Lemma 6.3 and $\bar{H} \triangleq \tilde{A}^\top \bar{W} \tilde{A}$; since $\bar{w}_i \geq \rho^{-1}$ by Lemma 6.1, the truncation error ρ^{-20} per entry of Equation (12) gives $\hat{W} \approx_{\rho^{-18}} \bar{W}$ and $H \approx_{\rho^{-18}} \bar{H}$. Lemma 6.3 gives the exact point $\tilde{x}^{\text{fin}} = \tilde{x} + \bar{W} \tilde{A} \bar{H}^{-1}(b - \tilde{A}^\top \tilde{x})$, and the algorithm forms $\tilde{x}' \triangleq \text{Trunc}(\tilde{x} + \hat{W} \tilde{A} \text{SOLVE}(\tilde{r}), 40\log\rho)$, $\tilde{r} \triangleq \text{Trunc}(b - \tilde{A}^\top \tilde{x}, 20\log\rho)$, in $\tilde{O}((\text{nnz}(A) + d^2)B^2)$ bit operations, the B^2 being the cost of the one SOLVE, the entries being truncated having magnitude at most $2\max(\|\tilde{l}\|_\infty, \|\tilde{u}\|_\infty) \leq \rho$ by Lemma 6.3. Writing $r \triangleq b - \tilde{A}^\top \tilde{x}$, each entry of $\tilde{r}$ is a sum of the second kind on stored numbers cut to $20\log\rho$ bits, so $\|\tilde{r} - r\|_2 \leq (n+k)^{1/2}(\rho^{-36} + \rho^{-20}) \leq \rho^{-19}$ and $\|H^{-1}\tilde{r} - H^{-1}r\|_2 \leq \rho^{-18}$. Writing $\delta \triangleq \text{SOLVE}(\tilde{r}) - H^{-1}\tilde{r}$, the guarantee of Equation (1) is $\|\delta\|_H \leq \rho^{-10}\|\tilde{r}\|_{H^{-1}} \leq 2\rho^{-10}$, by Item 4.3(3), $\gamma \leq 1$, $\tilde{T} \approx_{\gamma/100} T(\tilde{x})$, $H \approx_{\rho^{-18}} \bar{H}$ and $\|\tilde{r} - r\|_{H^{-1}} \leq \rho^{-18}$, so $\|\delta\|_2 \leq 2\rho^{-9}$. The

truncation of the weights contributes

$$\begin{aligned}
& \left\| \widehat{W} \tilde{A} H^{-1} r - \overline{W} \tilde{A} \overline{H}^{-1} r \right\|_2 \\
& \leq \left\| \left(\widehat{W} - \overline{W} \right) \tilde{A} H^{-1} r \right\|_2 + \left\| \overline{W} \tilde{A} \left(H^{-1} - \overline{H}^{-1} \right) r \right\|_2 \\
& \leq \rho^{-18} \rho \cdot \rho^3 \|r\|_2 + \rho^3 \cdot 2\rho^{-18} \rho \cdot \rho \|r\|_2 \\
& \leq \rho^{-11},
\end{aligned}$$

using $\|\widehat{W} - \overline{W}\| \leq \rho^{-18} \rho$, $\|H^{-1} - \overline{H}^{-1}\| \leq \|H^{-1}\| \|\overline{H} - H\| \|\overline{H}^{-1}\|$ with $\|\overline{H} - H\| \leq 2\rho^{-18} \rho$, the magnitude bounds $\|\widehat{W}\|, \|\overline{W}\|, \|\tilde{A}\|, \|H^{\pm 1}\|, \|\overline{H}^{\pm 1}\| \leq \rho$ of Lemma 6.1, and $\|r\|_2 \leq \|\overline{H}\|^{1/2} \|r\|_{\overline{H}^{-1}} \leq \rho$. Together with the solve error, the truncation of the right hand side, and the final truncation, which by Equation (12) moves each entry by at most ρ^{-20} ,

$$\begin{aligned}
\|\tilde{x}' - \tilde{x}^{\text{fin}}\|_2 & \leq \|\widehat{W} \tilde{A}\|_2 (\|\delta\|_2 + \rho^{-18}) + \rho^{-11} + (n+k)^{1/2} \rho^{-20} \\
& \leq \rho^{-6},
\end{aligned}$$

and

$$\begin{aligned}
\|\tilde{A}^\top \tilde{x}' - b\|_2 & = \|\tilde{A}^\top (\tilde{x}' - \tilde{x}^{\text{fin}})\|_2 \\
& \leq \rho^{-5},
\end{aligned}$$

using $\rho \geq n$ and, below, $\rho \geq \|\tilde{c}\|_2$, both part of the choice of ρ recorded after Equation (11). Finally clip $\tilde{x}'$ into the augmented box coordinatewise; since $\tilde{x}^{\text{fin}}$ is inside the box, clipping moves each coordinate by at most $|\tilde{x}'_i - \tilde{x}^{\text{fin}}_i|$ and the two displayed bounds persist with ρ^{-5} replaced by ρ^{-4} . Call the result $(x, y) \in [l, u] \times [0, 4\Delta]^k$, with x the returned point.

It remains to compare with the original instance. Let x^* be optimal for the primal, so that $(x^*, 0)$ is feasible for the augmented instance and the augmented optimum is at most $c^\top x^*$. Lemma 6.3 gives

$$\begin{aligned}
c^\top x + \Lambda \mathbf{1}^\top y & \leq c^\top x^* + 2\mu_f \|\tau(\tilde{x})\|_1 + \|\tilde{c}\|_2 \rho^{-4} \\
& \leq c^\top x^* + 10d\mu_f + \rho^{-3},
\end{aligned}$$

using $\|\tau(\tilde{x})\|_1 \leq 5d$ from Definition 4.2. Since both x and x^* lie in $[l, u]$, $c^\top x - c^\top x^* \geq -M$, so $\Lambda \mathbf{1}^\top y \leq M + 10d\mu_f + \rho^{-3} \leq M + 1$ by the choice of μ_f , and therefore

$$\begin{aligned}
\|A^\top x - b\|_2 & \leq \|E^\top y\|_2 + \rho^{-4} \\
& \leq \|E\|_2 \mathbf{1}^\top y + \rho^{-4} \\
& \leq \frac{\|E\|_2 (1 + M)}{\Lambda} + \rho^{-4} \\
& \leq 2^{-B-2} + \rho^{-4} \\
& \leq 2^{-B},
\end{aligned}$$

the first step because $A^\top x - b = -E^\top y + (\tilde{A}^\top(x, y) - b)$, and the last because $\rho \geq 2^B$. For the objective, $\Lambda \mathbf{1}^\top y \geq 0$ gives $c^\top x \leq c^\top x^* + 10d\mu_f + \rho^{-3} \leq c^\top x^* + 2^{-B}$ by $\mu_f = 2^{-B-7}/d$, which is the guarantee of Item 6.4(3). The existence of x^* is the only place the feasibility of the primal is used. $\square$

6.4 Proof of the main theorem

Proof of Theorem 1. Run STEP on the augmented instance of Lemma 6.4 from $(\tilde{x}, z = 0, \mu_0)$, so $s = s_{\text{init}}$, until $\mu \leq \mu_f$, with the initial $\tilde{\tau}$ from the static estimate of Lemma 4.1 at accuracy $\gamma/100$. By Lemma 5.10 and Item 6.4(4), this is

$$r = \tilde{O}\left(d^{1/2} \log(\mu_0/\mu_f)\right) = \tilde{O}\left(d^{1/2} B\right)$$

calls, at the end of which the potential is $n^{O(1)}$ and the point is ϵ -centered at μ_f ; Item 6.4(3) converts this, with one more SOLVE call, into the claimed feasibility and objective guarantees for the primal, the feasibility of that program, which Item 6.4(3) needs, being witnessed by $x^{(0)}$. It remains to account for the bit complexity of one call, all matrices involved having magnitude $\rho = 2^{O(B)}$ by Lemma 6.1, so that $\log \rho = O(B)$.

- The Lewis estimate of step 2 is one call of Lemma 4.2 against SOLVE, with $T_S = \tilde{O}((\text{nnz}(A) + d^2) \log^2 \rho)$ by Item 2(3) of Theorem 2 and $\epsilon = \gamma/100$, so that $\epsilon^{-2} \log^2(1/\epsilon) = O(\log^7 n)$ by Equation (11), at

$$\tilde{O}\left(\epsilon^{-2} (\text{nnz}(A) \log \rho + T_S) \log^2(1/\epsilon)\right) = \tilde{O}\left((\text{nnz}(A) + d^2) B^2\right)$$

bit operations per round, hence $\tilde{O}(d^{1/2}(\text{nnz}(A) + d^2)B^3)$ over the r rounds. The one INITIALIZE is a one-off $\tilde{O}((\text{nnz}(A) + \text{MATMULT}(d, d, d))B)$ by Item 2(1), and the static estimate of Lemma 4.1 that supplies the initial $\tilde{\tau}$ is a one-off

$$\tilde{O}\left((\epsilon^{-2} (\text{nnz}(A) + d^2) + \text{MATMULT}(d, d, d)) \log \rho \log^2(1/\epsilon)\right) = \tilde{O}\left((\text{nnz}(A) + \text{MATMULT}(d, d, d)) B\right),$$

as $d^2 \leq \text{MATMULT}(d, d, d)$.

- The weights $\hat{w}$ of step 3 are ζ -leverage stable up to noise with $\zeta = O(\gamma)$ by Lemma 6.2, so by Item 2(2), applied through Corollary 3.3 because the weights are computed from the solver's own outputs, all UPDATE calls over the r rounds cost

$$\tilde{O}\left(r (\text{nnz}(A) + d^2) \log^2 \rho\right) = \tilde{O}\left(d^{1/2} (\text{nnz}(A) + d^2) B^3\right)$$

in total.

- The two SOLVE calls of step 5 cost $\tilde{O}((\text{nnz}(A) + d^2) \log^2 \rho) = \tilde{O}((\text{nnz}(A) + d^2) B^2)$ per round by Item 2(3), including the $\tilde{O}(\text{nnz}(A) \log \rho)$ overhead of the noisy solver of Imported result 3.1; DIRECTION in step 4 costs $\tilde{O}(n \log \rho)$ by Lemma 5.3; and the explicit updates of steps 1, 3 and 6 apply A and $A^\top$ a constant number of times and touch each coordinate $O(1)$ times, on numbers with $40 \log \rho$ bits, so they cost $\tilde{O}(\text{nnz}(A)B)$ per round. Over the r rounds this is $\tilde{O}(d^{1/2}(\text{nnz}(A) + d^2)B^3)$.

Summing, and using $\text{MATMULT}(d, d, d) = \tilde{O}(d^{1+\sqrt{2}}) \leq \tilde{O}(d^{1/2} \cdot d^2)$ from Equation (2) to absorb the one-off terms, the total is

$$\tilde{O}\left(d^{1/2} (\text{nnz}(A) + d^2) B^3\right)$$

bit operations, the $\tilde{O}$ hiding factors polylogarithmic in n only: one factor B is the bit length of every number handled, one the iteration count $O(\log \rho)$ of a SOLVE, and one the number $\tilde{O}(d^{1/2}B)$ of rounds. The failure probability is n^{-7} from the static estimate, rn^{-7} from Lemma 5.10, on whose event Lemma 5.11 is deterministic, and $n^{-5} + 2Qn^{-3}$ from Theorem 2, the number Q of SOLVE

calls being $\tilde{O}(r) = \tilde{O}(d^{1/2}B)$: a constant number per round in STEP, $O(\log n)$ per round inside the structure, and $\tilde{O}(\gamma^{-2}) = \tilde{O}(1)$ per round in the Lewis estimate. The total failure probability is therefore $\tilde{O}(d^{1/2}B)n^{-3} \leq \tilde{O}(dn^{-3}) \leq \tilde{O}(n^{-2})$, the hidden factor being polylogarithmic in n , which is the “high probability” of the theorem; by the standing assumption $B \leq d^{1/2}$ the number of rounds is $r = \tilde{O}(d^{1/2}B) \leq d \log^{O(1)} n \leq n^2$ for d larger than a constant, which is the hypothesis $r \leq n^2$ of Theorem 2 and Corollary 3.3. No term of the form $\text{MATMULT}(n, d, d)$ survives: the only dense operations are on $d \times d$ matrices and on the $\tilde{O}(d)$ -row sampled matrix, and the full A is only ever touched by matrix vector products and by the single static Lewis estimate of Lemma 4.1. $\square$

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A Proof of Lemma 2.2

Proof. Write L_v for the constant 1 if operation v is $+$ or $-$, $2M$ if it is $\times$, $2M^3$ if it is $\div$, and M if it is a square root; in every case $L_v \leq 2M^3$. We show by induction along the formula that the error e_v of the computed value at operation v satisfies

$$e_v \leq 2^{-\ell} \sum_{u \preceq v} \prod_{u \prec t \preceq v} L_t,$$

where the sum runs over the operations u in the subformula of v , including v , and the product over the operations strictly after u on the unique path to v . Since at most q factors of each product exceed 1 and each is at most $2M^3$, the right hand side is at most $m(2M^3)^q 2^{-\ell}$, which is the claim for the output and is at most $1/(2M)$ for every intermediate result by the hypothesis.

Let v have inputs with exact values x, y and computed values $\tilde{x}, \tilde{y}$, with errors $e_x, e_y \leq 1/(2M)$ by induction. For $+$ and $-$ the exact operation on the computed inputs is off by at most $e_x + e_y$. For $\times$,

$$\begin{aligned} |\tilde{x}\tilde{y} - xy| &\leq |\tilde{x}| |\tilde{y} - y| + |y| |\tilde{x} - x| \\ &\leq \left(M + \frac{1}{2M}\right) e_y + M e_x \\ &\leq 2M (e_x + e_y). \end{aligned}$$

For $\div$ the divisor satisfies $|y| \geq 1/M$ and hence $|\tilde{y}| \geq 1/(2M)$, so

$$\begin{aligned} \left| \frac{\tilde{x}}{\tilde{y}} - \frac{x}{y} \right| &= \frac{|\tilde{x}y - x\tilde{y}|}{|y\tilde{y}|} \\ &\leq 2M^2 (|y| e_x + |x| e_y) \\ &\leq 2M^3 (e_x + e_y). \end{aligned}$$

For the square root the radicand satisfies $x \geq 1/M$ and $\tilde{x} \geq 1/(2M) > 0$, so

$$\left| \tilde{x}^{1/2} - x^{1/2} \right| = \frac{|\tilde{x} - x|}{\tilde{x}^{1/2} + x^{1/2}} \leq M^{1/2} e_x \leq M e_x.$$

In every case the exact operation on the computed inputs is within $L_v(e_x + e_y)$ of the exact result; its absolute value is at most $M + 1/(2M) < 2^\ell$, so it has at most ℓ bits before the decimal point and the truncation adds at most $2^{-\ell}$, giving $e_v \leq L_v(e_x + e_y) + 2^{-\ell}$, which is the inductive bound. $\square$